Instrumented Principal Component Analysis*

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Abstract

IPCA is a new approach to latent factor analysis that, besides the main panel of interest, incorporates vast additional information. It estimates a structural link between the time-varying instruments and the dynamic factor loadings, rather than estimating static loadings for each individual. It improves factor model estimation especially in settings with dynamic loadings and large cross sections, and helps to ascertain the economic relationships among factors and individuals via the observable instruments. The estimation is fast to calculate and accommodates unbalanced panels. We show consistency and asymptotic normality under general conditions. We demonstrate the advantages of IPCA in simulated data and in applications to equity asset pricing and international macroeconomics.

Keywords: IPCA, Latent Factors, Factor Loading Modeling, Identification, Large Panel

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1 Introduction

Latent factor analysis is an empirical workhorse in economics and finance. Its essence is parsimony: a small number of common factors drive the variation of a large cross section (Geweke, 1977; Sargent and Sims, 1977). A large literature has moved beyond using factor analysis merely for exploratory dimension-reduction and data pre-processing, to using it as an estimator for economic models that link hundreds, thousands, or millions of individual variables to a few aggregate economic forces. At the heart of these models are factor loadings—they quantify economic relationships as individuals' sensitivities to the aggregate factors.

High dimensional environments present an econometric challenge because the number of factor loadings to be estimated is increasing with the size of the cross section. The literature studies this issue predominantly in the case of static factor loadings (e.g., Bai and Ng, 2002; Stock and Watson, 2002; Bai and Li, 2012, 2016). But the estimation challenges are magnified when factor loadings are time-varying, as called for in many important economic applications. In this case, the number of factor loadings may even exceed the size of the data panel.[1]

We propose instrumented principal component analysis, or IPCA, to overcome the challenges of recovering accurate estimates of latent factor models when panel dimensions are large and/or when loadings are time-varying. Our solution draws on conditioning data, beyond that in the main panel of interest, to instrument for the factor loadings. This makes it possible to simultaneously bring more data to bear on the statistical model while introducing structure among factor loadings that *reduces* the model's degrees of freedom. In doing so, it also significantly expands the economic content of factor models.

For concreteness, let $t = 1, \ldots, T$ index time and $i = 1, \ldots, N$ index individuals. The IPCA model is

$$x_{i,t} = \beta_{i,t} f_t + \mu_{i,t}, \tag{1}$$

$$\beta_{i,t} = z_{i,t}\Gamma + \eta_{i,t}. \tag{2}$$

[1] Su and Wang (2017) achieve smoothly varying loadings by repeatedly estimating static ones in moving kernels. Recent advances in state-space models view factor loadings as latent dynamic processes themselves (Primiceri, 2005; Pruitt, 2012; Del Negro and Otrok, 2008, among others).

Equation (1) is the canonical linear factor model in which $x_{i,t}$ is scalar panel data, f_t and $\beta_{i,t}$ are the factors and their loadings ($K \times 1$ and $1 \times K$ vectors, respectively, where K is the number of factors), and $\mu_{i,t}$ is the idiosyncratic error. Equation (2) is the core of IPCA. It maps the observed information in an L-dimensional vector of instrumental variables ($z_{i,t}$) to the K-dimensional factor loadings $\beta_{i,t}$ ($L \geq K$). The key restriction giving content to IPCA is this mapping is fixed over time and across individuals. The idea follows the seminal model of Projected PCA (Fan, Liao, and Wang, 2016), who allow the mapping to be semi-parametric, but restrict both z and β to be constant over time. We parameterize the mapping linearly with the $L \times K$ matrix Γ. The mapping matrix Γ is our interest of estimation and asymptotic analysis, together with the latent factors f_t.

The error term $\eta_{i,t}$ allows for unobservable behavior of $\beta_{i,t}$ beyond what the observable instruments can capture. A key identifying assumption for consistent estimation is an orthogonality condition between $z_{i,t}$ and the structural error term, analogous to an exclusion restriction in IV models. If $\eta_{i,t}$ is constant over time (or contains a constant component), it can play the role of the constant loadings in traditional PCA, although this is not the focus of this paper.

IPCA draws not only on the main panel $x_{i,t}$, but also on auxiliary observed information $z_{i,t}$ about the panel units. These instruments can vary over time and across individuals and help sharpen the latent factor structure in $x_{i,t}$. Economic theory often suggests that individuals possess ancillary observable attributes that convey information about their factor loadings. For example, Fama and French (1993) note that firm-level observable characteristics "such as size and book-to-market equity must proxy for sensitivity to ... risk factors in returns". Kelly, Pruitt, and Su (2019, henceforth KPS19) use IPCA to formalize this idea and estimate a conditional asset-pricing model with an unprecedented degree of empirical success in equity returns. As another example, option "Greeks" are sensitivities to underlying risk factors and are mechanically linked to observable and dynamic quantities such as the option's moneyness and maturity (Büchner and Kelly, 2021). As a third example, some macroeconomic models describe inter-firm trade as interaction of nodes on a network. As a firm's centrality increases or decreases it becomes more or less integral to overall economic activity, thus implying dynamic loadings on an aggregate

growth factor (Acemoglu and Azar, 2020).

The common idea underlying each of these examples is that the individual's (stock, option, or firm's) constant identity becomes irrelevant once we condition on the appropriate attributes of the individual. These attributes (book-to-market ratio, moneyness, network centrality) that dictate the individual's exposure to aggregate fluctuations are often observable and thus ripe for harvesting in the estimation of latent factor models. IPCA operationalizes this logic through equation (2). The matrix Γ determines how each of the L attributes maps into each of the K factor loadings. It is a constraint imposed on the factor loadings in the generic factor model (1) by requiring that the Γ mapping applies uniformly for all individuals and in all time periods.

Despite the fact that IPCA brings more data to bear on the factor model, the constraint in equation (2) renders IPCA an especially parsimonious factor model. The idea of IPCA is that individual loadings are an unnecessary excess, requiring a divergent count of $N \times K$ parameters in the commonly-studied large cross section limit. IPCA displaces the need to estimate individual-specific loading parameters, instead requiring only an understanding of how individuals' attributes map into their factor loadings, summarized by the $L \times K$ parameters in Γ, whose size does not increase with either of the panel's dimensions. Accordingly, our theoretical analyses consider large panel asymptotics with N, T approaching infinity while holding L, K constant.[2] The ability to assimilate large panel data in a factor model with fixed parameter dimensions gives IPCA Γ estimation a convergence rate of $\sqrt{NT}$ (see Theorem 3), faster than PCA β's rate of $\sqrt{T}$ (see Bai and Ng, 2002).

At the same time, the constraint that IPCA imposes on factor loadings brings new economic content to factor modeling. It probes the economic content of a factor model by estimating which attributes in $z_{i,t}$ are important for capturing the individuals' sensitivities to aggregate shocks. In summary, two directions of improvements—expanded input data and decreased model freedom—combine to allow for deeper economic underpinnings than traditionally ascertained by

[2]Allowing for a larger number of characteristics (large L) is also relevant for many applications. Bybee et al. (2023) applies IPCA in a news-narrative asset-pricing setting and uses group lasso for characteristic selection. Fu et al. (2026) provides a formal treatment of the high-dimensional setting by combining group lasso with nuclear norm regularization and establishing asymptotic theory under jointly large N, T, and L.

static factor models.

Estimation: In the following analysis, we often plug $\beta_{i,t}$ in equation (2) into equation (1) and combine each equation's error term into a compound error $e_{i,t}$, yielding an equivalent representation of the IPCA model:

$$x_{i,t} = z_{i,t}\Gamma f_t + e_{i,t}, \qquad e_{i,t} := \eta_{i,t} f_t + \mu_{i,t}. \tag{3}$$

IPCA is estimated by solving the least squares problem that minimizes the sample sum of squared errors ($e_{i,t}$) over parameters Γ and $\{f_t\}$ jointly. This least squares estimation follows that of PCA, which minimizes the sum of squared errors over $\{f_t\}$ and static loading $\{\beta_i\}$.

The optimization does not admit an analytical solution in general, but is speedily solved by an alternating least squares (ALS) algorithm. It iterates between two sub-problems—minimizing over Γ while holding $\{f_t\}$ fixed and minimizing over $\{f_t\}$ while holding Γ fixed, until convergence. Importantly, the two partial optimization sub-problems are ordinary least squares. This has a number of practical benefits. For example, the procedure converges quickly and without the need for complicated numerical algorithms. Also, IPCA handles unbalanced panels as easily as pooled panel OLS, which is a great advantage in many applications.

Empirical Applications: The empirical applications demonstrate the two main benefits of IPCA—improved factor estimation and richer economic content revealed by the instrumenting relationship. Section 8.1 applies IPCA to the international economic growth data, using time-varying country attributes as instruments. It finds a greater proportion of country-level variations can be attributed to exposures to the global fluctuations than a PCA would find (and with only a fraction of parameters). Allowing for dynamic loadings as countries evolve is the key to improving the factor model fit. We find inflation and growth volatility are positively associated with sensitivity to global fluctuations. Exports in emerging countries and imports in developed countries are positively associated with the sensitivity to global fluctuations. Section 8.2 extends KPS19 to newly-listed firms. IPCA is applied to estimate systematic risk factors of stock returns by instrumenting factor exposures (β) with firm characteristics. The improvement in model fit compared to PCA is especially salient out of sample. We apply Γ estimated with incumbent firms to the observable characteristics of newly-listed firms, and find the new firms' risk exposures are

still recovered well. Effectively, we extrapolate the structural parameter (Γ) to firms with a short history of public data, which is useful to measure new firms' risk as well as expected return (cost of capital). This is not feasible with PCA or other methods that estimate β with firm-by-firm time-series regressions, since they would suffer from the short data history of the new firms.

Asymptotic Results We show consistency, the rate of convergence, and the limiting distributions for the Γ and f_t estimators within the framework of a large panel wherein N, T simultaneously approach infinity. Notably, Γ converges at the rate of $\sqrt{NT}$. As a comparison, PCA loading's (β) rate is $\sqrt{T}$ since it relies on each individual's time-series information. IPCA additionally picks up cross-sectional information to estimate Γ—the benefit of bringing the wealth of instrumental information into the problem. Meanwhile, the estimation of f_t achieves a convergence rate of $\sqrt{N}$, the same as PCA. This is because, in either case, estimating f_t relies on cross-sectional linear regressions at each t, which converges at rate of $\sqrt{N}$.

Rotational Identification: Rotational unidentification is a well-known issue in factor analysis. In the context of IPCA, the mapping matrix Γ and the factors $\{f_t\}$ cannot be separately identified without further assumptions because a parameter pair $\Gamma, \{f_t\}$ and any of its "rotations" $\Gamma R, \{R^{-1} f_t\}$ generate identical sample fits.

It is well understood that the choice of identifying assumption, or "normalization," is not unique and that different ones induce different economic interpretations. Our analysis is related to Bai and Li (2012, 2016), which analyze the asymptotics of PCA for different normalization conditions. The IPCA structure allows for a unified analysis that is general to any normalization choice. We show how the normalization choice affects the rate of convergence and the asymptotic distribution. The asymptotic distribution of parameter estimation is decomposed into a component arising from the estimation problem absent a normalization, and a separate component purely attributable to the normalization assumption. The asymptotic error only depends on the first component, so long as the true and estimated parameters are subject to the same normalization. The results discussed above are derived in a general fashion for *any* valid identifying assumption. We provide two example normalization specifications, where the asymptotic variances are evaluated straightforwardly given the general formula.

Relation to Existing Papers: The seminal method of Projected PCA (Fan, Liao, and Wang, 2016) also uses individual-level data to provide information about factor loadings. IPCA admits time-varying instruments, effectively extending the amount of instrumental information from a fixed cross section to a two-dimensional panel. Thereby, IPCA can model dynamic factor loadings, which is relevant in many economic applications as illustrated above. Fan et al. (2016) allow a non-linear relationship between instruments and factor loadings, which is estimated semi-parametrically, whereas we specify a linear relationship parameterized by Γ (Eq. 2).[3] The parameterization leads to IPCA's simultaneous estimation of Γ and $\{f_t\}$. Even in the special case of constant instruments, the estimation is different from Projected PCA's two-step procedure of first projecting and then conducting a traditional PCA. The asymptotic analysis, especially on the asymptotic distribution of the estimation, are also different.

IPCA is also related to the literature on panel data with interactive fixed effects, see Bai (2009) and Moon and Weidner (2015) among others. The connection is methodological, these papers also study least-squares estimation in panel models with latent factors. However, the models are not nested. In interactive fixed-effects regressions, the factor structure enters as an unrestricted latent component and the main parameter of interest is typically a regression coefficient, whereas in IPCA the loadings themselves are structured through observables via the common map Γ, which is the main object of interest.

This paper establishes the econometric theory for IPCA's applications. KPS19 applies IPCA to estimate systematic risk factors of stock returns, leveraging observable firm characteristics as instruments.[4] That paper revolves around the empirical asset pricing research question of whether average stock returns represent risk compensation or mispricing, which is different from the analysis in this econometric paper. The theories developed in this paper clarify the conditions for consistent estimation, establishes convergence rates and asymptotic normality, and explains how the parsimonious parameterization leads to the significantly improved factor model fits.

[3] IPCA can be extended to allow for a non-linear instrument-to-loading mapping too. Just like in Fan et al. (2016), one can expand the set of instruments by adding basis functions such as polynomials and splines of z.

[4] Besides KPS19, IPCA is also applied to study the factor structures of international equity returns (Langlois, 2021), option returns (Büchner and Kelly, 2021), corporate bonds (Kelly et al., 2020), momentum factors (Kelly et al., 2018), and news narrative-based factors (Bybee et al., 2023).

2 The Data Generating Process

Henceforth, we write Γ^0 and f_t^0 with the "0" superscript for to the true parameters, whereas Γ, f_t refers to the parameters in general discussion.

We work with two-dimensional panel data $x_{i,t}, z_{i,t}$ and study the asymptotics when both N and T increase simultaneously to infinity. Panel $x_{i,t}$ is generated according to (3) by primitive random variables $z_{i,t}, \eta_{i,t}, \mu_{i,t}$ (which are panel variables) f_t^0 (which is a time series variable, i.e., common to different individuals and no i subscript). This section provides the required assumptions on these variables, which are defined on probability space $(\Omega, \mathfrak{F}, Pr)$. Besides the factor structure of $x_{i,t}$ that is our target, z, η, μ could well have their own factor structures too, which we do not model concretely so as to leave it as unrestricted as possible. Following Andrews (2005) and Kuersteiner and Prucha (2013), we assume there exists a σ-algebra $\mathcal{C} \subset \mathfrak{F}$, such that a $\mathcal{C}$-measurable variable is common across individuals and is about the cross-sectional population distribution at each time. We assume the cross-sectional average of panel variables converges to the population counterpart—the expectation conditional on $\mathcal{C}$:

Assumption A.1 (Cross-sectional LLN)**.** *Let $y_{i,t} = g(z_{i,t}, \eta_{i,t}, \mu_{i,t}, f_t^0)$, where g is any continuous function. If $\mathbb{E}\left\|y_{i,t}\right\|^2 < \infty$, then $\frac{1}{N}\sum_{i=1}^N y_{i,t} \xrightarrow{L^2} \mathbb{E}\left[y_{i,t}|\mathcal{C}\right]$ as $N \to \infty$.*

For example, fixing t, the cross-sectional sample second moment $\frac{1}{N}\sum_i z_{i,t}^\top z_{i,t}$ converges to the population second moment $\Omega_t^{zz} := \mathbb{E}\left[z_{i,t}^\top z_{i,t}\middle|\mathcal{C}\right]$, which is $\mathcal{C}$-measurable common variable ($L \times L$ matrix) and rightfully index by only t but not i. Additionally, we assume a common variable's time-series average converges, and that the unconditional distribution of the cross section does not change over time:

Assumption A.2 (Common Variable's Times-series LLN)**.** *If ϕ_t is $\mathcal{C}$-measurable and $\mathbb{E}\left\|\phi_t\right\|^2 < \infty$, then $\frac{1}{T}\sum_{t=1}^T \phi_t \xrightarrow{L^2} \mathbb{E}\phi_t$ as $T \to \infty$.*

Assumption A.3 (Time-series Stationarity)**.** *The unconditional distribution of $\left[z_{1,t}, \ldots, z_{N,t}, \eta_{1,t}, \ldots, \eta_{N,t}, \mu_{1,t}, \ldots, \mu_{N,t}, f_t^0\right]$ is the same if t is shifted to $t+1, t+2, \ldots$.*

These conditions are about the general panel structure and are repeatedly applied in the proofs.[5] The role of Assumptions A.1–A.3 is to organize the asymptotic argument in this common-shocks environment. Assumption A.1 delivers a cross-sectional law of large numbers conditional on $\mathcal{C}$, Assumption A.2 governs time averaging of $\mathcal{C}$-measurable common variables, and Assumption A.3 keeps the cross-sectional environment stable over time. Economically, $\mathcal{C}$ can be viewed as the date-t common information generated by the latent factors and other aggregate shocks shared by all units.[6] Next, we lay out more specific assumptions, starting from the assumptions for consistency:

Assumption B (Instruments orthogonal to error)**.** $\mathbb{E}\left[z_{i,t}^{\top} e_{i,t} \middle| \mathcal{C}\right] = \mathbf{0}_{L\times 1}$.

This is the key condition giving content to IPCA, which is analogous to the exclusion restriction of instrumental variables (IV). In particular, by conditioning on $\mathcal{C}$, the orthogonality between $z_{i,t}$ and $e_{i,t}$ holds in the cross-sectional population *at each* t.[7] The key structural feature of IPCA is that the loadings are governed by a common matrix Γ that links observable characteristics to factor exposures across all dates. This common structure enables pooling across both i and t, delivers a parsimonious finite-dimensional parameterization of the loading process, and underlies the $\sqrt{NT}$ rate for estimating Γ. This conditioning is useful because $z_{i,t}$ and $e_{i,t}$ may comove unconditionally through the same aggregate shocks. Assumption B requires that, once those common shocks are held fixed, the remaining cross-sectional variation in $z_{i,t}$ be orthogonal to the residual term $e_{i,t}$. Alternatively, it is sufficient to impose the orthogonality with respect to each of the two primitive errors: $\mathbb{E}\left[z_{i,t}^{\top}\eta_{i,t}\middle|\mathcal{C}\right] = \mathbf{0}$ and $\mathbb{E}\left[z_{i,t}^{\top}\mu_{i,t}\middle|\mathcal{C}\right] = \mathbf{0}$.

Assumption C (Moment Existence)**.** (1) $\mathbb{E}\left\|f_t^0\right\|^4 < \infty$. (2) $\mathbb{E}\|z_{i,t}e_{i,t}\|^2 < \infty$.

Assumption D (Boundedness)**.** (1) Γ*'s parameter space* Ψ *is compact and bounded away from rank deficiency:* $\exists \epsilon > 0$*, s.t.* $\forall \Gamma \in \Psi$*,* $\det \Gamma^{\top}\Gamma > \epsilon$*.* (2) *Almost surely,* $z_{i,t}$ *is bounded.* (3) *Let* $\Omega_t^{zz} := \mathbb{E}\left[z_{i,t}^{\top} z_{i,t}\middle|\mathcal{C}\right]$*,* Ω_t^{zz} *is bounded away from singularity:* $\exists \epsilon > 0$*, s.t. almost surely* $\det \Omega_t^{zz} > \epsilon$*.*

[5] Appendix I provides a set of low-level conditions on the probability space and the panel structure that are sufficient for these conditions.

[6] Conditioning on $\mathcal{C}$ is a way to separate the common time-t realization from the cross-sectional heterogeneity that remains after that realization is taken as given.

[7] In Fan, Liao, and Wang (2016) where there is no time-variation, the equivalence to Assumption B is simply a cross-sectional orthogonality condition (see Fan et al., 2016, Page 222).

Assumption C lists the required finite moments that will be cited when applying Assumptions A.1, A.2. Assumption D guarantees the terms involving the second moment of instruments, $\frac{1}{N}\Gamma^\top\left(\sum_i z_{i,t}^\top z_{i,t}\right)\Gamma$, which frequently appear in the upcoming analysis, stay well behaved. This result is stated in Lemma 5, which is then cited repeatedly when proving other results. Notice that Assumption D(1) implies that Γ has full column rank K, and does so uniformly over the parameter space Ψ. A necessary condition is $L \geq K$: the number of instruments must be at least as large as the number of latent factors. This rank condition means that the instruments are informative. If full rank fails, the model is over-specified in K relative to the instruments' information content.

For asymptotic normality and deriving the asymptotic variance, we impose the following additional assumptions.

Assumption E (Central Limit Theorems)**.**

(1) *As* $N, T \to \infty$, $\frac{1}{\sqrt{NT}}\sum_{i,t}\text{vect}\left(z_{i,t}^\top e_{i,t} f_t^{0\top}\right) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \Omega^{zef}\right)$.[8]

(2) *As* $N \to \infty$, $\frac{1}{\sqrt{N}}\sum_i z_{i,t}^\top e_{i,t} \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \Omega_t^{ze}\right)$, *where* $\Omega_t^{ze} := \mathbb{V}\text{ar}\left[z_{i,t}^\top e_{i,t} \middle| \mathcal{C}\right]$. *Unconditionally,* $\mathbb{E}\left\|\frac{1}{\sqrt{N}}\sum_i z_{i,t}^\top e_{i,t}\right\|^4$ *is bounded, as* $N \to \infty$.

The asymptotic distribution of the term in Assumption E(1), $z_{i,t}^\top e_{i,t} f_t^{0\top}$, is important, as it eventually determines the asymptotic estimation errors (see Theorem 2). We are assuming a panel-wise central limit theorem given that $\mathbb{E}\left[z_{i,t}^\top e_{i,t} f_t^{0\top}\right] = \mathbf{0}$ (implied by Assumption B). The assumption nonetheless rules out an extreme case with constant loadings. Suppose $z_{i,t}$ and $\eta_{i,t}$ are constant over t and that $\eta_{i,t} \neq 0$, the term in Assumption E(1) would converge at the rate of $\sqrt{N}$, implying the estimator is only $\sqrt{N}$-consistent. The reduction from $\sqrt{NT}$ to $\sqrt{N}$ is understandable, as the time-constant instrumental relationship $\beta_i = z_i\Gamma + \eta_i$ would repeat over time, rendering a large T useless for the inference of Γ. PCA, rather than IPCA, can be more applicable in this constant β case, which nonetheless is $\sqrt{N}$-consistent too. Thus, Assumption E(1) is to be interpreted as requiring sufficiently time varying η_{it} for the benchmark $\sqrt{NT}$ rate. When the loading noise changes more slowly over time, the estimator can remain consistent,

[8] $\text{vect}(A) := \text{vec}(A^\top)$ is a vectorization operator. See Appendix A for details. $\Omega^{zef} := \text{plim}_{N,T\to\infty}\frac{1}{NT}\left(\sum_{i,t}\text{vect}\left(z_{i,t}^\top e_{i,t} f_t^{0\top}\right)\right)\left(\sum_{i,t}\text{vect}\left(z_{i,t}^\top e_{i,t} f_t^{0\top}\right)\right)^\top$.

but the effective convergence rate may fall below $\sqrt{NT}$ and move toward $\sqrt{N}$ as the degree of persistence increases to non time-varying η, see Appendix H.

Assumption E(2) is a cross-sectional central limit theorem fixing t (conditional on $\mathcal{C}$). It appears in the asymptotic distribution of the factor estimation (see Theorem 4.b), and works in a similar way as Bai's (2003) Assumption F3.

Assumption F (Constant Instrument Second Moment)**.** *Ω_t^{zz}, which is defined as $\mathbb{E}\left[z_{i,t}^\top z_{i,t}\middle|\mathcal{C}\right]$, is constant. (When this assumption is imposed, we drop the t subscript in Ω_t^{zz} and write Ω^{zz}.)*

Assumption F shuts down the variation of Ω_t^{zz} in order to retain a concise expression for the asymptotic variance for the estimation error of Γ (in Theorem 3). The essence of the asymptotic theory and other theorems do not depend on it.

3 Estimator as Normalized Optimizer

Estimation consists of two steps: optimizing the objective function to find the set of equivalent solutions, followed by a normalization of the solutions that selects a unique element of this set to serve as the parameter estimate. The second step is necessary because IPCA has a rotational unidentification issue well-known from other latent factor settings. While our identification approach in some sense follows convention in the literature, we more formally construct identification conditions. This formality is not just for defining the estimator, but for more rigorously expounding the notion of a "true" parameter and the corresponding measurement of estimation errors. As a result, we develop a general asymptotic analysis framework for estimators normalized by *any* valid identifying assumption, which is also applicable in contexts beyond IPCA (such as in the asymptotic analysis of PCA).

3.1 Optimization

IPCA is estimated as a least squares problem that minimizes the sample sum of squared errors (SSE) over parameters Γ and $\{f_t\}$:

$$\min_{\Gamma,\{f_t\}} \sum_{i,t} \left(x_{i,t} - z_{i,t}\Gamma f_t\right)^2. \tag{4}$$

This objective is inspired by PCA which also optimizes the sample SSE but over parameters $\{\beta_i\}$ and $\{f_t\}$.

Unlike PCA, the IPCA optimization problem does not have an analytical solution through eigen-decomposition and must be solved numerically.[9] We adopt the Alternating Least Squares (ALS) algorithm. Notice, the SSE target in (4) is quadratic in either Γ or $\{f_t\}$ when the other is fixed, allowing for closed-form optimizations over Γ or $\{f_t\}$ one at a time. Given any Γ, factors $\{f_t\}$ are t-separable and solved with cross-sectional OLS for each t:

$$\widehat{f}_{t(\Gamma)} := \arg\min_{f_t} \sum_i \left(x_{i,t} - z_{i,t}\Gamma f_t\right)^2 = \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1} \Gamma^\top Z_t^\top x_t.^{10} \tag{5}$$

Symmetrically, given $\{f_t\}$, the optimizing Γ (vectorized as γ) is solved with a pooled panel OLS of $x_{i,t}$ onto LK regressors $z_{i,t} \otimes f_t^\top$:

$$\arg\min_{\gamma} \sum_{i,t} \left(x_{i,t} - z_{i,t}\Gamma f_t\right)^2 = \left(\sum_{i,t} \left(z_{i,t}^\top \otimes f_t\right)\left(z_{i,t} \otimes f_t^\top\right)\right)^{-1} \left(\sum_{i,t} \left(z_{i,t}^\top \otimes f_t\right) x_{i,t}\right). \tag{6}$$

Throughout the paper, the lower case γ represents the row-wise vectorized Γ, $\gamma := \text{vect}\left(\Gamma\right)$ (likewise for γ^0, $\widehat{\gamma}$, and so forth).

The ALS algorithm begins with an initial guess and iterates between updates of Γ and $\{f_t\}$ while holding the other fixed according to the above first-order conditions. The program stops when the optimality conditions are satisfied up to a predetermined tolerance. In practice, it converges after a few hundred iterations in a matter of seconds in our empirical and simulated datasets. The speedy calculation is a notable feature that allows for widespread application of IPCA. For example, it is easy to repeatedly estimate the model for a large number of generated samples, as required in simulation-based inference procedures. Another practical advantage of the numerical method is it adapts to unbalanced panels, because both the partial optimizations are regressions which tolerate missing values. In particular, we can rewrite Equations (4)-(6) by changing all the summation signs to summing over *only* the observed i, t entries, and the same estimation procedure follows through.[11]

[9] In the special case that $Z_t^\top Z_t$ is constant across t, the solution of (4) reduces to the SVD of the interaction of $x_{i,t}$ and $z_{i,t}$, which is similar to Fan et al.'s (2016) Projected PCA.

[10] Here, x_t is the $N \times 1$ vector of $x_{i,t}$, Z_t is the $N \times L$ matrix of $z_{i,t}$ stacked up.

[11] In applications with missing values, we first "fill up" data $\{x_{i,t}, z_{i,t}\}$ at any unobserved i, t entry with zeros, and then use the same ALS program for the completed panel. It is easy to show this process is equivalent to minimizing the summation over only the observed entries.

Because the least-squares problem (4) is not necessarily jointly convex in $(\Gamma, \{f_t\})$, it is useful to record the computational guarantee delivered by the ALS iterations. Appendix D shows that each F-step and Γ-step is an exact least-squares minimization over one block while holding the other fixed. Consequently, the sequence of objective values is nonincreasing and converges to a finite limit. It also shows that any limit point of the ALS iterations is a stationary point of the unconstrained least-squares objective; see Proposition 4. Thus ALS is guaranteed to monotonically decrease the objective and to approach stationary solutions, although no claim is made that it converges to a global optimum.

The asymptotic analysis proceeds with Γ first by concentrating out f_t as an intermediate object. Define the *target function* $\widehat{G}(\Gamma)$ as the concentrated SSE objective:

$$\widehat{G}(\Gamma) = \frac{1}{2NT} \sum_{i,t} \left(x_{i,t} - z_{i,t} \Gamma \widehat{f}_{t(\Gamma)} \right)^2, \tag{7}$$

where $\widehat{f}_{t(\Gamma)}$ is the optimal factor given any Γ according to (5). Define the *score function* as the derivative: $\widehat{S}(\Gamma) = \frac{\partial \widehat{G}(\Gamma)}{\partial \gamma}$. Therefore, the two-argument joint minimization problem (Eq. 4) is equivalent to minimizing $\widehat{G}(\Gamma)$, or solving the first order condition $\widehat{S}(\Gamma) = \mathbf{0}_{LK \times 1}$, with respect to Γ only. The asymptotic analysis proceeds by first analyzing $\widehat{S}$ to characterize the asymptotics of $\widehat{\Gamma}$ while treating the factor estimate $\widehat{f}_t(\widehat{\Gamma})$ as an implicit intermediate. Once the asymptotic analysis of $\widehat{\Gamma}$ is done, Section 6 finishes up the factor asymptotics by plugging $\widehat{\Gamma}$ into $\widehat{f}_t(\Gamma)$.

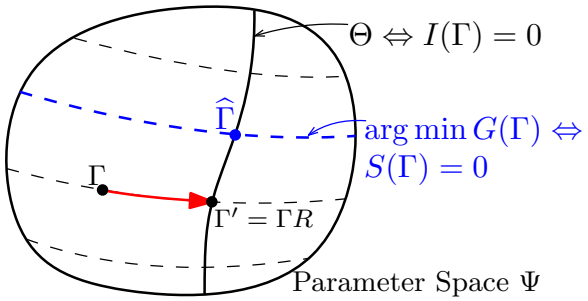
3.2 Normalization

In population, a class of "rotated" true parameters $\Gamma^0 R$ yields exactly the same data generating process for $x_{i,t}, z_{i,t}$, so long as true factors are rotated inversely as $R^{-1} f_t^0$ (for any full rank $K \times K$ matrix R). The counterpart sample issue is that the target and score are invariant to rotation: if $\widehat{\Gamma}$ solves $\widehat{S}(\Gamma) = \mathbf{0}$, then any rotation $\widehat{\Gamma} R$ is also a solution.

IPCA estimation deals with the issue by following the latent factor literature convention—it selects a normalization from the set of many rotational equivalent optimizers as the unique estimator. The choice of normalization is often made based on economic interpretability, algebraic elegance, or computational convenience.

The convention in the literature is to derive asymptotic theory of an estimator conditional

Figure 1: Identification Condition, Normalization, and Estimation



Notes: The dashed lines are sets of rotationally equivalent parameters. A particular one (in blue) minimizes the target function. The black curve represents the identification condition Θ. The intersection of sets Θ and $\arg\min \widehat{G}(\Gamma)$ defines the estimator $\widehat{\Gamma}$. The red arrow represents the normalization of a parameter to Θ.

on a *specific* normalization. In contrast with this convention, we provide a general method to characterize asymptotic properties under *any* valid normalization. To this end, we construct the following concepts for a generic normalization. We will discuss two specific normalization examples towards the end of this section. The asymptotic analysis in the following sections are first derived with the generic normalization, and thereafter the specific cases become straightforward evaluations.

To define key concepts, let *parameter space* Ψ be the set of all possible parameters Γ. A parameter Γ is called *rotationally equivalent* to Γ' if there exists an invertible matrix R such that $\Gamma = \Gamma' R$. Define an *identification condition* as a subset $\Theta \subset \Psi$, such that for any $\Gamma \in \Psi$ there is a unique $\Gamma' \in \Theta$ which is rotationally equivalent to Γ. Associated with an identification condition, the *normalization* is the mapping from any Γ to the (unique) equivalent $\Gamma' \in \Theta$. Define *identification function* $I(\Gamma)$ as a (vector-valued) function on the parameter space such that its solution set constitutes an identification condition: $\Theta = \{\Gamma \mid I(\Gamma) = \mathbf{0}\}$. The identification function operationalizes a normalization, and is analyzed in tandem with the score function in our asymptotic analysis.

Figure 1 illustrates these concepts. Dashed lines through the parameter space Ψ represent sets of rotationally equivalent parameters. The data generating process is unchanged for parameters in the same set. In addition, the target function $\widehat{G}$ is invariant within each set. The particular set that minimize $\widehat{G}$ is drawn in blue.

We have plotted a generic identification condition Θ as the solid black line which intersects

each dashed line at only one point. There are many possible normalizations, representable as different Θ's that "cut through" all of the rotationally equivalent parameter sets *each only once*. We will analyze two concrete examples of Θ further blow.

3.3 Estimator Defined as Normalized Optimizer

The *estimator* is defined as the optimizer of target function $\widehat{G}(\Gamma)$ that is normalized by Θ:

$$\widehat{\Gamma}(\Theta) = \arg\min_{\Gamma \in \Theta} \widehat{G}(\Gamma). \tag{8}$$

When there is no emphasis on the specific normalization choice, we omit "(Θ)" and simply write $\widehat{\Gamma}$.

Despite first appearances, (8) is not a constrained optimization because the constraint "$\Gamma \in \Theta$" never restrains the target from achieving its global optimum—it only picks a unique representation out of the (rotationally equivalent) set of Γ's that all achieve the same minimum. Equivalently, $\widehat{\Gamma}$ is the solution of the simultaneous equations $\widehat{S}(\Gamma) = \mathbf{0}$ and $I(\Gamma) = \mathbf{0}$, shown as the intersection of the two corresponding curves in Figure 1. Based on this representation, we will characterize the asymptotics of $\widehat{\Gamma}$ by analyzing both the score and the identification functions.

In addition, we note that a normalization Θ must be known to the econometrician, meaning it can depend on the sample but cannot depend on the underlying population parameters.

3.4 Two Normalization Examples

We give two specific normalization examples that are convenient and interpretable, called Θ_{X} and Θ_{Y}. We will keep returning to these two examples for concrete asymptotic analysis, and always use "X" and "Y" subscripts when referring to the constructions. For example, $I_{\mathrm{X}}(\Gamma)$ and $I_{\mathrm{Y}}(\Gamma)$ are identification functions for the two cases.

Γ has LK degrees of freedom in total, within which K^2 degrees are unidentified since the rotation R is $K \times K$. So an identification condition must restrict K^2 degrees of freedom. The first example pins down the top $K \times K$ block of Γ as the identity matrix, and leaves the lower $L - K$ rows free. Specifically, define $\Theta_{\mathrm{X}} := \{\Gamma \in \Psi \mid Block_{1:K}(\Gamma) = \mathbb{I}_K\}$, where $Block_{1:K}(\cdot)$ is a function of a matrix that cuts out its first K rows. The associated normalization mapping is

$\Gamma' = \Gamma Block_{1:K}(\Gamma)^{-1}$.

This identification condition forces a one-to-one correspondence between the K factors and the first K instruments, which gives the factors an economic interpretation dictated by those instruments.[12] Using a familiar asset pricing example Fama and French (1993), suppose there are three factors ($K = 3$), that $x_{i,t}$ consists of monthly stock returns, and that the first three instruments are lagged size, book-to-market ratio, and momentum of a stock ($z_{i,t} = [size_{i,t-1}, bm_{i,t-1}, mom_{i,t-1}, ...]$). We know that $\beta_{i,t}$ is given by $z_{i,t}\Gamma$. Under Θ_{X}, the first loading is $\beta_{i,t,1} = 1 \cdot size_{i,t-1} + 0 \cdot bm_{i,t-1} + 0 \cdot mom_{i,t-1} +$ *effects of other instruments*—loadings on the first factor are driven one-for-one by the first instrument (size), but are unaffected by book-to-market and momentum, giving the first factor a direct interpretation as a size factor analogous to the "SMB" factor of Fama and French (1993). Likewise, the second and third factors are book-to-market and momentum factors. The lower $L - K$ rows of Γ tell us how the remaining $L - K$ instruments affect loadings. Hence, this normalization is reminiscent of, and a less ad-hoc alternative to, the popular characteristics-sorted portfolios that the empirical asset pricing literature has adopted for understanding systematic risks.[13]

For the second normalization, define Θ_{Y} as the set of $\Gamma \in \Psi$ such that [1] Γ is orthonormal: $\Gamma^\top\Gamma = \mathbb{I}_K$; and [2] $\widehat{f}_{t(\Gamma)}$ is orthogonal over t: $\frac{1}{T}\sum_t \widehat{f}_{t(\Gamma)}\widehat{f}_t^{\top}{}_{(\Gamma)}$ is diagonal, with distinct and descending entries. Appendix B.1 verifies Θ_{Y} is indeed an identification condition that satisfies the uniqueness condition, and writes out the associated normalization procedure.[14] This identification condition is a familiar PCA convention and is implemented for IPCA in KPS19. It chooses a set of orthogonal factors to represent the factor space. Notice [1] and [2] pin down $\frac{1}{2}K(K+1)$ and $\frac{1}{2}K(K-1)$ degrees of freedom, respectively, for a total of K^2. Importantly,

[12] Without loss of generality, we can reorder the instruments however we wish.

[13] This identification condition Θ_{X} appears similar to identifying restriction PC3 in Bai and Ng (2013), but carries an important distinction due to IPCA's utilization of instruments. In the static-loading PCA estimator, the PC3 restriction says that the first K *individuals* have a direct correspondence to the K factors. Continuing the asset-pricing example, it would imply that the factors are aligned with the first K individual stocks which, from a finance perspective, is arbitrary and not intuitive for understanding the systematic risk structure.

[14] An additional nuance is that the signs of the K Γ columns (and the corresponding K factors) need one more restriction to pin down. If the additional restriction is not suitable, it brings problems in finite sample simulations (for example, the one Stock and Watson (2002) proposed). Appendix G.2 explains the issue in detail and constructs a sign restriction [3] that is theoretically sound and practically easy to use in simulations.

identification condition Θ_{Y} depends on the sample, because $\widehat{f}_{t(\Gamma)}$ in [2] is a sample-dependent function. This sample-dependence brings some complexity in the asymptotic analysis which we will address in the following sections.

4 Consistency of Γ Estimation

This section demonstrates that IPCA estimates the mapping matrix Γ consistently. Since the estimate $\widehat{\Gamma}$ is the simultaneous solution of the first order condition (score function) and the identification function, its consistency is based on the (uniform) convergence of these two functions, following the standard strategy for analyzing M-estimators (Newey and McFadden, 1994). Relative to that classical framework, we additionally address the identification issue with constructions in Section 3.2.

4.1 Score Function Convergence

We show the score function uniformly converges to a deterministic limiting function:

Proposition 1 (Uniform Convergence of the Score Function)**.**

Under Assumptions A–D, the score function converges uniformly in probability:

$$\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S^0(\Gamma)\right\| \xrightarrow{p} 0, \qquad\qquad N,T\to\infty.^{15}$$

Moreover, we verify that the limiting function S^0 is solved *only* by Γ^0 and its unidentified rotations:

Proposition 2 (Only True Parameters Solve Limiting Score)**.** *Under Assumptions A–D, $S^0(\Gamma)=\mathbf{0}$ if and only if Γ is rotationally equivalent with Γ^0.*

These two results are the foundation of IPCA consistency. When combined, they imply a sense of *set* convergence regardless of the identification issue—the set of SSE minimizers $\left\{\Gamma \text{ s.t. } \widehat{S}(\Gamma)=0\right\}$ converges to the set of unidentified true parameters. Once we build the counterparts of these results for the identification function in the next subsection, score and identification functions together will lead to unique estimator convergence.

[15] The expression of S^0 is in Appendix B.3 together with the proof. An intermediary result in the proof can be used to show finite-T large-N consistency besides simultaneous convergence.

But before that, let us explain the intuition behind Proposition 1, which is essential for IPCA's large sample theory. By taking the derivative of the target, we find $\widehat{S}(\Gamma) = \frac{1}{NT}\sum_t \text{vect}\left(Z_t^\top \widehat{e}_{t(\Gamma)}\widehat{f}_t^\top{}_{(\Gamma)}\right)$, where $\widehat{f}_{t(\Gamma)}$ and $\widehat{e}_{t(\Gamma)}$ are the coefficients and errors, respectively, of the cross-sectional K-variate OLS regression of $x_{i,t}$ onto $z_{i,t}\Gamma$.[16] The sample OLS estimate $\widehat{f}_{t(\Gamma)}$ misses the true f_t^0 for two reasons. For one, it uses a Γ that is not the true Γ^0, and thereby the instrumented factor loadings $z_{i,t}\Gamma$ are off. Second, even if we knew the true Γ^0 and thus the sample β's were right, OLS in the finite cross section does not exactly reveal the true f_t^0. To formalize this intuition, we construct $\widetilde{f}_{t(\Gamma)}$ as the "population" counterpart of the same $x_{i,t}$-on-$z_{i,t}\Gamma$ regression. By "population" we mean conditional on $\mathcal{C}$, that is, conditional on all the common time-series variables from which one knows the current cross-sectional distribution.

$$\widetilde{f}_{t(\Gamma)} := \mathbb{E}\left[\Gamma^\top z_{i,t}^\top z_{i,t}\Gamma\middle|\mathcal{C}\right]^{-1}\mathbb{E}\left[\Gamma^\top z_{i,t}^\top x_{i,t}\middle|\mathcal{C}\right] = \left(\Gamma^\top \Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0 f_t^0, \tag{9}$$

with the shorthand $\Omega_t^{zz} := \mathbb{E}\left[z_{i,t}^\top z_{i,t}\middle|\mathcal{C}\right]$.[17] Therefore, the aforementioned two sources of $\widehat{f}_{t(\Gamma)}$ error are formally represented with the decomposition

$$\widehat{f}_{t(\Gamma)} = \widetilde{f}_{t(\Gamma)} + \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}, \tag{10}$$

where $\widetilde{e}_{t(\Gamma)} := x_t - Z_t\Gamma\widetilde{f}_{t(\Gamma)}$ is the corresponding population OLS error. Importantly, given a rotation R, $\widetilde{f}_t\left(\Gamma^0 R\right) = R^{-1}f_t^0$ and $\widetilde{e}_t\left(\Gamma^0 R\right) = e_t$. This means if one knew the true Γ^0 (even if not knowing the particular rotation), functions $\widetilde{f}_{t(\Gamma)}$ and $\widetilde{e}_{t(\Gamma)}$ would be able to reveal the true factor structure.

With this knowledge, the score function can be broken down by representing $\widehat{f}_{t(\Gamma)}, \widehat{e}_{t(\Gamma)}$ with $\widetilde{f}_{t(\Gamma)}, \widetilde{e}_{t(\Gamma)}$. We save the exact expression to Appendix B.3, except noting that the score function inherits the same decomposition: a part that is only "off" due to the "wrong" Γ, and another part due to finite cross section. Intuitively, the first part converges to the probability limit $S^0(\Gamma)$, while the the second vanishes in the large-N,T limit.[18] Finally, the limiting score $S^0(\Gamma)$ can recover the true Γ^0 (up to a rotation) according to Proposition 2.

[16] Eq. (5) defined $\widehat{f}_{t(\Gamma)}$, and $\widehat{e}_{t(\Gamma)} := x_t - Z_t\Gamma\widehat{f}_{t(\Gamma)}$.

[17] Assumption B is used at second equal sign in (9).

[18] The proof is in Appendix B.3, which deals with additional complication of *uniform* large-N,T convergence across Γ, which is necessary for the convergence discussed next.

4.2 Identification Function Convergence

As defined in Section 3.2, an identification function is solved by the normalized parameters: $\Theta = \{\Gamma \text{ s.t. } I(\Gamma) = \mathbf{0}\}$. We define the following identification functions for the the two examples Θ_{X} and Θ_{Y}, respectively:

$$I_{\mathrm{X}}(\Gamma) := \mathrm{vect}\left(Block_{1:K}(\Gamma) - \mathbb{I}_K\right), \qquad I_{\mathrm{Y}}(\Gamma) := \begin{bmatrix} \mathrm{veca}\left(\Gamma^\top\Gamma - \mathbb{I}_K\right) \\ \mathrm{vecb}\left(\frac{1}{T}\sum_t \widehat{f}_{t(\Gamma)}\widehat{f}_t^{\top}(\Gamma) - V^{\mathrm{ff}}\right) \end{bmatrix}.^{19}$$

Notice $\widehat{f}_{t(\Gamma)}$ involves sample data, and hence I_{Y} is a random function.

We show I_{Y} converges uniformly to a deterministic function I^0_{Y}, which also constitutes an identification function. The same claim trivially applies to I_{X} as it is deterministic to begin with. Formally, Lemma 8 in Appendix B.5 verifies both $I_{\mathrm{X}}, I_{\mathrm{Y}}$ satisfy the following conditions about a generic identification function I:

IF.1 **Uniform convergence:** There exists a deterministic function $I^0(\Gamma)$ such that

$$\sup_{\Gamma\in\Psi} \left\| I(\Gamma) - I^0(\Gamma) \right\| \xrightarrow{p} 0, \qquad\qquad N, T \to \infty.$$

IF.2 **Limit is Identification Condition:**

Set $\Theta^0 := \{\Gamma \in \Psi \mid I^0(\Gamma) = \mathbf{0}\}$ is an identification condition.

In addition, we maintain the identification assumption about the true Γ^0:

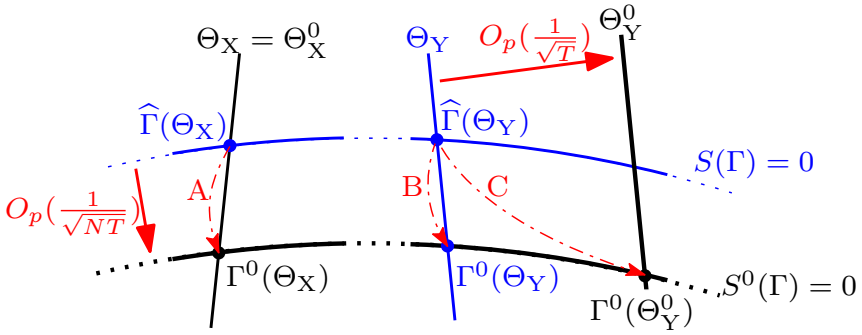
IA **Identification Assumption:** $\Gamma^0 \in \Theta^0$, i.e., $I^0(\Gamma^0) = \mathbf{0}$.

IA implies Γ^0 solves I^0. In addition, IF.2 implies Γ^0 is the *only* solution among all of its equivalent rotations.

The above convergence and uniqueness conditions about I are symmetric to Propositions 1 and 2 about $\widehat{S}$. Since the estimator is the intersection of $\widehat{S}$ and I, these conditions together form the premise of estimator consistency, which will be formally stated in Section 4.4. Conditions

[19] Mappings veca and vecb vectorize the upper triangular entries of a square matrix. The difference is veca includes the diagonal entries, and vecb does not. See details with an example in Appendix A. $V^{\mathrm{ff}} := \mathbb{E}\left[f^0_t f^{0\top}_t\right]$ is the *population* second moment matrix of f^0_t. In the case of Θ_{Y}, we have assumed that V^{ff} is a diagonal matrix. Hence, it is indeed redundant to write "$-V^{\mathrm{ff}}$" in the expression of I_{Y}, since vecb only picks up the off-diagonal zeros. We still leave it in the expression in order to illustrate that our method allows Θ_{Y} to be easily amended by changing V^{ff} at the applied researcher's discretion, and that the asymptotic analysis would mostly be unchanged. For example, one can specify $V^{\mathrm{ff}} = \mathbb{I}_K$ and switch "veca" and "vecb" to standardize factors instead of Γ.

Figure 2: Estimated and True Parameters in the Two Normalization Cases



Notes: The top horizontal curve is the set of optimizers. The lower curve is the set of rotationally equivalent true parameters. The three vertical lines are identification conditions. All the black objects are deterministic, the blue ones are sample-dependent. The two solid red arrows point from the random sets to their deterministic limits, labeled with the rates of convergence. The dashed red arrows A, B, C mark three specific estimation errors, whose asymptotic distributions are in Theorems 3.a, 3.b, 5, respectively.

IF.1–2 can be used to establish a normalization other than Θ_X or Θ_Y yields consistent parametric estimation.

We note IA is an innocuous assumption because the true parameter can always be normalized to satisfy any identification condition, including Θ^0, without changing the data generating process. Next, we sort out some ambiguity brought about by having different, but rotationally equivalent, true parameters.

4.3 Which True Parameter?

This subsection clarifies some subtle issues involved in normalizing the true parameter. Let us first review how the points talked about so far are shown in Figure 2. Based on this visual representation, we explain the roles played by the different true parameters in asymptotic analysis.

Proposition 1 establishes that the set of optimizers (shown as the upper "$\widehat{S}(\Gamma) = \mathbf{0}$" curve) converges to the set of rotationally-equivalent true parameters (the lower "$S^0(\Gamma) = \mathbf{0}$" curve). Meanwhile, IF.1 says the identification conditions are also converging. The blue vertical line describes the normalization Θ_Y for a specific sample. It varies across samples but converges to the limit Θ_Y^0, shown as the vertical line on the right. On the other hand, the limit of Θ_X is itself (since it is deterministic), shown as the vertical line on the left. The estimators $\widehat{\Gamma}(\Theta_X)$ and $\widehat{\Gamma}(\Theta_Y)$ are at the intersections of the optimizer set and the identification-condition set, where

the parenthesis clarifies which identification condition it is in.

We make two main points. First, assumption IA means that the true Γ^0 under the two normalization cases Θ_{X} and Θ_{Y} are *not* the same. This can be seen by noting that the limiting identification conditions Θ^0 are different for the two cases—one is Θ_{X} itself and the other is Θ^0_{Y}. To avoid ambiguity, we write the true parameter under the two cases as $\Gamma^0(\Theta_{\mathrm{X}})$ and $\Gamma^0(\Theta^0_{\mathrm{Y}})$, while Γ^0 is reserved as the shorthand when there is no emphasis on the specific case. The two are rotationally equivalent, but satisfy different limiting identification conditions as clarified in the parentheses. The true parameter is deterministic, since it is the intersection of two deterministic sets: $S^0(\Gamma) = \mathbf{0}$ and Θ^0, and is the parameter in the IPCA data generating process (Eq. 2). Later on, it will serve as the fixed reference point in asymptotic expansion. Finally, the asymptotic variance is expressed with regard to the generic Γ^0, which is subsequently evaluated at either $\Gamma^0(\Theta_{\mathrm{X}})$ or $\Gamma^0(\Theta^0_{\mathrm{Y}})$ depending on the specific case.

Our second main point is that we measure estimation error against the true parameter normalized in the same way as the estimate is normalized. We call this the *normalized true* parameter and denote it as $\Gamma^0(\Theta)$, where Θ is the normalization used for estimation.[20] Arrows A and B in Figure 2 are examples of this choice of estimation error. The two arrows represent the two estimation errors involved in our asymptotic expansions: $\widehat{\Gamma}(\Theta_{\mathrm{X}}) - \Gamma^0(\Theta_{\mathrm{X}})$ and $\widehat{\Gamma}(\Theta_{\mathrm{Y}}) - \Gamma^0(\Theta_{\mathrm{Y}})$. Each pair is an estimate minus a true parameter, both under a common normalization.

A subtle distinction is that the normalized truth can be *made* sample-dependent by the normalization. This is the case of Arrow B in particular, where the reference point $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ is sample-dependent. On the other hand, in the case of Θ_{X} there is no distinction for the normalized true because Θ_{X} and its limit are one and the same.

Measuring estimation errors against the normalized true is justified, although perhaps peculiar at first glance that a "true" parameter can be random. Indeed, it follows and formalizes the convention of asymptotic analysis in factor analysis.[21] Following previous literature, our primary

[20] Since Θ is an identification condition, per its definition, such a $\Gamma^0(\Theta)$ is unique.

[21] Bai and Ng (2002) and Bai (2003) are prominent examples that estimation errors are measured against the normalize true in the literature. They study PCA estimation error $\widehat{\lambda}_i - H^{-1}\lambda^0_i$, where H is a *sample-dependent* rotation imposed on the true λ^0_i. (In their notation, λ_i is factor loadings, equivalent to our β_i.) Although in Bai and Ng (2013) such a rotation matrix H is avoided, the true parameters used as comparison targets are

interest is on Arrows A and B, since they are essential about the accuracy of IPCA estimation in recovering a true parameter, albeit a rotated true parameter. In contrast, the asymptotic analysis of $\widehat{\Gamma}(\Theta_Y) - \Gamma^0(\Theta_Y^0)$ (Arrow C) would reveal the complete asymptotic distribution of the estimator $\widehat{\Gamma}(\Theta_Y)$ since the reference point $\Gamma^0(\Theta_Y^0)$ is fixed. We will show that Arrow C converges at a slower rate than Arrows A or B due to an additional stochastic wedge. The wedge exists because the estimate is normalized according to Θ_Y while the reference point is normalized by its limit Θ_Y^0.[22] This slows down the convergence of Arrow C, because it also relies on the convergence of Θ_Y to Θ_Y^0 (which happens at the slower rate $\sqrt{T}$).

We derive these and other facts rigorously in the analysis below. We focus our asymptotic analysis on the errors labeled as Arrows A and B, and relegate theoretical and simulation results for Arrow C to Appendix C.

To recap, we have defined the following notation. The estimate is $\widehat{\Gamma}(\Theta)$, or $\widehat{\Gamma}$ in short when there is no emphasis on the specific normalization Θ; Similarly, the true parameter is $\Gamma^0(\Theta^0)$, or Γ^0 in short. The normalized true parameter is $\Gamma^0(\Theta)$, against which the estimation errors are referenced. Lastly, the same notations with a small γ represent the vectorized counterparts.

4.4 Consistency of Γ Estimation

Now we combine the previous work to show the consistency of Γ estimation.

Per the estimator definition, $\widehat{\Gamma}$ solves the simultaneous equations $\left[\widehat{S}; I\right](\Gamma) = \mathbf{0}$.[23] Proposition 1 and IF.1 combine to imply $\left[\widehat{S}; I\right](\Gamma)$ uniformly converges to $[S^0; I^0](\Gamma)$. Proposition 2, IF.1, and IA together imply Γ^0 is the unique solution of $[S^0; I^0](\Gamma) = \mathbf{0}$. Since the solution of a uniformly converging function converges to the limit's *unique* solution (Newey and McFadden, 1994), we have $\widehat{\Gamma} \xrightarrow{p} \Gamma^0$. Meanwhile, $\Gamma^0(\Theta)$ is the solution of $[S^0; I](\Gamma)$, and we can similarly show it approaches the same fixed limit Γ^0. Bridged by the common limit Γ^0, the difference between the estimation and the normalized true must converge.

still normalized according to sample realizations. For example, they directly restrict the *sample* second moment matrix of the true factors in their PC1 and PC2. This is comparable to our Θ_Y condition [2].

[22] Normalizing the estimate by Θ_Y^0 is not practical, because Θ_Y^0 is a population quantity not available from finite data. It is a theoretical construction for large sample theory.

[23] $\left[\widehat{S}; I\right](\Gamma)$ means the vector-valued function by vertically stacking up $\widehat{S}(\Gamma)$ and $I(\Gamma)$. Same for a few other stacked functions below.

Theorem 1 (Γ Estimation Consistency – Generic Normalization)**.** *Under Assumptions A–D, and if the identification condition* Θ *has an associated identification function* $I(\Gamma)$ *that satisfies IF.1–2, then as* $N, T \to \infty$, $\widehat{\Gamma} - \Gamma^0(\Theta) \xrightarrow{p} \mathbf{0}$.

The theorem above is the consistency result for a generic identification condition Θ. For the two specific cases (Θ_{X} and Θ_{Y}), Lemma 8 verifies that both identification functions $I_{\text{X}}(\Gamma)$ and $I_{\text{Y}}(\Gamma)$ satisfy Condition IF.1–2. Therefore, the estimators in these two specific cases are consistent as well.

Corollary 1 (Γ Estimation Consistency — Specific Cases)**.** *Under Assumptions A–D, IPCA* $\widehat{\Gamma}$ *estimated under normalization* Θ_{X} *or* Θ_{Y} *are consistent with respect to the accordingly normalized true parameters: as* $N, T \to \infty$, $\widehat{\Gamma}(\Theta_{\text{X}}) - \Gamma^0(\Theta_{\text{X}}) \xrightarrow{p} \mathbf{0}$, *and* $\widehat{\Gamma}(\Theta_{\text{Y}}) - \Gamma^0(\Theta_{\text{Y}}) \xrightarrow{p} \mathbf{0}$.

5 Asymptotic Distributions of Γ Estimation Error

This section analyzes the asymptotic distribution of estimation error $\widehat{\gamma} - \gamma^0(\Theta)$.[24] We first provide the general results for a generic identification condition Θ, and then calculate the asymptotic distributions under the two specific normalization cases.

5.1 Asymptotic Error — Generic Normalization

Theorem 2 (Asymptotic Error — Generic Normalization)**.** *Under the conditions of Theorem 1, and assuming both* $\widehat{S}(\Gamma)$ *and* $I(\Gamma)$ *are continuously differentiable in a neighborhood around* Γ^0, *where* Γ^0 *satisfy IA, then as* $N, T \to \infty$,

$$\widehat{\gamma} - \gamma^0 = -\left(H^{0\top}H^0 + J^{0\top}J^0\right)^{-1}\left(H^{0\top}\widehat{S}(\Gamma^0) + J^{0\top}I(\Gamma^0)\right) + o_p\left(\widehat{S}(\Gamma^0) + I(\Gamma^0)\right), \tag{2.a}$$

$$\widehat{\gamma} - \gamma^0(\Theta) = -\left(H^{0\top}H^0 + J^{0\top}J^0\right)^{-1}H^{0\top}\widehat{S}(\Gamma^0) + o_p\left(\widehat{S}(\Gamma^0)\right), \tag{2.b}$$

where $H^0 := \left.\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}$ *and* $J^0 := \left.\frac{\partial I^0(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}$.

The proof is built on Newey and McFadden's (1994) analysis of an M-estimator by linearizing the score function. We extend that result to the not-locally-identified situation by appending

[24] Recall that we use the lower case letters $\gamma, \gamma^0, \widehat{\gamma}$ etc. to denote the parameters vectorized in an $LK \times 1$ column. The vectorized version allows us to represent the *joint distribution* of the LK parameter estimates , e.g. with a variance-covariance matrix.

the identification function to the score and linearizing both symmetrically.[25] This analytical method, and the results it affords, could more broadly be used to analyze other estimators that require a normalization step.

The theorem offers a clear decomposition of the sources of randomness in parameter estimation. On the right-hand sides of 2.a and 2.b, the only random terms are the score and identification functions (both evaluated at Γ^0), representing the sample inaccuracies emanating from optimization and normalization. The theorem says $\widehat{\gamma} - \gamma^0$ loads on both sources of randomness while $\widehat{\gamma} - \gamma^0(\Theta)$ loads only on the randomness of the score.

The intuition can be illustrated by Figure 2. Look at the small neighborhood around $\Gamma^0(\Theta_{\mathrm{Y}}^0)$, and imagine both the "$\widehat{S}(\Gamma) = \mathbf{0}$" curve and the Θ_{Y} curve are "wobbling" around their deterministic limits: they represent the two sources of randomness affecting estimation. Since the estimate $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ is the intersection of the two curves, it must load on the randomness of both (shown as Arrow C). The asymptotic distribution of Arrow C has a different convergence rate (see Appendix C, Bai and Li 2016 report a similar result for PCA). However, since both $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ and $\Gamma^0(\Theta_{\mathrm{Y}})$ are under the same identification Θ_{Y}, their difference does not depend on the randomness of Θ_{Y} (shown as Arrow B).

5.2 Asymptotic Error — Specific Cases

This section brings the general asymptotic result in Theorem 2.b to the specific cases denoted by Arrows A and B. For the reasons discussed in Section 4.3, asymptotic analysis of Arrow C, which is a special case of 2.a, is relegated to Appendix C.

To evaluate 2.b, it only remains to calculate the three right-hand side terms: H^0, J^0, and the asymptotic distribution of $\widehat{S}(\Gamma^0)$. As we have discussed in Section 4.3, the general expressions in Theorem 2 need to be evaluated locally at either $\Gamma^0(\Theta_{\mathrm{X}})$ or $\Gamma^0(\Theta_{\mathrm{Y}}^0)$. As mentioned before, we use use subscripts X and Y to denote the values calculated for their corresponding normalizations.

The Hessian-like matrices H_{X}^0, J_{X}^0 and H_{Y}^0, J_{Y}^0 are calculated by taking the derivatives of the limiting functions, which are detailed in Appendix B.9–B.10. The more interesting result is the asymptotic distribution of $\widehat{S}(\Gamma^0)$. We know $\widehat{S}(\Gamma^0) \xrightarrow{p} \mathbf{0}$ from Propositions 1 and 2. The

[25] The proof that details our way of analysis is in Appendix B.7.

following result says that the convergence happens at the rate of $\sqrt{NT}$ and gives the asymptotic distributions.

Proposition 3 (Asymptotic Distribution of the Score Evaluated at Γ^0)**.**

Under Assumptions A–F, as $N, T \to \infty$ such that $T/N \to 0$,

$$\sqrt{NT}S\left(\gamma^0(\Theta_{\mathrm{X}})\right) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{X}}^{[1]}\right), \qquad \sqrt{NT}S\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{Y}}^{[1]}\right).^{26}$$

This proposition implies the convergence rate of Γ estimation error is $\sqrt{NT}$, regardless of the normalization choice. The $\sqrt{NT}$-convergence rate highlights IPCA's ability to harness not just the time-series, but also the cross-sectional information. The ability ultimately comes from the assumption that the instrumental mapping is common across individuals. In contrast, even were factors observed, unconstrained individual-specific loading estimates could only rely on time-series information and thereby converge at rate $\sqrt{T}$.

From the perspective of the panel literature, the $\sqrt{NT}$ convergence rate can be understood by viewing Γ as a common structural parameter and f_t as a time fixed-effects. This view raises the concern that estimation could be asymptotically biased in the presence of "incidental parameters" whose number increases with the sample size (Neyman and Scott, 1948; Lancaster, 2000). Gagliardini and Gourieroux (2014) note the incidental parameter problem is much less pronounced in the case of time fixed-effects with large cross sections. We follow them and focus on the $T/N \to 0$ case, in which the asymptotic distribution is centered around zero and no asymptotic bias correction is needed. Should T/N converge to a positive number, we conjecture the asymptotic distribution is still normal but with a non-zero mean—exact analysis is left to future research. The simulation results we report below show no noticeable bias.

Finally, assembling the previous calculations into 2.b, we have the asymptotic distributions for Arrows A and B. The expressions are symmetric for the two cases and the specific values are numerically verified by Monte Carlo simulation in Section 7.

Theorem 3 (Asymptotic Error — Specific Cases)**.** *Under Assumptions A–F, as $N, T \to \infty$*

[26] The expressions of $\mathbb{V}_{\mathrm{X}}^{[1]}$ and $\mathbb{V}_{\mathrm{Y}}^{[1]}$ is with the proof in appendix B.8.

such that $T/N \to 0$,

$$\sqrt{NT}\left(\widehat{\gamma}(\Theta_{\mathrm{X}}) - \gamma^0(\Theta_{\mathrm{X}})\right) \xrightarrow{d} -\left(H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1} H_{\mathrm{X}}^{0\top} \mathrm{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{X}}^{[1]}\right), \tag{3.a}$$

$$\sqrt{NT}\left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}})\right) \xrightarrow{d} -\left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} H_{\mathrm{Y}}^{0\top} \mathrm{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{Y}}^{[1]}\right). \tag{3.b}$$

The asymptotic distributions in Theorem 3 for Arrows A and B suggest feasible inference for linear combinations of $\gamma = \mathrm{vect}\,(\Gamma)$. We develop this implication in the Appendix, where Section F constructs plug-in sandwich estimators for the asymptotic covariance under both Θ_{X} and Θ_{Y}, and shows their consistency, delivering asymptotically valid Wald confidence intervals and tests for entries of Γ. In the Θ_{X} case (Arrow A), the argument is especially convenient because the identification Jacobian J_{X}^0 is known exactly rather than estimated, which simplifies the map. Under the sample-dependent normalization Θ_{Y} (Arrow B), the same appendix section extends the construction by estimating the Jacobian of the identification function and incorporating it into the sandwich map.

In particular, the plug-in covariance estimators are of the form

$$\widehat{\Sigma}_{\gamma}^{\mathrm{X}} = \widehat{m}_{\mathrm{X}}\, \widehat{\mathbb{V}_{\mathrm{X}}^{[1]}}\, \widehat{m}_{\mathrm{X}}^{\top}, \qquad \widehat{m}_{\mathrm{X}} = \left(\widehat{H}_{\mathrm{X}}^{\top}\widehat{H}_{\mathrm{X}} + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1} \widehat{H}_{\mathrm{X}}^{\top},$$

for Arrow A, and

$$\widehat{\Sigma}_{\gamma}^{\mathrm{Y}} = \widehat{m}_{\mathrm{Y}}\, \widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}}\, \widehat{m}_{\mathrm{Y}}^{\top}, \qquad \widehat{m}_{\mathrm{Y}} = \left(\widehat{H}_{\mathrm{Y}}^{\top}\widehat{H}_{\mathrm{Y}} + \widehat{J}_{\mathrm{Y}}^{\top} \widehat{J}_{\mathrm{Y}}\right)^{-1} \widehat{H}_{\mathrm{Y}}^{\top},$$

for Arrow B. The plug-in outer-product term $\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}}$ (and its Θ_{Y} analogue $\widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}}$) is defined in Section F.

6 Asymptotic Analysis of Factor Estimation

We have so far treated factor estimation $\widehat{f}_t(\widehat{\Gamma})$ implicitly, because we concentrated-out f_t in the target function (Eq. 7). Now, with the asymptotics of $\widehat{\Gamma}$ in hand, we come back to factor estimation and lay out its asymptotic distribution with relative ease. Given $\Gamma^0(\Theta)$ is the normalized true Γ, the correspondingly normalized true factor is $\widetilde{f}_t\left(\Gamma^0(\Theta)\right)$.[27] This factor estimation error

[27] Given a (sample-dependent) R, such that the normalized true Γ is represented as $\Gamma^0(\Theta) = \Gamma^0 R$, the correspondingly normalized true factor is inversely rotated as $R^{-1} f_t^0 = R^{-1}\widetilde{f}_t\left(\Gamma^0\right) = \widetilde{f}_t\left(\Gamma^0 R\right) = \widetilde{f}_t\left(\Gamma^0(\Theta)\right)$. In other words, the two pairs, $\left\{\Gamma^0, f_t^0\right\}$ and $\left\{\Gamma^0(\Theta), \widetilde{f}_t\left(\Gamma^0(\Theta)\right)\right\}$, are rotationally equivalent to each other.

can be decomposed into two sources following the discussion around Eq. (10),

$$\widehat{f}_t(\widehat{\Gamma}) - \widetilde{f}_t\left(\Gamma^0(\Theta)\right) = \left(\widetilde{f}_t(\widehat{\Gamma}) - \widetilde{f}_t\left(\Gamma^0(\Theta)\right)\right) + \left(\widehat{\Gamma}^\top Z_t^\top Z_t \widehat{\Gamma}\right)^{-1} \widehat{\Gamma}^\top Z_t^\top \widetilde{e}_t(\widehat{\Gamma}). \tag{11}$$

The first term captures the part of factor estimation error only due to the inaccuracy of Γ estimation, which converges at the rate of $\sqrt{NT}$. The second term captures the error from the cross-sectional regressions, which converges at the dominant rate of $\sqrt{N}$.

Theorem 4 (Factor Estimation Asymptotics)**.**

Under Assumptions A–F, with an identification Θ that satisfies IF.1–2,

(4.a) *Factor estimation is consistent: as $N, T \to \infty$, $\widehat{f}_t(\widehat{\Gamma}) - \widetilde{f}_t\left(\Gamma^0(\Theta)\right) \xrightarrow{p} \mathbf{0}$.*

(4.b) *Factor estimation error centered against the normalized true factor converges to a normal distribution at the rate of $\sqrt{N}$: as $N, T \to \infty$, $\forall t$,*

$$\sqrt{N}\left(\widehat{f}_t(\widehat{\Gamma}) - \widetilde{f}_t\left(\Gamma^0(\Theta)\right)\right) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \mathbb{V}_t^{[2]}\right).^{28}$$

The theorem gives the factor's asymptotic distribution under a generic normalization Θ. Similar to Γ estimation, we can evaluate $\mathbb{V}_t^{[2]}$ at either the $\Gamma^0(\Theta_{\text{X}})$ or $\Gamma^0(\Theta_{\text{Y}}^0)$ normalizations.[29]

7 Simulations

Simulation exercises illustrate the finite-sample behavior of the IPCA estimator and assess whether the asymptotic theory derived above provides an accurate approximation at moderate sample sizes. We find that estimation errors are well-approximated by the asymptotic normal distribution of Theorem 3, and that the feasible standard errors developed in Section F are accurate enough in moderate samples to support reliable inference.

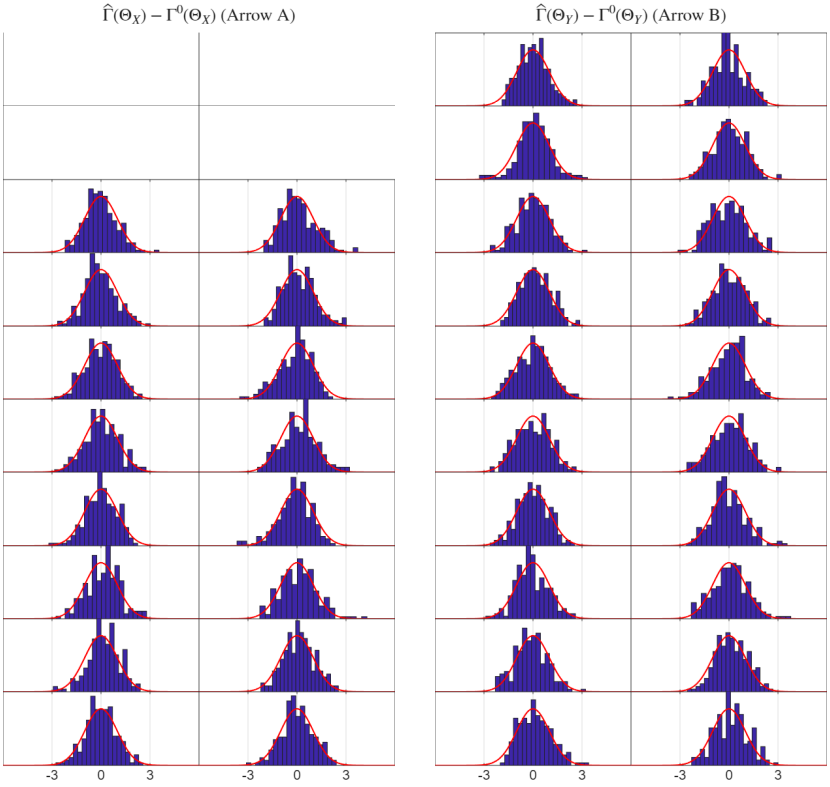
We simulate data panels $\{x_{i,t}, z_{i,t}\}$ from a model calibrated with the empirical data of KPS19 (details in Appendix G.1). The model has $L = 10$ characteristics and $K = 2$ factors. For each simulated sample, we estimate Γ via ALS and normalize under both Θ_{X} (Arrow A) and Θ_{Y} (Arrow B). We also compute the covariance matrices $\widehat{\Sigma}_\gamma^{\text{X}}$ and $\widehat{\Sigma}_\gamma^{\text{Y}}$.

Figure 3 reports, for each element of Γ, the cross-simulation histogram of the feasibly stan-

[28] The expression of the asymptotic variance $\mathbb{V}_t^{[2]}$ is with the proof in Appendix B.11.

[29] Appendix C.3 contains the asymptotics when centered by the original f_t^0. This situation corresponds to Arrow C for Γ estimation, and the the additional stochastic wedge introduced by a sample-based normalization affects the asymptotic distribution.

Figure 3: Standardized Γ Estimation Errors vs. Normal $(0, 1)$



Note: Monte Carlo simulations with $N = 200$, $T = 200$. Each subplot corresponds to one element of the matrix Γ. The bars show the histogram of the estimation error standardized by each draw's own standard error $\widehat{\sigma}_k^{(s)}$; the red curve is the Normal $(0, 1)$ density. The left panel shows the normalization Arrow A (Θ_X); the four top entries are blank because those K^2 entries of Γ are pinned down by Θ_X. The right panel shows the normalization Arrow B (Θ_Y).

dardized estimation error

$$t_{s,k}^{X} = \frac{\sqrt{NT}\left(\widehat{\gamma}_k(\Theta_X)^{(s)} - \gamma_k^0(\Theta_X)\right)}{\widehat{\sigma}_k^{X,(s)}}, \qquad t_{s,k}^{Y} = \frac{\sqrt{NT}\left(\widehat{\gamma}_k(\Theta_Y)^{(s)} - \gamma_k^0(\Theta_Y)\right)}{\widehat{\sigma}_k^{Y,(s)}},$$

where $\widehat{\sigma}_k^{X,(s)}$ and $\widehat{\sigma}_k^{Y,(s)}$ are the plug-in standard errors from draw s's own estimated covariance matrix. Each histogram is overlaid with the standard normal density (red curve) as the asymptotic benchmark.

In both normalizations, the histograms are centered near zero and track the standard normal density closely, confirming that the asymptotic normal approximation of Theorem 3 is accurate at $N = T = 200$, and that the feasible standard errors are well-estimated so that the standardized errors are indeed approximately Normal $(0, 1)$. [30]

[30] In untabulated simulations with $N = T = 1000$, the distributions become visually indistinguishable from a

8 Applications

This section describes two empirical applications of IPCA to demonstrate its broad usefulness for analyzing economic data.

8.1 International Macroeconomics

Country-level macroeconomic fluctuations are globally connected (Backus, Kehoe, and Kydland, 1992). Gregory, Head, and Raynauld (1997) document this using a static state-space model and maximum likelihood for the G7 countries. Likewise, Kose, Otrok, and Whiteman (2003) use a static state-space model estimated with Bayesian methods to disentangle global from regional and country-specific growth factors for a panel of countries over 30 years. Recently Kose, Otrok, and Prasad (2012) (henceforth KOP) used data from the World Development Indicators and estimated their static state-space model for a panel of countries before and after 1985 in order to analyze the convergence or decoupling of global business cycles. In essence, they ask: have countries' relationships to global growth changed as the countries themselves have evolved? This question is ideally suited for investigation with IPCA.

We use IPCA to analyze the global factor structure in GDP growth using data from the World Development Indicators database. We include as many countries as possible from the "industrial/developed" and "emerging" country groups studied by KOP. After reasonable data filters on indicators and countries, which we detail in Appendix G.3, we are left with 45 countries. Within this sample there are nine variables, that are available for most countries, so we use these as our instruments. The first two instruments are the import and export share of GDP—natural indicators of a country's economic connectedness with the rest of the world. Next, we use the proportion of GDP relative to world GDP to measure the nation's relative size. We account for dynamics in capital intensity using gross capital formation, and we use population growth to account for growth in the labor force. To account for recent economic growth and risks, we include the 5-year rolling mean and volatility of the nation's GDP growth and its rate of inflation. Finally, we include a constant characteristic. In the revised specification we use these

standard normal.

nine baseline instruments directly, without interacting them with the country-group indicator. Our annual data run from 1961 to 2015 so that $T = 55$, and we demean the growth rates following KOP.

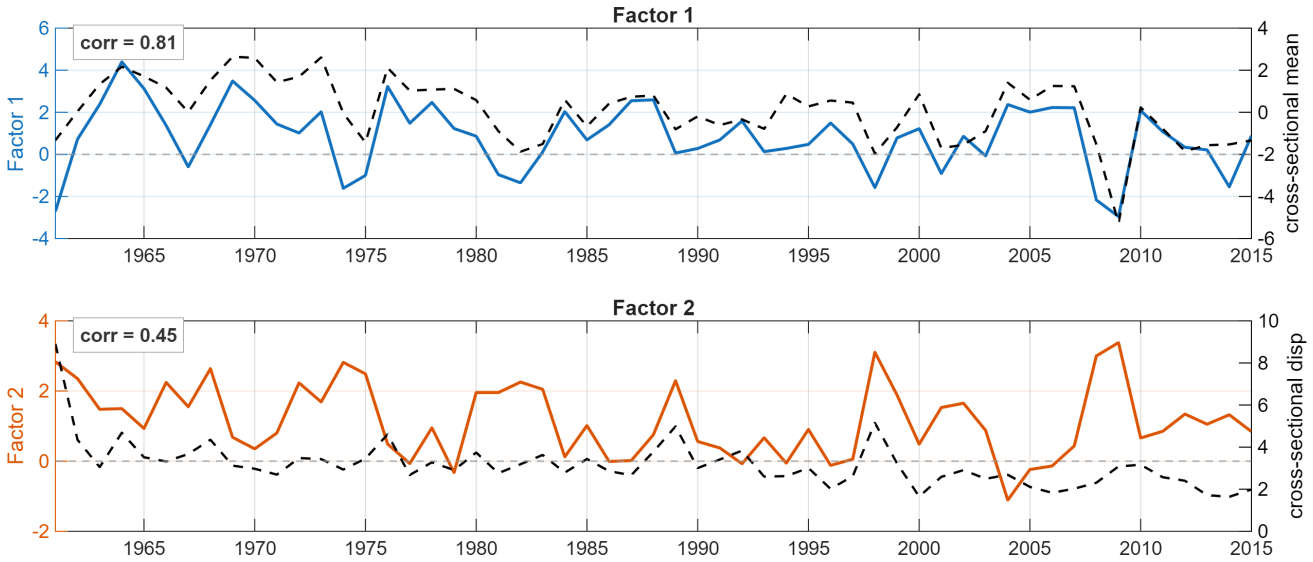
The latent dimension is selected by the cross-sectional cross-validation exercise described in Appendix E. In the macroeconomic panel, countries are partitioned into ten folds and the pooled out-of-sample R^2 is computed over candidate dimensions $K \in \{1, \ldots, 5\}$. As reported in Appendix Figure 10, the validation fit increases from one factor to two factors and reaches its maximum at $K = 2$, after which it declines. We therefore use a two-factor IPCA model in what follows.

With $K = 2$, the panel-wise R^2 of the IPCA model is 48.6% when measured relative to the predictor zero, whereas the corresponding two-factor PCA benchmark delivers an R^2 of only 34.7%. More generally, the IPCA fit remains well above the fit delivered by the static-loading PCA alternatives considered in the companion numerical exercise. The gain from IPCA is therefore not simply a consequence of allowing for more latent variation. Rather, the structured, characteristic-driven loading map provides a materially better description of international growth comovement than static-loadings PCA.

Figure 4 reports the two estimated factor series. The first factor closely tracks the equal-weighted cross-sectional mean of country growth rates, indicating that it captures the common level component in the panel. The second factor instead comoves primarily with the cross-sectional dispersion of country growth. This second factor therefore appears to summarize how unevenly global shocks are transmitted across countries, rather than a second mean-growth component.

The two-factor loading estimates are reported in Figure 5. 90% confidence intervals are reported using the inference results in Appendix F. The most significantcoefficients for the first factor are inflation and population growth, both of which enter negatively. For the second factor, growth volatility enters negatively and mean growth enters positively, with both effects statistically significant. Taken together, the loading estimates reinforce the interpretation that the first factor acts as a broad common-growth component, while the second factor is tied more

Figure 4: Estimated Factors in the Revised International Macroeconomic Specification



Notes: The top panel reports Factor 1 together with the equal-weighted cross-sectional mean of GDP growth. The bottom panel reports Factor 2 together with the cross-sectional dispersion of GDP growth. Both factors are from the $K = 2$ IPCA model.

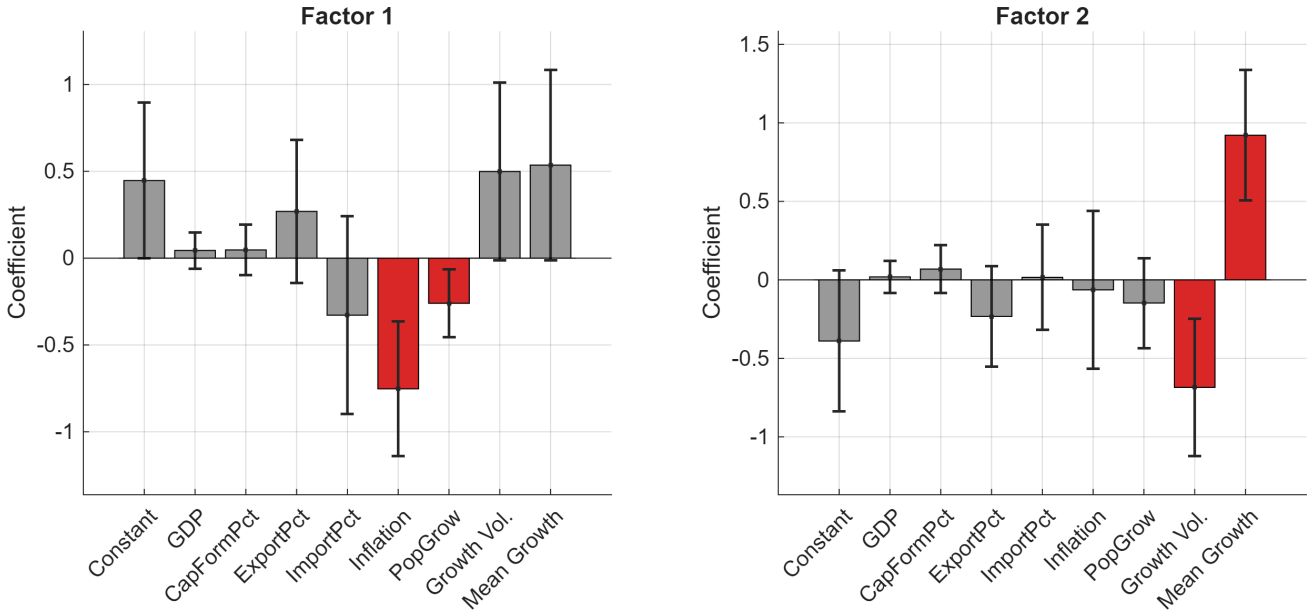
closely to cross-country dispersion in macroeconomic performance.

Finally, Figure 6 summarizes the model's time-varying fit and average loading behavior. The left panel reports the time series of cross-sectional mean absolute error together with a five-year moving average. The right panel reports the cross-sectional mean of $z_{it}\Gamma$ for each factor. The gradual decline in MAE indicates that the model becomes increasingly successful at describing the panel of country growth rates over time. At the same time, the mean loading on the first factor varies substantially, indicating meaningful dynamics in exposure to the common-growth component. The mean loading on the second factor is more oscillatory, consistent with a factor that captures changing dispersion rather than average world growth. Overall, the results point to an important role for dynamic loadings in global business cycle models, while the second factor reveals an additional dispersion channel that is missed by one-factor specifications and by static-loadings PCA.

8.2 Asset Pricing

KPS19 applies IPCA to estimate systematic risk factors of US stocks, leveraging firm characteristics as instruments. Here, we expand upon their asset pricing context and use IPCA to calculate

Figure 5: Estimated Loading Coefficients with 90% Confidence Intervals



Notes: Bars report the estimated loading coefficients in the $K = 2$ IPCA model. Error bars are 90% confidence intervals based on analytical standard errors from the feasible sandwich procedure in Appendix F. Highlighted coefficients are those whose 90% confidence intervals exclude zero.

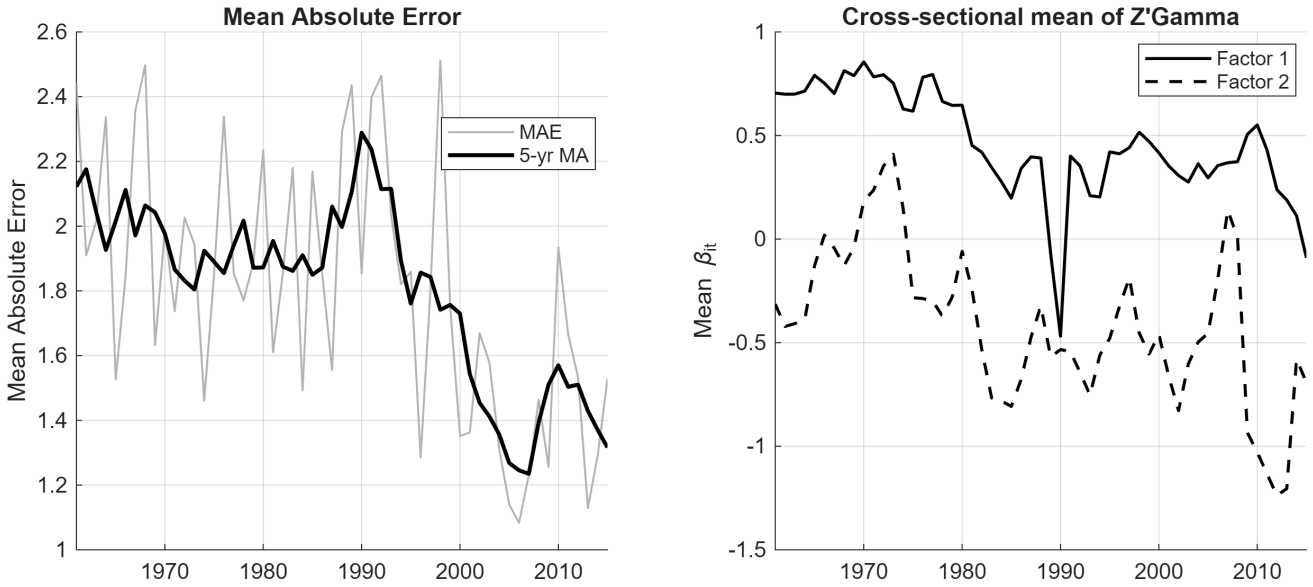
the risk exposure (β) and associated risk premium (cost of capital) for *newly-listed* firms that the model has not seen before. This is the challenging question of how to value private firms and if it is possible to use information in publicly traded equity prices for this purpose. Like the macroeconomic example, IPCA is ideally suited to address this question because the model parameterizes risk and cost of capital as a function of firm characteristics. IPCA estimates the link between risk exposures and observable characteristics of traded firms, which is then extrapolated to newly-listed firms to approximate their risk exposures. Note that this is not possible with standard empirical methods that require the history of returns for individual assets to run time-series regressions.

Our data consists of over 2.9 million stock-month observations of excess stock returns (x) and 93 associated firm characteristics (z) from 1965-2018.[31] We use lagged firm characteristics to instrument for the conditional systematic risk loadings ($z_{i,t} = [size_{i,t-1}, bm_{i,t-1}, ...]$).[32] The out-

[31]The dataset is from Gu, Kelly, and Xiu (2020). Firm characteristics are transformed into ranks on the interval $[-0.5, 0.5]$ as in KPS19.

[32]In finance applications, the $t-1$ subscript in $z_{i,t-1}$ indicates that instruments are observable at the end of month $t-1$, as opposed to the returns which are available and the end of month t. In this paper, the content expressed with these notation is formalized by Assumption B, which indicates idiosyncratic returns are orthogonal

Figure 6: Mean Absolute Error and Average Loading



Notes: The left panel reports the annual cross-sectional mean absolute error of the fitted values together with a five-year moving average. The right panel reports the cross-sectional mean of the fitted loading index $z_{it}\Gamma$ for each of the two factors. Both panels are based on the $K = 2$ IPCA model.

of-sample evaluation is performed for newly-listed firms, defined as the first 12 months following a firm's initial public offering (IPO). In our data, 21,275 firms have an IPO, comprising 249,414 out-of-sample stock-month observations. In-sample estimation is based on the complementary 2.6 million observations of incumbent stocks that exclude those in the test sample of new listings.

We calculate the total R^2 and predictive R^2 to evaluate the estimates of stocks' systematic risk and expected returns, respectively. The procedure starts by first estimating $\widehat{\Gamma}$ within the in-sample data set of incumbent stocks. Then, the estimates are brought out of the estimation sample to the data of new listings to calculate fitted values:

$$\widehat{x}_{i,t}^{\text{Tot}} := z_{i,t}\widehat{\Gamma}\widehat{f}_t(\widehat{\Gamma}), \qquad \widehat{x}_{i,t}^{\text{Pred}} := z_{i,t}\widehat{\Gamma}\widehat{\lambda}, \qquad \forall i,t \in \text{new listings}.$$

The first term $\widehat{x}_{i,t}^{\text{Tot}}$ is the IPCA factor model's fitted value of realized returns. The second is a fitted value for the risk premium. It replaces the factor realization with the factor premium, $\widehat{\lambda}$, which is estimated as the in-sample time series mean of $\widehat{f}_t(\widehat{\Gamma})$. The Total and Predictive R^2's are then calculated with respect to these two kinds of fitted values, receptively.[33]

to instruments.

[33]Total $R^2 := 1 - \frac{\sum_{i,t}(x_{i,t}-\widehat{x}_{i,t}^{\text{Tot}})^2}{\sum_{i,t} x_{i,t}^2}$, Predictive $R^2 := 1 - \frac{\sum_{i,t}(x_{i,t}-\widehat{x}_{i,t}^{\text{Pred}})^2}{\sum_{i,t} x_{i,t}^2}$. When i,t is summed over the sample *outside* of the parameter estimation sample, the R^2 is called "out-of-sample".

Table 1: Explained Variation of Stock Returns

	Total R^2	Predictive R^2
Incumbent stocks (in-sample)	15.66	0.25
New listings (out-of-sample)	13.44	0.22

Notes: R^2 in percentage. Based on IPCA with $K = 4$ for the 1965-2018 US stock-month panel.

Table 1 reports the estimation result. The close similarity of the in-sample and out-of-sample total R^2 indicates the same characteristics that determine the systematic riskiness of incumbent stocks also determine the riskiness of new listings. In other words, once we condition on firm characteristics, the model finds a highly similar description for the common variation among returns on newly-listed stocks compared to the common variation in returns on incumbents.

While the total R^2 is indicative of the model's ability to describe systematic risks of new listings, the predictive R^2 summarizes the model's description of their expected returns (or, in equivalent terms, cost of capital or discount rates). That is, the predictive R^2 is especially informative about the usefulness of the model for asset valuation. The close similarity of the in-sample and out-of-sample predictive R^2 indicates that the IPCA model is as effective at "pricing" new listings as it is for pricing incumbent stocks. The most important takeaway is that the model does this without using the individual return history of the new firms (which is, of course, unavailable and is the crux of the research question), but manages to price them nonetheless by extrapolating what it learns from data on incumbent stocks.

9 Conclusion

This paper has introduced a new approach of modeling and estimating the latent factor structure of panel data, called Instrumented Principal Component Analysis (IPCA). The key innovation is using additional panel data to instrument for the dynamic factor loadings. Mainly, each individual's time-varying factor loading is related to instrumental data according a structural mapping, which is the key interest of estimation alongside the latent factors.

Estimating this mapping, rather than the factor loadings directly, has many econometric advantages compared to other latent variable estimators like PCA. On one hand, the mapping's parameterization tends to be more parsimonious. Its degrees of freedom are fixed and not in-

creasing with the size of the cross section, which improves the rate of convergence and tends to avoid the over-fitting problem in high-dimensional applications. At the same time, IPCA brings broad possibilities of economic discovery by harnessing the wealth of additional panel information, as well as relying on economic theories that indicate relationships between observable instruments and factor exposures. These advantages are exemplified with two applications to stock returns and international macroeconomics, respectively.

Our main theoretical contribution is to derive the large sample properties of the IPCA estimator. We show consistency and asymptotic normality under general data generating processes, by treating IPCA as an M-estimator. A key condition necessary for consistent estimation is the orthogonality between the instruments and errors in the cross section at each time, which is analogous to the exclusion restriction in the familiar instrumental variables (IV) regression. Our method deals with the well-known rotational unidentification issue in latent factor analysis in general terms rather than under specific normalization assumptions. We show how the choice of normalization affects the rate of convergence and the asymptotic distribution. In particular, the sampling noise introduced by the normalization does not matter for asymptotic noise so long as the estimated and true parameters are under the same identification condition.

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Appendix

A Notation Details

Matrix Operations: $\text{vect}(A)$ vectorizes matrix A to a column by going right first, then down. Throughout the paper, γ is always $\text{vect}(\Gamma)$, same for all the other decorated $\widehat{\Gamma}, \Gamma^0(\Theta)$, etc. Related, $\text{veca}(A)$ stacks A's upper triangle entries, including the diagonal, in a vector by going right first, then down; $\text{vecb}(A)$ stacks A's upper triangle entries, *not* including the diagonal, in a vector by going right first, then down. For example, let

$$A = \begin{bmatrix} 1, 2, 3 \\ 4, 5, 6 \\ 7, 8, 9 \end{bmatrix}, \tag{12}$$

then $\text{vect}(A) = [1, 2, 3, 4, 5, 6, 7, 8, 9]^\top$, $\text{veca}(A) = [1, 2, 3, 5, 6, 9]^\top$, $\text{vecb}(A) = [2, 3, 6]^\top$.

$[A; B]$ means the two matrices vertically stacked.

$\|\cdot\|$ is the Euclidean norm of a vector, or the Frobenius norm if the input is a matrix. $\|\cdot\|^2$ is the square of $\|\cdot\|$, or the sum of squares across all entries of the vector or matrix.

Summations like $\sum_i, \sum_t$ are always in the range of the panel sample: $i = 1$ to N, $t = 1$ to T, without writing out the details.

B Proofs of Results in the Main Text

B.1 Verify Θ_{Y} is an identification condition

Lemma 1. *For any $\Gamma \in \Psi$, there exists a rotationally equivalent representative $\Gamma' \in \Theta_{\text{Y}}$. This representative is unique once the diagonal entries in [2] are distinct and the column signs are fixed by the sign convention in Appendix G.2.*

Proof. We proof by first constructing the normalization Γ', and then characterizing the conditions under which it is unique. Before that, we notice a relationship:

$$\frac{1}{T}\sum_t \widehat{f}_t(\Gamma R)\,\widehat{f}_t^{\top}(\Gamma R) = R^{-1}\frac{1}{T}\sum_t \widehat{f}_{t(\Gamma)}\widehat{f}_t^{\top}{}_{(\Gamma)}R^{-1\top} \tag{13}$$

Normalization procedure: Starting from a Γ, we look for a $\Gamma R \in \Theta_{\mathrm{Y}}$.

1. Find Cholesky decomposition: $[\mathrm{Chol}]^\top [\mathrm{Chol}] = \Gamma^\top \Gamma$.

2. Calculate: $[\mathrm{OldV}] = \frac{1}{T} \sum_t \widehat{f}_{t(\Gamma)} \widehat{f}_t^{\top}{}_{(\Gamma)}$.

3. Find Eigen-decomposition: $[\mathrm{Chol}] [\mathrm{OldV}] [\mathrm{Chol}]^\top = [\mathrm{Orth}] [\mathrm{Diag}] [\mathrm{Orth}]^\top$ such that $[\mathrm{Diag}]$ has descending diagonal entries.

4. Finally, define the normalization as $\Gamma' = \Gamma [\mathrm{Chol}]^{-1} [\mathrm{Orth}]$.

5. If one also wishes to pin down the signs of the K columns of Γ', align them according to the sign convention in Appendix G.2.

To verify the normalization procedure is correct, we check that $\Gamma' \in \Theta_{\mathrm{Y}}$. We have $\Gamma'^\top \Gamma' = [\mathrm{Orth}]^\top [\mathrm{Chol}]^{-1\,\top} \Gamma^\top \Gamma [\mathrm{Chol}]^{-1} [\mathrm{Orth}] = \mathbb{I}_K$. And, based on Eq. 13,

$$\frac{1}{T} \sum_t \widehat{f}_t (\Gamma') \widehat{f}_t^{\top} (\Gamma') = [\mathrm{Orth}]^{-1} [\mathrm{Chol}] [\mathrm{OldV}] [\mathrm{Chol}]^\top [\mathrm{Orth}]^{-1\,\top} = [\mathrm{Diag}] \tag{14}$$

which is diagonal with descending entries. Hence there exists a rotationally equivalent representative $\Gamma' \in \Theta_{\mathrm{Y}}$.

Uniqueness:

We now characterize when such a Γ' is unique. Since $\Gamma' = \Gamma R$, it is enough to characterize the admissible R.

First, given Θ_{Y} restriction [1], $R^\top \Gamma^\top \Gamma R = \mathbb{I}_K$. The possible R must therefore have decomposition $R = [\mathrm{Chol}]^{-1} [\mathrm{Orth}]$, where $[\mathrm{Orth}]$ is an orthonormal matrix ($[\mathrm{Orth}]^\top [\mathrm{Orth}] = \mathbb{I}$).

Plugging this decomposition into Θ_{Y} restriction [2], we need an $[\mathrm{Orth}]$ that also satisfies $[\mathrm{Orth}]^{-1} [\mathrm{Chol}] [\mathrm{OldV}] [\mathrm{Chol}]^\top [\mathrm{Orth}]^{-1\,\top} = [\mathrm{Diag}]$. Thus $[\mathrm{Orth}]$ must be an eigenvector matrix of $[\mathrm{Chol}] [\mathrm{OldV}] [\mathrm{Chol}]^\top$. If the diagonal entries of $[\mathrm{Diag}]$ are distinct, then this eigenvector matrix is unique up to columnwise sign changes, and the descending-order convention rules out permutations. After additionally imposing the sign convention in Appendix G.2, Γ' is unique. If some diagonal entries are repeated, then the corresponding eigenspace is identified only up to orthogonal rotation within that eigenspace, so uniqueness holds only up to such within-eigenspace

rotations. □

B.2 General Results for Uniform Large-N, T Convergence

This subsection gives some general results for uniform large-N, T convergence that are used as building blocks for proving the subsequent propositions. These results are stated with a general term like $y_{N,t}(\Gamma)$ rather than of specific expression involving z, μ etc.

Let $y_{N,t}(\Gamma)$ represent a sequence (indexed by N) of stochastic processes that are functions of Γ and are of finite dimensions. Define several large N limiting conditions that it can be subject to. These conditions are all *uniformly* across Γ, hence prefixed with "U-".

$$\lim_{N\to\infty} \mathbb{E}\left[\sup_{\Gamma\in\Psi} \|y_{N,t}(\Gamma)\|\right] = 0 \qquad \text{(U-mean converging)}$$

$$\lim_{N\to\infty} \mathbb{E}\left[\sup_{\Gamma\in\Psi} \|y_{N,t}(\Gamma)\|^2\right] = 0 \qquad \text{(U-mean square converging)}$$

$$\exists M, N^*, \text{s.t.} \quad \mathbb{E}\left[\sup_{\Gamma\in\Psi} \|y_{N,t}(\Gamma)\|^2\right] < M \quad \forall N > N^* \qquad \text{(U-mean square bounded)}$$

$$\exists M, N^*, \text{s.t.} \quad Pr\left\{\sup_{\Gamma} \|y_{N,t}(\Gamma)\| < M\right\} = 1, \quad \forall N > N^* \qquad \text{(U-a.s. bounded)}$$

Lemma 2 is critical to establish large N, T simultaneous convergence. It shows U-mean converging is important since it is the necessary condition for large-N, T simultaneous convergence. Notice in the conclusion statement of the Lemma, $N^{[1]}$ does not depend on T but only on ϵ, δ. Hence, this statement implies large N, T simultaneous convergence, a property that will be used in the proof of Proposition 1 later.

Lemma 2. *Given Assumption A.3 and let* $y_{N,t}(\Gamma) := g_N(\Gamma, z_{1,t}, \ldots, z_{N,t}, \eta_{1,t}, \ldots, \eta_{N,t}, \mu_{1,t}, \ldots, \mu_{N,t}, f_t^0)$ *be a continuous vector-valued function of* Γ *and the variables at time* t*. If* $y_{N,t}(\Gamma)$ *is U-mean converging, then its time-series average is converging in the large-*N *probability limit uniformly for any* T*. That is,* $\forall \epsilon, \delta > 0, \exists N^{[1]}$ *s.t.* $\forall T$ *and* $\forall N > N^{[1]}$

$$Pr\left\{\sup_{\Gamma\in\Psi} \left\| \frac{1}{T}\sum_t y_{N,t}(\Gamma) \right\| > \epsilon \right\} < \delta. \xrightarrow{p} 0, \qquad N \to \infty. \tag{15}$$

Proof. Start by an inequality exchanging the order of $\sup \| \ \|$ and $\sum$:

$$\sup_{\Gamma\in\Psi}\left\|\frac{1}{T}\sum_t y_{N,t}(\Gamma)\right\| \leq \sup_{\Gamma\in\Psi}\frac{1}{T}\sum_t \|y_{N,t}(\Gamma)\| \leq \frac{1}{T}\sum_t \sup_{\Gamma\in\Psi}\|y_{N,t}(\Gamma)\|. \tag{16}$$

Apply expectation on both sides. By Assumption A.3, $\sup_{\Gamma\in\Psi}\|y_{N,t}(\Gamma)\|$ is stationary.

$$\mathbb{E}\sup_{\Gamma\in\Psi}\left\|\frac{1}{T}\sum_t y_{N,t}(\Gamma)\right\| \leq \mathbb{E}\frac{1}{T}\sum_t \sup_{\Gamma\in\Psi}\|y_{N,t}(\Gamma)\| = \mathbb{E}\sup_{\Gamma\in\Psi}\|y_{N,t}(\Gamma)\|. \tag{17}$$

The last term $\mathbb{E}\sup_{\Gamma\in\Psi}\|y_{N,t}(\Gamma)\|$ is irrelevant of T, and it converges to zero as $N\to\infty$, according to the definition of U-mean converging. Hence, the left-hand side term is T-uniform large N-converging. That is, $\forall\epsilon>0, \exists N^{[1]}$ s.t. $\forall T$ and $\forall N > N^{[1]}$

$$\mathbb{E}\sup_{\Gamma\in\Psi}\left\|\frac{1}{T}\sum_t y_{N,t}(\Gamma)\right\| < \epsilon. \tag{18}$$

This implies, by Chebyshev's inequality, the required T-uniform N-convergence in probability.

□

The next lemma gives some calculation rules to reach a U-mean converging sequence.

Lemma 3. *If $x^{[1]}_{N,t}(\Gamma)$ is U-mean square converging, $x^{[2]}_{N,t}(\Gamma)$ is U-mean square bounded, and $x^{[3]}_{N,t}(\Gamma)$ is U-a.s. bounded, then*

1. *$x^{[1]}_{N,t}(\Gamma)x^{[2]}_{N,t}(\Gamma)$ is U-mean converging, which implies*
2. *$x^{[1]}_{N,t}(\Gamma)$ by itself is also U-mean converging.*
3. *$x^{[1]}_{N,t}(\Gamma)x^{[3]}_{N,t}(\Gamma)$ is still U-mean square converging.*
4. *$x^{[2]}_{N,t}(\Gamma)x^{[3]}_{N,t}(\Gamma)$ is still U-mean square bounded.*

Proof. 1. For each $\omega\in\Omega$, we have

$$\sup_{\Gamma}\|y_{N,t}(\Gamma)\| = \sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)x^{[2]}_{N,t}(\Gamma)\right\| \leq \sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|\sup_{\Gamma}\left\|x^{[2]}_{N,t}(\Gamma)\right\|. \tag{19}$$

So, put inside expectation:

$$\mathbb{E}\sup_{\Gamma}\|y_{N,t}(\Gamma)\| \leq \mathbb{E}\left[\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|\sup_{\Gamma}\left\|x^{[2]}_{N,t}(\Gamma)\right\|\right] \tag{20}$$

$$\leq \left(\mathbb{E}\left[\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\right]\mathbb{E}\left[\sup_{\Gamma}\left\|x^{[2]}_{N,t}(\Gamma)\right\|^2\right]\right)^{1/2} \tag{21}$$

by Cauchy-Schwarz inequality. Then it is easy to wrap up the proof with deterministic limit analysis. Namely the product of a sequence converging to zero and a bounded sequence is also converging to zero, and the square root of a converging sequence converges to the square root.

2. Trivial.

3. By a matrix version of the Cauchy-Schwarz inequality:

$$\|y_{N,t}(\Gamma)\|^2 = \left\|x^{[1]}_{N,t}(\Gamma)x^{[3]}_{N,t}(\Gamma)\right\|^2 \leq \left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\left\|x^{[3]}_{N,t}(\Gamma)\right\|^2 \tag{22}$$

Apply $E\sup_{\Gamma}$ on both sides:

$$\mathbb{E}\sup_{\Gamma}\|y_{N,t}(\Gamma)\|^2 \leq \mathbb{E}\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\left\|x^{[3]}_{N,t}(\Gamma)\right\|^2 \tag{23}$$

$$\leq \mathbb{E}\left[\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\sup_{\Gamma}\left\|x^{[3]}_{N,t}(\Gamma)\right\|^2\right] \tag{24}$$

For any ω, if $\sup_{\Gamma\in\Psi}\left\|x^{[3]}_{N,t}(\Gamma)\right\| < M$, then $\sup_{\Gamma\in\Psi}\left\|x^{[3]}_{N,t}(\Gamma)\right\|^2 < M^2$. That means $\left\|x^{[3]}_{N,t}(\Gamma)\right\|^2$ is also U-a.s. bounded for large enough N's. Plug the bound, M^2, back into the expectation calculation for any finite N we had above:

$$\mathbb{E}\sup_{\Gamma}\|y_{N,t}(\Gamma)\|^2 \leq \mathbb{E}\left[\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\right]M^2 \tag{25}$$

Take large-N limits on both sides:

$$\lim_N \mathbb{E}\sup_{\Gamma}\|y_{N,t}(\Gamma)\|^2 \leq \lim_N \mathbb{E}\left[\sup_{\Gamma}\left\|x^{[1]}_{N,t}(\Gamma)\right\|^2\right]M^2 = 0. \tag{26}$$

4. Almost the same the as the previous proof. Just change $x^{[1]}_{N,t}(\Gamma)$ to $x^{[2]}_{N,t}(\Gamma)$ everywhere until

the last three lines. In the last three lines, just change "$\lim_N = 0$" to "$\limsup_N < \infty$". □

Lemma 4 builds the necessary conditions for U-mean square converging from Assumption A.1.

Lemma 4. *Given Assumption A.1, and that* $\mathbb{E}[y|\mathcal{C}] = \mathbf{0}$ *and* $y_{i,t}$ *is irrelevant of* Γ*, then its cross-sectional average* $y_{N,t}(\Gamma) = \frac{1}{N}\sum_i y_{i,t}$ *is U-mean square converging.*

Proof. Given Assumption A.1, and that $\mathbb{E}[y_{i,t}|\mathcal{C}] = \mathbf{0}$:

$$\frac{1}{N}\sum_{i=1}^{N} y_{i,t} \xrightarrow{L^2} \mathbf{0}, \forall t. \tag{27}$$

Because $y_{i,t}$ is irrelevant of Γ, it is trivial that the mean squared convergence is uniform, and that $y_{N,t}(\Gamma) = \frac{1}{N}\sum_i y_{i,t}$ is U-mean square converging. □

B.3 Proposition 1

B.3.1 Complete Statement of Proposition 1

We first state the complete version of the Proposition 1 with the addition of an intermediate large-N result and writing the expressions of the probability limits.

Proposition 1 complete (Uniform Convergence of the Score Function)**.**

Under Assumptions A–D, the score function converges uniformly in probability:

$$\sup_{\Gamma\in\Psi} \left\| \widehat{S}(\Gamma) - S_T(\Gamma) \right\| \xrightarrow{p} 0, \qquad N \to \infty, \forall T, \tag{28}$$

$$\sup_{\Gamma\in\Psi} \left\| \widehat{S}(\Gamma) - S^0(\Gamma) \right\| \xrightarrow{p} 0, \qquad N, T \to \infty, \tag{29}$$

where

$$S_T(\Gamma) := \frac{1}{T}\sum_t \operatorname{vect}\left(\Omega_t^{zz}\Pi_{t(\Gamma)} f_t^0 \widetilde{f}_{t(\Gamma)}^{\top}\right), \tag{30}$$

$$S^0(\Gamma) := \mathbb{E}\operatorname{vect}\left(\Omega_t^{zz}\Pi_{t(\Gamma)} f_t^0 \widetilde{f}_{t(\Gamma)}^{\top}\right), \tag{31}$$

$$\Pi_{t(\Gamma)} := \left(\mathbb{I}_L - \Gamma\left(\Gamma^{\top}\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^{\top}\Omega_t^{zz}\right)\Gamma^0. \tag{32}$$

B.3.2 Preparations for the Proof of Proposition 1

The proof of Proposition 1 is quite involved. The first part manipulates the expression of the score function to a form consisting of primitive random variables. Second, Lemma 6 deals with the cross-sectional convergence. Then, Lemma 7 deals with the time-series dimension. In the final step, the results are put together to finish the proof of for the large N, T convergence in Proposition 1.

The first part of proof manipulates the expression of the score function to a form consisting of primitive random variables. Then, we analyze its uniform convergence. We already have

$$\widehat{S}(\Gamma) = \frac{1}{NT}\sum_t \left(Z_t^\top \otimes \widehat{f}_{t(\Gamma)}\right)\left(x_t - Z_t\Gamma\widehat{f}_{t(\Gamma)}\right) = \frac{1}{NT}\sum_t \operatorname{vect}\left(Z_t^\top \widehat{e}_{t(\Gamma)}\widehat{f}_{t\,(\Gamma)}^\top\right), \tag{33}$$

and

$$\widehat{f}_{t(\Gamma)} = \widetilde{f}_{t(\Gamma)} + \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}. \tag{34}$$

Population OLS error:

$$\widetilde{e}_{t(\Gamma)} = x_t - Z_t\Gamma\widetilde{f}_{t(\Gamma)} = e_t + Z_t\Pi_{t(\Gamma)}f_t^0,$$

with the shorthand $\Pi_{t(\Gamma)} := \left(\mathbb{I}_L - \Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\right)\Gamma^0$. Combined with Eq. (10), we have

$$\begin{aligned}\widehat{e}_{t(\Gamma)} &= e_t + Z_t\Pi_{t(\Gamma)}f_t^0 - Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top\widetilde{e}_{t(\Gamma)}\\ \widehat{f}_{t(\Gamma)} &= \widetilde{f}_{t(\Gamma)} + \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top\widetilde{e}_{t(\Gamma)}\end{aligned} \tag{35}$$

Plug those back to the score. Each summand in equation (33) yields $3 \times 2 = 6$ terms:

$$\begin{aligned}&Z_t^\top \widehat{e}_{t(\Gamma)}\widehat{f}_{t\,(\Gamma)}^\top\\ =&Z_t^\top e_t \qquad\qquad \widetilde{f}_{t\,(\Gamma)}^\top\end{aligned}$$

$$
\begin{aligned}
&+Z_t^\top Z_t \Pi_{t(\Gamma)} f_t^0 && \widetilde{f}_t^\top{}_{(\Gamma)} \\
&-Z_t^\top Z_t \Gamma \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1} \Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)} && \widetilde{f}_t^\top{}_{(\Gamma)} \\
&+Z_t^\top e_t && \widetilde{e}_t^\top{}_{(\Gamma)} Z_t \Gamma \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1} \\
&+Z_t^\top Z_t \Pi_{t(\Gamma)} f_t^0 && \widetilde{e}_t^\top{}_{(\Gamma)} Z_t \Gamma \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1} \\
&-Z_t^\top Z_t \Gamma \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1} \Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)} && \widetilde{e}_t^\top{}_{(\Gamma)} Z_t \Gamma \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1}. \qquad (36)
\end{aligned}
$$

Call the six terms $S_t^{[1]}(\Gamma)$ to $S_t^{[6]}(\Gamma)$, so that

$$
\widehat{S}(\Gamma) = \frac{1}{NT} \sum_t \text{vect}\left(S_t^{[1]}(\Gamma) + \cdots + S_t^{[6]}(\Gamma)\right). \qquad (37)
$$

Before jumping into the rigorous proof, we give a loose description of the rationale. Given the expression of score function in Eq. (37), Proposition 1 states the score function's uniform probability limit. First, taking $N \to \infty$ at a fixed t, we have the three modular results, $\frac{1}{N} Z_t^\top e_t \to \mathbf{0}$, $\frac{1}{N} \Gamma Z_t^\top \widetilde{e}_{t(\Gamma)} \to \mathbf{0}$, and $\frac{1}{N} Z_t^\top Z_t = \mathcal{O}_p(1)$, by cross-sectional LLN. Plugging these into the score expression (36), we find that $\frac{1}{N} S_t^{[p]}(\Gamma) \to \mathbf{0}$ for $p = 1, 3, 4, 5, 6$. The second term is an exception as the only one not involving an $\widetilde{e}_{t(\Gamma)}$ or e_t error term. It correspond to the first source of $\widehat{f}_t$ decomposition purely from a "wrong" Γ rather than the finite sample. We see $S_t^{[2]}$ increases with N. We will show $\frac{1}{N} S_t^{[2]}(\Gamma) \to \Omega_t^{zz} \Pi_{t(\Gamma)} f_t^0 \widetilde{f}_t^\top{}_{(\Gamma)}$, where cross-sectional limit is $\mathcal{C}$-measurable as in Assumption A.1. Taking a finite time-series average yields the finite-T large-N convergence of the score given by Eq. (28).[34] Finally, taking $T \to \infty$ delivers the score's convergence to the unconditional expectation, as reported in Eq. (29).

The proof, to a large extent, follows these steps and uses the arguments in the proof of 16, with the complication of uniform convergence across Γ, which is necessary for solution convergence (Theorem 1) proved further below.

Lemma 5 (Terms only involving instruments)**.** *Given Assumption D, let $y_{N,t}(\Gamma)$ be a function*

[34] This result could be used to construct finite-T large-N inference – we leave that for future work.

involving the following terms:

$$y_{N,t}(\Gamma) := g\left(\Gamma, \frac{1}{N}Z_t^\top Z_t, \left(\frac{1}{N}\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}, \Omega_t^{zz}, \left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right), \tag{38}$$

where g is a vector-valued continuous function, then $y_{N,t}(\Gamma)$ is U-a.s. bounded.

Proof. Given that g is a continuous function, it only remains to show that the input terms are all U-a.s. bounded. Γ is U-a.s. bounded by Assumption D(1). $\frac{1}{N}Z_t^\top Z_t$ is U-a.s. bounded by Assumption D(2). According to Assumption A.1, $\frac{1}{N}Z_t^\top Z_t - \Omega_t^{zz} \xrightarrow{p} \mathbf{0}$, and $\left(\frac{1}{N}\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1} - \left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1} \xrightarrow{p} \mathbf{0}$, as $N \to \infty$. Combining Assumptions D(1) and D(3), matrix $\Gamma^\top\Omega_t^{zz}\Gamma$ is invertible and bounded. As a result, the last three terms $\left(\frac{1}{N}\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}, \Omega_t^{zz}, \left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}$ must also be U-a.s. bounded. □

Lemma 6 (Large N Cross-sectional Convergence at each t)**.** *Under the conditions of Proposition 1,*

$$\frac{1}{N}\left(S_t^{[1]}(\Gamma) + \cdots + S_t^{[6]}(\Gamma)\right) - \Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top{}_{(\Gamma)} \tag{39}$$

is U-mean converging.

Proof. We proceed by analyzing the six terms one by one. We list out the statements in each step and provide in-line proofs of the statements.

1. $\frac{1}{N}Z_t^\top e_t$ is U-mean square converging.

   This is by Lemma 4, treating $z_{i,t}e_{i,t}$ as the $y_{i,t}$ in the lemma. The conditions are met given Assumptions B and C.

2. $\widetilde{f}_t^\top{}_{(\Gamma)}$ is U-mean square bounded.

   $\widetilde{f}_{t(\Gamma)} = \left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0 f_t^0$, in which $\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0$ is U-a.s. bounded by Lemma 5; and f_t^0 is trivially U-mean square bounded by assumption C(1). Then apply Lemma 3.4 to combine the two terms.

3. $\frac{1}{N}S_t^{[1]}(\Gamma)$ is U-mean converging.

   Notice $\frac{1}{N}S_t^{[1]}(\Gamma) = \left[\frac{1}{N}Z_t^\top e_t\right]\left[\widetilde{f}_t^\top(\Gamma)\right]$. Then use the two previous statements about the two parts and apply Lemma 3.1.

4. $\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma)$ is U-mean square bounded.

$$\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma) = \left[\left(\mathbb{I}_L - \Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\right)\Gamma^0\right]\left[f_t^0 f_t^{0\top}\right]\left[\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right]$$

   The first and third terms are U-a.s. bounded according to Lemma 5. The middle term is U-mean square bounded by assumption C(1). Then by Lemma 3.4, the product of the three is U-mean square bounded.

5. $\frac{1}{N}\sum_i\left(z_{i,t}^\top z_{i,t} - \Omega_t^{zz}\right)$ is U-mean square converging.

   The argument is the same as statement number 1 above: treat $z_{i,t}^\top z_{i,t} - \Omega_t^{zz}$ as the $y_{i,t}$ and apply Lemma 4. The required conditions of Lemma 4 are met: we know $\mathbb{E}\left[z_{i,t}^\top z_{i,t} - \Omega_t^{zz}\middle|\mathcal{C}\right] = \mathbf{0}$ per the definition of Ω_t^{zz}; and that the term's expectation exists by Lemma 5.

6. $\frac{1}{N}S_t^{[2]}(\Gamma) - \Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma)$ is U-mean converging.

$$\frac{1}{N}S_t^{[2]}(\Gamma) = \frac{1}{N}Z_t^\top Z_t\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma) \tag{40}$$

$$\frac{1}{N}S_t^{[2]}(\Gamma) - \Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma) = \left[\frac{1}{N}\sum_i\left(z_{i,t}^\top z_{i,t} - \Omega_t^{zz}\right)\right]\left[\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^\top(\Gamma)\right] \tag{41}$$

   Then, straightforward application of the previous two statements on Lemma 3.1.

7. $Z_t^\top Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}$ is U-a.s. bounded.

   The term equals $\left[\frac{1}{N}Z_t^\top Z_t\Gamma\right]\left[\left(\Gamma^\top\frac{1}{N}Z_t^\top Z_t\Gamma\right)^{-1}\right]$. Then apply Lemma 5.

8. $\frac{1}{N}\Gamma^\top Z_t^\top\widetilde{e}_{t(\Gamma)}$ is U-mean square converging.

   We have previously defined:

$$\widetilde{e}_{t(\Gamma)} = e_t + Z_t\Pi_{t(\Gamma)}f_t^0 \tag{42}$$

$$\Pi_{t(\Gamma)} = \left(\mathbb{I}_L - \Gamma \left(\Gamma^\top \Omega_t^{zz} \Gamma \right)^{-1} \Gamma^\top \Omega_t^{zz} \right) \Gamma^0 \tag{43}$$

Now define:

$$Q_{t(\Gamma)} := \mathbb{I}_L - \Omega_t^{zz} \Gamma \left(\Gamma^\top \Omega_t^{zz} \Gamma \right)^{-1} \Gamma^\top. \tag{44}$$

Therefore,

$$\frac{1}{N} \Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)} = \frac{1}{N} \Gamma^\top Z_t^\top e_t + \frac{1}{N} \Gamma^\top Z_t^\top Z_t Q_{t\,(\Gamma)}^\top \Gamma^0 f_t^0. \tag{45}$$

The first term is made up $\Gamma^\top$, which is U-a.s. bounded according to Lemma 5, and $\frac{1}{N} Z_t^\top \widetilde{e}_{t(\Gamma)}$, which is U-mean square converging, according to Statement 1. Therefore, the first term itself is also U-mean square converging, according to Lemma 3.3.

For the second term, we know $Q_{t(\Gamma)} \Omega_t^{zz} \Gamma = \mathbf{0}$. Hence, the second term changes to:

$$\frac{1}{N} \Gamma^\top Z_t^\top Z_t Q_{t\,(\Gamma)}^\top \Gamma^0 f_t^0 = \Gamma^\top \left(\frac{1}{N} Z_t^\top Z_t - \Omega_t^{zz} \right) Q_{t\,(\Gamma)}^\top \Gamma^0 f_t^0 \tag{46}$$

Rearrange it as,

$$\Gamma^\top \left[\frac{1}{N} \sum_i \left(z_{i,t}^\top z_{i,t} - \Omega_t^{zz} \right) \otimes f_t^{0\top} \right] \text{vect} \left(Q_{t\,(\Gamma)}^\top \Gamma^0 \right). \tag{47}$$

The first and third part is U-a.s. bounded per Lemma 5. The middle part is U-mean square converging by the same argument as Statement 1 and 5. To apply the same argument, using Lemma 4, we can verify that

$$\mathbb{E} \left[\left(z_{i,t}^\top z_{i,t} - \Omega_t^{zz} \right) \otimes f_t^{0\top} \middle| \mathcal{C} \right] = \mathbf{0}; \tag{48}$$

and that the moment $\mathbb{E} \left\| \left(z_{i,t}^\top z_{i,t} - \Omega_t^{zz} \right) \otimes f_t^{0\top} \right\|^2$ exists per Assumption C(1) and D(2). Combining the three parts by applying Lemma Lemma 3.3, the second term is also U-

mean square converging.

9. $\frac{1}{N}S_t^{[3]}(\Gamma)$ is U-mean converging.

$$\frac{1}{N}S_t^{[3]}(\Gamma) = \left[Z_t^\top Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\right]\left[\frac{1}{N}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}\right]\left[\widetilde{f}_{t(\Gamma)}^\top\right] \tag{49}$$

The three term are U-a.s. bounded, U-mean square converging, and U-mean square bounded, per above analysis. Then apply Lemma 3 to combine them.

10. $\frac{1}{N}S_t^{[4]}(\Gamma)$, $\frac{1}{N}S_t^{[5]}(\Gamma)$, $\frac{1}{N}S_t^{[6]}(\Gamma)$ are all U-mean converging.

For the last three terms, given the similarities to the situations above, we just write out the decompositions. The remaining arguments that involve repeated applications of Lemma 3 are omitted.

$$\frac{1}{N}S_t^{[4]}(\Gamma) = \left[\frac{1}{N}Z_t^\top e_t\right]\left[\frac{1}{N}\widetilde{e}_{t(\Gamma)}^\top Z_t\Gamma\right]\left(\Gamma^\top\frac{1}{N}Z_t^\top Z_t\Gamma\right)^{-1} \tag{50}$$

$$\frac{1}{N}S_t^{[5]}(\Gamma) = \left[\frac{1}{N}Z_t^\top Z_t\Pi_{t(\Gamma)}\right]\left[f_t^0\right]\left[\frac{1}{N}\widetilde{e}_{t(\Gamma)}^\top Z_t\Gamma\right]\left(\Gamma^\top\frac{1}{N}Z_t^\top Z_t\Gamma\right)^{-1} \tag{51}$$

$$\begin{aligned}\frac{1}{N}S_t^{[6]}(\Gamma) = &\left[\frac{1}{N}Z_t^\top Z_t\Gamma\left(\Gamma^\top\frac{1}{N}Z_t^\top Z_t\Gamma\right)^{-1}\right]\left[\frac{1}{N}\Gamma^\top Z_t^\top\widetilde{e}_{t(\Gamma)}\right]\left[\frac{1}{N}\widetilde{e}_{t(\Gamma)}^\top Z_t\Gamma\right]\\ &\left(\Gamma^\top\frac{1}{N}Z_t^\top Z_t\Gamma\right)^{-1}\end{aligned} \tag{52}$$

Finally, given the analysis above of $S_t^{[1]}\cdots S_t^{[6]}$, we can conclude the required statement. □

Lemma 7 (S_T Convergence)**.** $\sup_{\Gamma\in\Psi}\|S_T(\Gamma) - S^0(\Gamma)\| \xrightarrow{p} 0, T\to\infty$.

Proof. As seen in the definition,

$$S_T(\Gamma) := \frac{1}{T}\sum_t \text{vect}\left(\Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_{t(\Gamma)}^\top\right), \tag{53}$$

$S_T(\Gamma)$ is a time-series average, in which we make $\varkappa_t(\Gamma)$ be the shorthand of the summand,

$\varkappa_t(\Gamma) := \text{vect}\left(\Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_{t(\Gamma)}^{\top}\right)$. In the summand, recall that

$$\Pi_{t(\Gamma)} := \left(\mathbb{I}_L - \Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\right)\Gamma^0, \tag{54}$$

$$\widetilde{f}_t = \left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0 f_t^0. \tag{55}$$

Therefore, the random variables that make up $\varkappa_t(\Gamma)$ are only f_t^0 and Ω_t^{zz} (which is $\Omega_t^{zz} := \mathbb{E}\left[z_{i,t}^\top z_{i,t}\middle|\mathcal{C}\right]$). Both of the two random variables are $\mathcal{C}$-measurable, we can apply Assumption A.3 for the convergence at each Γ.

The additional twist is the Lemma requires uniform convergence over $\Gamma \in \Psi$. We proceed by applying Lemma 2.4 in Newey and McFadden (1994). It requires to construct the Γ-irrelevant random variable d_t, and verify it dominates $\varkappa_t$ and has a finite expectation. To do that, we have

$$\|\varkappa_t(\Gamma)\| = \left\|\left[\Omega_t^{zz}\Pi_{t(\Gamma)}\right]\left[f_t^0 f_t^{0\top}\right]\left[\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right]\right\| \tag{56}$$

$$\leq \|\Omega_t^{zz}\Pi_{t(\Gamma)}\|\left\|\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right\|\left\|f_t^0 f_t^{0\top}\right\| \tag{57}$$

$$\leq M\left\|f_t^0 f_t^{0\top}\right\| := d_t, \forall\Gamma \in \Psi. \tag{58}$$

The first inequality is due to the matrix version of the Cauchy-Schwarz inequality. In the second inequality, the bound M exists because $\|\Omega_t^{zz}\Pi_{t(\Gamma)}\|\left\|\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right\|$ is only constructed of Γ, Ω_t^{zz}, and $\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}$. Hence this term is U-a.s. bounded according to Lemma 5. That is say a finite M exists such that

$$Pr\left\{\sup_{\Gamma\in\Psi}\|\Omega_t^{zz}\Pi_{t(\Gamma)}\|\left\|\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\right\| < M\right\} = 1. \tag{59}$$

We have thus constructed d_t, and shown that $\|\varkappa_t(\Gamma)\| \leq \|d_t\|, \forall\Gamma \in \Psi$. Finally, it is easy to see $\mathbb{E}d_t < \infty$ given Assumption C(1). And thereby the proof is completed. □

B.3.3 The Main Proof of Proposition 1

After preparing the lemmas above, we are finally ready for the main proof of Proposition 1.

Proof. According to Lemma 6, $\frac{1}{N}\left(S_t^{[1]}(\Gamma)+\cdots+S_t^{[6]}(\Gamma)\right)-\Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^{\top}{}_{(\Gamma)}$ is U-mean converging. Apply Lemma 2, that implies $\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S_T(\Gamma)\right\|\xrightarrow{p}0, N\to\infty, \forall T$. That is to say, $\forall\epsilon,\delta>0$, $\exists N^{[1]}$, s.t. $\forall T$ and $\forall N>N^{[1]}$

$$Pr\left\{\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S_T(\Gamma)\right\|<\delta\right\}>1-\epsilon, \tag{60}$$

where we have already defined

$$S_T(\Gamma):=\frac{1}{T}\sum_t \text{vect}\left(\Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_t^{\top}{}_{(\Gamma)}\right). \tag{61}$$

Separately, by Lemma 7, $\sup_{\Gamma\in\Psi}\|S_T(\Gamma)-S^0(\Gamma)\|\xrightarrow{p}0, T\to\infty$. That is to say, $\forall\epsilon,\delta>0$, $\exists T^{[1]}$, s.t. irrelevant of N, $\forall T>T^{[1]}$

$$Pr\left\{\sup_{\Gamma\in\Psi}\left\|S_T(\Gamma)-S^0(\Gamma)\right\|<\delta\right\}>1-\epsilon. \tag{62}$$

Combined, $\forall N>N^{[1]}, T>T^{[1]}$

$$\begin{aligned}
&Pr\left\{\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S^0(\Gamma)\right\|<2\delta\right\} &(63)\\
\geq&Pr\left\{\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S_T(\Gamma)\right\|<\delta \text{ AND } \sup_{\Gamma\in\Psi}\left\|S_T(\Gamma)-S^0(\Gamma)\right\|<\delta\right\} &(64)\\
\geq&1-2\epsilon. &(65)
\end{aligned}$$

That is equivalent to the required conclusion: $\sup_{\Gamma\in\Psi}\left\|\widehat{S}(\Gamma)-S^0(\Gamma)\right\|\xrightarrow{p}0$, as $N,T\to\infty$ □

B.4 Proof of Proposition 2

Proof. "If": It is easy to verify that Γ^0 and all of its rotations solve $S^0(\Gamma)=\mathbf{0}$, because they solve $\Pi_t(\Gamma)=\mathbf{0}$, $\forall\omega$.

"Only if": All we need to show is that $\forall\Gamma$ not rotationally equivalent to Γ^0, $S^0(\Gamma)\neq\mathbf{0}$. The

random term in S^0:

$$\begin{aligned}\Omega_t^{zz}\Pi_{t(\Gamma)}f_t^0\widetilde{f}_{t(\Gamma)}^{\top} &= \Omega_t^{zz}\left(\mathbb{I}_L - \Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\right)\Gamma^0 f_t^0 f_t^{0\top}\Gamma^{0\top}\Omega_t^{zz}\Gamma\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\\ &:= ARf_t^0 f_t^{0\top}R^\top B, \end{aligned} \tag{66}$$

where R is rotation s.t. $\mathbb{E}Rf_t^0 f_t^{0\top}R$ is diagonal with positive entries, and A and B are the shorthands of $L\times K$ and $K\times K$ respectively. Notice $\forall\Gamma$ not rotationally equivalent to Γ^0, $\Pi_{t(\Gamma)}\neq\mathbf{0}$, so A, B are full rank K w.p. 1, by Assumption D. One can construct constant p,q of lengths L,K s.t. the signs of each entry in $p^\top A$ and Bq are always the same. As a result, $p^\top\mathbb{E}\left[ARf_t^0 f_t^{0\top}RB\right]q>0$. □

B.5 Specific Identification Functions Satisfy IF.1–2

Lemma 8 (Verify Condition IF.1–2)**.** *The identification functions $I_{\mathrm{X}}(\Gamma)$ and $I_{\mathrm{Y}}(\Gamma)$ both satisfy Condition IF.1–2.*

Proof. It is obvious for Θ_{X}, because it is deterministic. The identification function and its limit are the same $I_{\mathrm{X}}(\Gamma)=I_{\mathrm{X}}^0(\Gamma)=\mathrm{vect}\left(Block_{1:K}(\Gamma)-\mathbb{I}_K\right)$.

For Θ_{Y}, construct I_{Y}^0 as

$$I_{\mathrm{Y}}^0(\Gamma) := \begin{bmatrix}\mathrm{veca}\left(\Gamma^\top\Gamma-\mathbb{I}_K\right)\\ \mathrm{vecb}\left(\mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}_{t(\Gamma)}^{\top}\right]-V^{\mathrm{ff}}\right)\end{bmatrix}.^{35} \tag{67}$$

Next, we need to verify the two parts of the Condition.

Verify IF.1:

The top $\frac{1}{2}K(K+1)$ rows of I_{Y} and I_{Y}^0 are the same. Next, we mostly need to work on the lower part of I_{Y} and I_{Y}^0. Define

$$I_{[2]}(\Gamma) := \frac{1}{T}\sum_t \widehat{f}_{t(\Gamma)}\widehat{f}_{t(\Gamma)}^{\top} - V^{\mathrm{ff}} \tag{68}$$

[35] We are constructing I_{Y}^0 that will serve in IA to pick the true $\Gamma^0(\Theta_{\mathrm{Y}}^0)$. So in defining I_{Y}^0, function $\widetilde{f}_{t(\Gamma)}$ relies on a Γ^0 that is not pinned down yet. The factors needs to be accordingly rotated by "re-defining" $f_t^0=\widetilde{f}_t\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right)$.

$$I^0_{[2]}(\Gamma) := \mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)}\right] - V^{\text{ff}} \tag{69}$$

Given $\widehat{f}_{t(\Gamma)} = \widetilde{f}_{t(\Gamma)} + \left(\Gamma^\top Z_t^\top Z_t \Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}$, we have

$$\begin{aligned} I_{[2]}(\Gamma) - I^0_{[2]}(\Gamma) = \frac{1}{T}\sum_t \Big(& \widetilde{f}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)} - \mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)}\right] \\ & + \widetilde{f}_{t(\Gamma)}\widetilde{e}^{\top}_{t(\Gamma)} Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1} \\ & + \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)} \\ & + \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}\widetilde{e}^{\top}_{t(\Gamma)} Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Big). \end{aligned} \tag{70}$$

The analysis from here is very similar to the proof of Proposition 1.

For the first term, mimic Lemma 7 to draw the conclusion that

$$\sup_{\Gamma\in\Psi}\left\|\frac{1}{T}\sum_t\left(\widetilde{f}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)}\right) - \mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)}\right]\right\| \xrightarrow{p} 0, \qquad T\to\infty. \tag{71}$$

Define the sum of the second, third, and forth term inside the summation as $\Xi_t(\Gamma)$:

$$\Xi_t(\Gamma) = \widetilde{f}_{t(\Gamma)}\widetilde{e}^{\top}_{t(\Gamma)} Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1} \tag{72}$$

$$+ \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}\widetilde{f}^{\top}_{t(\Gamma)} \tag{73}$$

$$+ \left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}\widetilde{e}^{\top}_{t(\Gamma)} Z_t\Gamma\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1} \tag{74}$$

Repeat the arguments in Lemma 6 and we arrive at the conclusion that $\Xi_t(\Gamma)$ is U-mean converging.

Combining these two conclusions, following the same arguments in the proof of Proposition 1, IF.1 is verified.

Verify IF.2:

For any parameter Γ and its rotation $\Gamma' = \Gamma R$, we find the relationship that mirrors Eq. 13

in Lemma 1.

$$\mathbb{E}\left[\widetilde{f}_t\left(\Gamma R\right)\widetilde{f}_t^{\top}\left(\Gamma R\right)\right] = R^{-1\top}\mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}_{t(\Gamma)}^{\top}\right]R^{-1} \tag{75}$$

Given this property, we find I_{Υ}^{0} and I_{Υ} behave in the same way when Γ is rotated. Therefore, the procedure of the normalization is the same as that in that Lemma 1, which in turn proves the normalization is unique, that verifies IF.2.

A difference to point out is that here the input of the normalization procedure $\mathbb{E}\left[\widetilde{f}_{t(\Gamma)}\widetilde{f}_{t(\Gamma)}^{\top}\right]$ is an object about the underlying true and unknown to econometrian. □

B.6 Proof of Theorem 1

Proof. We stack the score and identification functions together and treat the pair as a single function. By their definitions, $\widehat{\Gamma}$ solves $\left[S;I\right]\left(\Gamma\right)=\mathbf{0}$ and $\Gamma^{0}(\Theta)$ solves $\left[S^{0};I\right]\left(\Gamma\right)=\mathbf{0}$. Proposition 1 and Condition IF.1 imply that the stacked function $[S;I]$ is uniformly converging:

$$\sup_{\Gamma\in\Psi}\left\|\left[S;I\right]\left(\Gamma\right)-\left[S^{0};I^{0}\right]\left(\Gamma\right)\right\|\xrightarrow{p}0, \qquad\qquad N,T\to\infty$$

Based on the previous analysis, Proposition 2 and IF.1-2 combined imply Γ^{0} is the unique solution to the function limits' equation system: $\left[S^{0};I^{0}\right]\left(\Gamma\right)=\mathbf{0}$. We know the solution of the uniformly converging function converges to the limit's *unique* solution, according to Newey and McFadden (1994). Therefore, $\widehat{\Gamma}\xrightarrow{p}\Gamma^{0}$. Similarly, $[S^{0};I]$ uniformly converges to $[S^{0};I^{0}]$, implying $\Gamma^{0}(\Theta)\xrightarrow{p}\Gamma^{0}$. These two convergence results combined imply $\widehat{\Gamma}-\Gamma^{0}(\Theta)$ converges to zero. □

Corollary 1 is a direct application of Theorem 1.

B.7 Proof of Theorem 2

Notation explanation: Function $\widehat{S}(\Gamma)$ and $\widehat{S}(\gamma)$ are the same regardless of whether the input is vectorized or not. In the proof below, we use the vectorized version. The same goes for functions I, S^{0}, I^{0}. The reason is for convenience in interpreting the function's Jacobian (e.g. $\frac{\partial\widehat{S}(\gamma)}{\partial\gamma^{\top}}$), which has LK columns that are derivatives of each entries of the input γ, and is arranged according to

the ordering in γ. The proof is general for any M-estimator that requires an identification, and therefore does not depend on the specific matrix structure of Γ.

Proof. We first consider $\widehat{\gamma} - \gamma^0$. Later, $\widehat{\gamma} - \gamma^0(\Theta)$ has almost the same derivation, except for one important difference.

Per Lagrange's Mean Value Theorem, at each realization, there exists $\bar{\gamma}$ in-between $\widehat{\gamma}$ and γ^0 such that

$$\widehat{S}\left(\widehat{\gamma}\right) = \widehat{S}(\gamma^0) + \left.\frac{\partial \widehat{S}\left(\gamma\right)}{\partial \gamma^\top}\right|_{\gamma=\bar{\gamma}} \left(\widehat{\gamma} - \gamma^0\right). \tag{76}$$

Notice $\widehat{S}\left(\widehat{\gamma}\right) = \mathbf{0}$. That means

$$\bar{H}\left(\widehat{\gamma} - \gamma^0\right) = -\widehat{S}(\gamma^0), \tag{77}$$

where $\bar{H} := \left.\frac{\partial \widehat{S}(\gamma)}{\partial \gamma^\top}\right|_{\gamma=\bar{\gamma}}$ is shorthand for the sample Hessian matrix at $\bar{\gamma}$.[36]

Usually, without the unidentification problem, one would divide by the Hessian to get $\widehat{\gamma} - \gamma = -\bar{H}^{-1}\widehat{S}(\gamma^0)$, and then take the limit of $\bar{H}$ to express the asymptotic distribution. But that is not possible here when $\widehat{S}$ has non-unique solutions.

The unidentification problem manifests as a singular $\bar{H}$. Specifically, the score should have zero gradients on the K^2 directions of unidentification. Intuitively, think in the IPCA case, where the rotation matrix R is $K \times K$. So there are K^2 directions to marginally perturb Γ^0 without changing the score — it would be constantly $\mathbf{0}$. This implies the score has zero gradient on those K^2 directions, meaning $\bar{H}$, although an LK-square matrix, has a rank of only $LK - K^2$.

A rank-deficient $\bar{H}$ leads to the non-uniqueness of the $\widehat{\gamma}$ solutions in equation (77). Given one $\widehat{\gamma} - \gamma^0$ that solves the linear equation, there are a family of other vectors that all solve the equation, forming a K^2-dimensional sub-linear space of solutions. This is the unidentification issue manifested as the same non-unique solution problem but now for a *linearized* equation.

[36] Lagrange's Mean Value Theorem only applies to function of multiple inputs and scalar output, but not to vector-valued functions. So each row of $\bar{H}$ are evaluated at a different $\bar{\gamma}$. Same for the additional K^2 rows of $\bar{J}$ below. But importantly, the different $\bar{\gamma}$'s are all in between γ^0 and $\widehat{\gamma}$ (in a linear-combination sense). This guarantees later we can take the limit of the Hessian (see Newey and McFadden, 1994, footnote 25).

To solve the problem, we resort to the identification function to pin down the indeterminacy. We linearize I similarly and then append it to the linearized score (remember $I\left(\widehat{\gamma}\right) = \mathbf{0}$ too):

$$I\left(\widehat{\gamma}\right) = I(\gamma^0) + \left.\frac{\partial I\left(\gamma\right)}{\partial \gamma^\top}\right|_{\gamma=\bar{\gamma}} \left(\widehat{\gamma} - \gamma^0\right). \tag{78}$$

$$\bar{J}\left(\widehat{\gamma} - \gamma^0\right) = -I(\gamma^0). \tag{79}$$

where $\bar{J} := \left.\frac{\partial I(\gamma)}{\partial \gamma^\top}\right|_{\gamma=\bar{\gamma}}$ is I's counterpart of the Hessian. Notice I is $K^2 \times 1$ vector and J is $K^2 \times LK$ Jacobian. So, these additional K^2 equation pins down the addition K^2 degrees of freedom. Append the K^2 equations in (79) below (77) and form a single linearization of the estimator,

$$\left[\bar{H}; \bar{J}\right]\left(\widehat{\gamma} - \gamma^0\right) = -\left[\widehat{S}(\gamma^0); I(\gamma^0)\right].$$

Now this linear equation system has a unique solution at $(\widehat{\gamma} - \gamma^0)$, because the stacked $\left[\bar{H}; \bar{J}\right]$ matrix of size $(LK + K^2) \times LK$ is now full rank LK. To solve it, left multiply both sides by the pseudoinverse $\left(\left[\bar{H}; \bar{J}\right]^\top \left[\bar{H}; \bar{J}\right]\right)^{-1} \left[\bar{H}; \bar{J}\right]^\top$:

$$\widehat{\gamma} - \gamma^0 = -\left(\bar{H}^\top \bar{H} + \bar{J}^\top \bar{J}\right)^{-1} \left(\bar{H}^\top \widehat{S}(\gamma^0) + \bar{J}^\top I(\gamma^0)\right). \tag{80}$$

With the estimator thus linearized, the rest of asymptotic derivation follows canonical M-estimator analysis. Given, $\widehat{\gamma} \xrightarrow{p} \gamma^0$, then $\bar{\gamma}$, the mean value between $\widehat{\gamma}$ and γ^0, goes to γ^0 as well. By the continuous mapping theorem, we find the limits of the Hessians:

$$\operatorname*{plim}_{N,T\to\infty} \bar{H} = H^0 := \left.\frac{\partial S^0\left(\gamma\right)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}, \quad \operatorname*{plim}_{N,T\to\infty} \bar{J} = J^0 := \left.\frac{\partial I^0\left(\gamma\right)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}.^{37} \tag{81}$$

Then, taking the probability limit of equation (80) leads to line 2.a in Theorem 2.

The derivation of line 2.b is similar, save for an important difference. Let us redo this proof, but now start by applying the Lagrange's Mean Value Theorem to $(\widehat{\gamma} - \gamma^0(\Theta))$ instead of $(\widehat{\gamma} - \gamma^0)$. Denote the requisite mean value as $\bar{\bar{\gamma}}$ and denote the Hessians evaluated at $\bar{\bar{\gamma}}$ as $\bar{\bar{H}}, \bar{\bar{J}}$. Then, the

same logic applies till we find the counterpart of equation (80) as

$$\widehat{\gamma} - \gamma^0(\Theta) = -\left(\bar{\bar{H}}^\top \bar{\bar{H}} + \bar{\bar{J}}^\top \bar{\bar{J}}\right)^{-1} \left(\bar{\bar{H}}^\top \widehat{S}\left(\gamma^0(\Theta)\right) + \bar{\bar{J}}^\top I\left(\gamma^0(\Theta)\right)\right). \tag{82}$$

Not only $\widehat{\gamma}$ but also $\gamma^0(\Theta)$ converge to γ^0. Hence the new mean value $\bar{\bar{\gamma}}$ between them converges to γ^0 as well. So, the limits of $\bar{\bar{H}}$ and $\bar{\bar{J}}$ are still H^0 and J^0, which are evaluated at the same deterministic γ^0.[38]

The important difference in this case is that $I\left(\gamma^0(\Theta)\right) = \mathbf{0}$ by construction, which eliminates the I term in (82) and leads to line 2.b in the Theorem. This difference carries an important intuition that is discussed below.

The remainder of the proof only needs to shows $\widehat{S}\left(\gamma^0(\Theta)\right) = \widehat{S}(\gamma^0) + o_p\left(\widehat{S}(\gamma^0)\right)$. Apply Mean Value for another time in between $\gamma^0(\Theta)$ to γ^0:

$$\widehat{S}\left(\Gamma^0(\Theta)\right) - \widehat{S}(\Gamma^0) = \left.\frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^{[3]}} \left(\gamma^0(\Theta) - \gamma^0\right) \tag{83}$$

$$= \left(\left.\frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^{[3]}} - \left.\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^{[3]}}\right) \left(\gamma^0(\Theta) - \gamma^0\right) + \left.\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^{[3]}} \left(\gamma^0(\Theta) - \gamma^0\right) \tag{84}$$

Notice the first term is $\mathcal{O}_p\left(S^0(\Gamma)\right) o_p\left(1\right) = o_p\left(S^0(\Gamma)\right)$. The second term is $\mathbf{0}$ because S^0 is constant at $\mathbf{0}$ from $\gamma^0(\Theta)$ to γ^0. □

B.8 Proposition 3

Proposition 3 is stated for the special cases of $\Gamma^0 = \Gamma^0(\Theta_{\mathrm{X}})$ and $\Gamma^0 = \Gamma^0(\Theta_{\mathrm{Y}})$. Here, we prove Lemma 9, which is a general version of Proposition 3. It is stated with respect to a generic Γ^0 and gives the general formula for the asymptotic variance $\mathbb{V}^{[1]}$. To calculate the specific values of the asymptotic variance $\mathbb{V}^{[1]}_{\mathrm{X}}, \mathbb{V}^{[1]}_{\mathrm{Y}}$, one simply plugs in $\Gamma^0(\Theta_{\mathrm{X}})$ or $\Gamma^0(\Theta_{\mathrm{Y}})$. (Notice if one wants to evaluate two rotationally equivalent versions of the same true model, then the correspond true f^0_t also needs to be rotated. As a result, the values of Ω^{zef}, which appears in the formula, are also different in the two cases.)

[38] This shows the importance of keeping the deterministic γ^0 as the limiting reference point in the linearizion, even though we are ultimately interested in the result about $\gamma^0(\Theta)$ and $\widehat{\gamma}$.

Lemma 9 (Asymptotic Distribution of the Score Evaluated at Γ^0 — Generic Identification)**.** *Under Assumptions A–F, if the identification condition Θ has an associated identification function $I(\Gamma)$ that satisfies IF.1–2, and γ^0 is under IA, then as $N, T \to \infty$ such that $T/N \to 0$,*

$$\sqrt{NT}\widehat{S}(\Gamma^0) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \mathbb{V}^{[1]}\right).$$

where $\mathbb{V}^{[1]} = \left(Q^0 \otimes \mathbb{I}_K\right) \Omega^{zef} \left(Q^{0\top} \otimes \mathbb{I}_K\right)$ *and* $Q^0 := \mathbb{I}_L - \Omega^{zz}\Gamma^0 \left(\Gamma^{0\top}\Omega^{zz}\Gamma^0\right)^{-1} \Gamma^{0\top}$.

Remark: $Q^0 := Q_{t(\Gamma^0)}$. Given the definition that $Q_{t(\Gamma)} := \mathbb{I}_L - \Omega_t^{zz}\Gamma \left(\Gamma^\top \Omega_t^{zz}\Gamma\right)^{-1} \Gamma^\top$ and that $\Omega_t^{zz} = \Omega^{zz}$ is constant over t (Assumption F), we know $Q_{t(\Gamma^0)}$ is constant, hence we dropped the t subscript in Q^0.

Proof. We evaluate the score function at $\Gamma = \Gamma^0$ and decompose it into the six parts following Eq. 37. We find two of the six terms drops out, $S_t^{[2]}(\Gamma^0) = S_t^{[5]}(\Gamma^0) = \mathbf{0}$, because the expressions of both $S_t^{[2]}(\Gamma)$ and $S_t^{[5]}(\Gamma)$ contain $\Pi_{t(\Gamma)}$. And, when evaluated at Γ^0, $\Pi_t(\Gamma^0) = \mathbf{0}$. (It is also true that $\Pi_{t(\Gamma)} = Q_{t(\Gamma)}{}^\top \Gamma^0$.)

For the remaining four terms, we split $S_t^{[1]} + S_t^{[3]}$ into two terms:

$$S_t^{[1]}(\Gamma^0) + S_t^{[3]}(\Gamma^0) = \left[\mathbb{I}_L - Z_t^\top Z_t \Gamma^0 \left(\Gamma^{0\top} Z_t^\top Z_t \Gamma^0\right)^{-1} \Gamma^{0\top}\right] Z_t^\top e_t f_t^{0\top} \tag{85}$$

$$= \left(Q_{t(\Gamma^0)} - \epsilon_{N,t}\right) Z_t^\top e_t f_t^{0\top} \tag{86}$$

where we made a shorthand:

$$\epsilon_{N,t} := Z_t^\top Z_t \Gamma^0 \left(\Gamma^{0\top} Z_t^\top Z_t \Gamma^0\right)^{-1} \Gamma^{0\top} - \Omega_t^{zz}\Gamma^0 \left(\Gamma^{0\top}\Omega_t^{zz}\Gamma^0\right)^{-1} \Gamma^{0\top}. \tag{87}$$

So we have the left-hand side of the Lemma in three parts:

$$\sqrt{NT}\widehat{S}(\Gamma^0) = \frac{1}{\sqrt{NT}} \sum_t \text{vect}\left(Q_{t(\Gamma^0)} Z_t^\top e_t f_t^{0\top} - \epsilon_{N,t} Z_t^\top e_t f_t^{0\top} + S_t^{[4]}(\Gamma^0) + S_t^{[6]}(\Gamma^0)\right) \tag{88}$$

And we show the following three results, one for each part, to proof the Lemma. As $N, T \to \infty$,

$$\frac{1}{\sqrt{NT}} \sum_t \text{vect}\left(Q_{t(\Gamma^0)} Z_t^\top e_t f_t^{0\top}\right) \xrightarrow{d} \text{Normal}\left(0, \mathbb{V}^{[1]}\right), \tag{89}$$

$$\frac{1}{\sqrt{NT}} \sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right) \xrightarrow{p} \mathbf{0}, \tag{90}$$

$$\frac{1}{\sqrt{NT}} \sum_t \text{vect}\left(S_t^{[4]}(\Gamma^0) + S_t^{[6]}(\Gamma^0)\right) \xrightarrow{p} \mathbf{0}. \tag{91}$$

Equation 89: Notice $Q_{t(\Gamma^0)} = Q^0$ according to Assumption F. Then, $\text{vect}\left(Q^0 Z_t^\top e_t f_t^{0\top}\right) = (Q^0 \otimes \mathbb{I}_K)\,\text{vect}\left(Z_t^\top e_t f_t^{0\top}\right)$. Then it is obvious applying a continuous mapping theorem to the central limiting statement in Assumption E(1).

Equation 90:

On the left-hand side

$$\frac{1}{\sqrt{NT}} \sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right) = \frac{\sqrt{T}}{\sqrt{N}} \frac{1}{T} \sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right). \tag{92}$$

Since we have imposed the restriction that $T/N \to 0$, it would be sufficient to show the time-average, $\frac{1}{T}\sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right)$, is bounded in probability. That is to say: $\forall \epsilon, \exists M^{[1]}, N^{[1]}, T^{[1]}$ s.t. $\forall N > N^{[1]}, T > T^{[1]}$,

$$Pr\left(\left\|\frac{1}{T} \sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right)\right\| > M^{[1]}\right) < \epsilon. \tag{93}$$

With an application of the Markov inequality, we know bounded in expectation implies bounded in probability. Therefore, it remains to show the expectation is bounded, i.e., $\exists M^{[2]}, N^{[2]}, T^{[2]}$, s.t. $\forall N > N^{[2]}, T > T^{[2]}$,

$$\mathbb{E}\left\|\frac{1}{T} \sum_t \text{vect}\left(\epsilon_{N,t} Z_t^\top e_t f_t^{0\top}\right)\right\| < M^{[2]}. \tag{94}$$

By a triangular inequality, we know

$$\mathbb{E}\left\|\frac{1}{T}\sum_{t}\text{vect}\left(\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right)\right\| \leq \frac{1}{T}\sum_{t}\mathbb{E}\left\|\text{vect}\left(\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right)\right\| \tag{95}$$

Given the stationarity Assumption A.3,

$$\frac{1}{T}\sum_{t}\mathbb{E}\left\|\text{vect}\left(\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right)\right\| = \mathbb{E}\left\|\text{vect}\left(\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right)\right\| \tag{96}$$

Therefore, it only remains to show that $\mathbb{E}\left\|\text{vect}\left(\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right)\right\|$ is bounded as N increases. We have,

$$\mathbb{E}\left\|\epsilon_{N,t}Z_t^\top e_t f_t^{0\top}\right\| \leq \mathbb{E}\left[\|\epsilon_{N,t}\|\left\|Z_t^\top e_t f_t^{0\top}\right\|\right] \leq \left(\mathbb{E}\|\epsilon_{N,t}\|^2\,\mathbb{E}\left\|Z_t^\top e_t f_t^{0\top}\right\|^2\right)^{1/2} \tag{97}$$

(The first inequity uses the Cauchy-Schwarz inequality on the Euclidean norm of matrix multiplications. The second inequality uses the Cauchy-Schwarz inequality on the expectation of the product of two random variables.)

Examining the definition of $\epsilon_{N,t}$ in Eq. 87, $\epsilon_{N,t}$ is of a form of $\epsilon_{N,t} = g\left(\frac{1}{N}Z_t^\top Z_t\right) - g\left(\Omega_t^{zz}\right)$, where g is a continuously differentiable function. We know $\frac{1}{N}Z_t^\top Z_t - \Omega_t^{zz} = \mathcal{O}_p\left(1/\sqrt{N}\right)$. By the delta method, the first term $\mathbb{E}\|\epsilon_{N,t}\|^2 = \mathcal{O}\left(1/N\right)$, as $N \to \infty$. Therefore, it would be suffice to show that the second term $\mathbb{E}\left\|Z_t^\top e_t f_t^{0\top}\right\|^2 = \mathcal{O}\left(N\right)$. To show that, we have

$$\mathbb{E}\left\|Z_t^\top e_t f_t^{0\top}\right\|^2 \leq \mathbb{E}\left[\left\|Z_t^\top e_t\right\|^2\left\|f_t^{0\top}\right\|^2\right] \leq \left(\mathbb{E}\left\|Z_t^\top e_t\right\|^4\,\mathbb{E}\left\|f_t^0\right\|^4\right)^{1/2} \tag{98}$$

According to Assumption E(2), the first term $\mathbb{E}\left\|Z_t^\top e_t\right\|^4 = \mathcal{O}\left(N^2\right)$ as $N \to \infty$. According to Assumption C(1), $\mathbb{E}\left\|f_t^0\right\|^4$ is finite. Therefore, $\mathbb{E}\left\|Z_t^\top e_t f_t^{0\top}\right\|^2 = \mathcal{O}\left(N\right)$. That completes the proof of (90).

Equation 91:

According to the definitions of the components in Eq. 36 and 37, the left-hand side of Eq. 91

becomes

$$\frac{1}{\sqrt{NT}}\sum_t \text{vect}\left(S_t^{[4]}(\Gamma^0)+S_t^{[6]}(\Gamma^0)\right) \tag{99}$$

$$=\frac{\sqrt{T}}{\sqrt{N}}\frac{1}{T}\sum_t \text{vect}\left(Q_t^N(\Gamma^0)\frac{1}{N}\left(Z_t^\top e_t\right)\left(e_t{}^\top Z_t\right)\Gamma^0\left(\frac{1}{N}\Gamma^{0\top}Z_t^\top Z_t\Gamma^0\right)^{-1}\right) \tag{100}$$

$$=\frac{\sqrt{T}}{\sqrt{N}}\frac{1}{T}\sum_t \left(Q_t^N(\Gamma^0)\otimes\Gamma^0\left(\frac{1}{N}\Gamma^{0\top}Z_t^\top Z_t\Gamma^0\right)^{-1}\right)\text{vect}\left(\frac{1}{N}\left(Z_t^\top e_t\right)\left(e_t{}^\top Z_t\right)\right) \tag{101}$$

where $Q_t^N(\Gamma):=\left(\mathbb{I}_L-\frac{1}{N}Z_t^\top Z_t\Gamma\left(\frac{1}{N}\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top\right)$.

Following the same analysis in the proof of Eq. 90 above, it only remains to show that the expectation of the summand's norm

$$\mathbb{E}\left\|\left(Q_t^N(\Gamma^0)\otimes\Gamma^0\left(\frac{1}{N}\Gamma^{0\top}Z_t^\top Z_t\Gamma^0\right)^{-1}\right)\text{vect}\left(\frac{1}{N}\left(Z_t^\top e_t\right)\left(e_t{}^\top Z_t\right)\right)\right\| \tag{102}$$

is bounded as N increases.

We notice the first term is a function of only $\frac{1}{N}Z_t^\top Z_t$ and $\left(\frac{1}{N}\Gamma^{0\top}Z_t^\top Z_t\Gamma^0\right)^{-1}$. Hence, the first term is a.s. bounded, according to Lemma 5.[39] The second term is mean square bounded according to Assumption E(2). Therefore, by Lemma 3.4, the expectation in Eq. 102 is indeed bounded. (In fact, when citing Lemma Lemma 3.4, we only need the downgraded version without considering uniformity across Γ, since we now have fixed the analysis at Γ^0.) And thus we have competed the proof of Equation 91. □

Additional comment: based on this analysis, we conjecture that if T/N converges to a fixed positive number, then Eq. 90 and 91 have non-zero probability limits. As a result, the score's asymptotic distribution is still normal but with a bias. That means Γ estimation would by asymptotically biased, but still consistent, as the "incidental parameter problem" would have suggested. We leave this scenario for future research.

[39] Lemma 5 indeed yields the stronger result of U-a.s. bounded. Here, the term is irrelevant of Γ, so it is meaningless to talk about "uniformly...". The term is just a.s. bounded

B.9 Calculate $H^0_{\mathrm{X}}, H^0_{\mathrm{Y}}$

Similarly to the situation in B.8, we first state and prove the general result about H^0, and then discuss how to evaluate the general expression at Θ_{X} and Θ_{Y} specifically.

Lemma 10 (Calculate H^0 - General)**.** *Under Assumption F,*

$$H^0 = \left(\Omega^{zz} \otimes V^{\mathrm{ff}}\right) \frac{\partial}{\partial \gamma} \mathrm{vect}\left(\Pi_{(\Gamma)}\right)\Big|_{\gamma=\gamma^0}.^{40}$$

Similar to the evaluation of $\mathbb{V}^{[1]}$ in Subsection B.8, there are two differences between H^0_{X} and H^0_{Y}. First, the expression of $\Pi_{(\Gamma)}$ and its derivative is only about Ω^{zz} which does not depend on the rotation. But the derivative needs to be evaluated at either $\Gamma^0(\Theta_{\mathrm{X}})$ or $\Gamma^0(\Theta^0_{\mathrm{Y}})$. Second, the correspond true f^0_t also need to be rotated. This will result in differences in the assumed asymptotic variance V^{ff}.

Next is the proof of the general Lemma 10.

Proof. Start from the definition

$$H^0 := \frac{\partial S^0(\Gamma)}{\partial \gamma^\top}\Big|_{\gamma=\gamma^0} = \mathbb{E}\left[\frac{\partial \mathrm{vect}\left(\Omega^{zz}_t \Pi_{t(\Gamma)} f^0_t \widetilde{f}^\top_{t\,(\Gamma)}\right)}{\partial \gamma^\top}\Big|_{\gamma=\gamma^0}\right] \tag{103}$$

Notice $\Pi_t(\Gamma^0) = \mathbf{0}$. Therefore, the terms involving $\nabla \widetilde{f}_t$ after taking derivative will drop out, only $\nabla \Pi_t(\Gamma^0)$ terms survive. We write the H^0 column by column. The p'th column, or the derivative w.r.t. the p'th entry γ_p simplifies as

$$H^0{}_p = \mathbb{E}\left[\mathrm{vect}\left(\Omega^{zz}_t \frac{\partial \Pi_{t(\Gamma)}}{\partial \gamma_p}\Big|_{\gamma=\gamma^0} f^0_t f^{0\top}_t\right)\right] \tag{104}$$

This result above does not require Assumption F, and can be calculated by LLN simulation if

[40] Under Assumption F, $\Pi_{t(\Gamma)}$ is also time-constant and deterministic, so we drop its t subscript. We choose do not write out the deterministic derivatives $\frac{\partial}{\partial \gamma}\mathrm{vect}\left(\Pi_{(\Gamma)}\right)$ analytically for conciseness, which is easily calculated numerically in the simulated and empirical exercises.

one is interested in the general case. Now, we impose the constant Ω_t^{zz} assumption F:

$$H^0{}_p = \text{vect}\left(\Omega^{zz}\left.\frac{\partial \Pi(\Gamma)}{\partial \gamma_p}\right|_{\gamma=\gamma^0} V^{\text{ff}}\right) = \left(\Omega^{zz}\otimes V^{\text{ff}}\right)\left.\frac{\partial \text{vect}\left(\Pi(\Gamma)\right)}{\partial \gamma_p}\right|_{\gamma=\gamma^0} \tag{105}$$

Finally, append the columns together, we have $H^0 = \left(\Omega^{zz}\otimes V^{\text{ff}}\right)\left.\frac{\partial \text{vect}(\Pi(\Gamma))}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}$ □

B.10 Calculate $J_{\text{X}}^0, J_{\text{Y}}^0$

J_{X}^0 is much easier. The identification function I_{X} is deterministic, i.e. $I_{\text{X}} = I_{\text{X}}^0$. According to J^0's definition (Eq. 81), we calculating I_{X}^0's derivative and get

$$J_{\text{X}}^0 = [\mathbb{I}_{K^2\times K^2}, \mathbf{0}_{K^2\times(L-K)K}]. \tag{106}$$

J_{Y}^0 is much more involved and we summarize the calculation in the following lemma.

Lemma 11 (Calculate J_{Y}^0)**.** *Under Assumption F: J_{Y}^0 is stacked up by two parts, the top $\frac{1}{2}K(K+1)$ rows are $J_{\text{Y}}^0{}_{\left[1:\left(\frac{1}{2}K(K+1)\right),\ :\ \right]} = \left.\frac{\partial \text{veca}\left(\Gamma^\top\Gamma\right)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0}$. For the bottom $\frac{1}{2}K(K-1)$ rows, the p'th column is*

$$J_{\text{Y}}^0{}_{\left[\left(\frac{1}{2}K(K+1)+1:K^2\right),\ p\ \right]} = \text{vecb}\left(D_p\left(\Gamma^0,\Omega^{zz}\right)V^{\text{ff}} + V^{\text{ff}}D_p^\top\left(\Gamma^0,\Omega^{zz}\right)\right), \tag{107}$$

where $D_p\left(\Gamma^0,\Omega_t^{zz}\right) := \left.\frac{\partial}{\partial \gamma_p}\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0\right|_{\gamma^0}$.

(This expression of D_p involves matrix derivative, which can be simply calculated with numerical derivatives in simulation exercises)

Proof. The first part of I is deterministic, so the first part of J^0 is easy. Lower part of J_{Y}^0:

$$\begin{aligned}
J_{\text{Y}}^0{}_{\left[\left(\frac{1}{2}K(K+1)+1:K^2\right),\ :\ \right]} &= \operatorname*{plim}_{N,T\to\infty}\frac{\partial}{\partial \gamma^\top}\text{vecb}\left(I_{[2]}\right)\Big|_{\gamma^0} \\
&= \operatorname*{plim}_{N,T\to\infty}\frac{\partial}{\partial \gamma^\top}\text{vecb}\left(\frac{1}{T}\sum_t\left(\widetilde{f}_{t(\Gamma)}\widetilde{f}_t^\top(\Gamma)\right)\right)\Big|_{\gamma^0} \\
\frac{\partial}{\partial \gamma_p}\text{vecb}\left(\widetilde{f}_{t(\Gamma)}\widetilde{f}_t^\top(\Gamma)\right)\Big|_{\gamma^0} &= \text{vecb}\left(\left.\frac{\partial}{\partial \gamma_p}\widetilde{f}_{t(\Gamma)}\right|_{\gamma^0} f_t^{0\top} + f_t^0\left.\frac{\partial}{\partial \gamma_p}\widetilde{f}_t^\top(\Gamma)\right|_{\gamma^0}\right)
\end{aligned}$$

$$\frac{\partial}{\partial\gamma_p}\widetilde{f}_{t(\Gamma)}\Big|_{\gamma^0} = \frac{\partial}{\partial\gamma_p}\left(\Gamma^\top\Omega_t^{zz}\Gamma\right)^{-1}\Gamma^\top\Omega_t^{zz}\Gamma^0\Big|_{\gamma^0} f_t^0 := D_p\left(\Gamma^0,\Omega_t^{zz}\right)f_t^0$$

$$\frac{\partial}{\partial\gamma_p}\mathrm{vecb}\left(\widetilde{f}_{t(\Gamma)}\widetilde{f}_t^\top{}_{(\Gamma)}\right)\Big|_{\gamma^0} = \mathrm{vecb}\left(D_p\left(\Gamma^0,\Omega_t^{zz}\right)f_t^0 f_t^{0\top} + f_t^0 f_t^{0\top} D_p^\top\left(\Gamma^0,\Omega_t^{zz}\right)\right)$$

$$J^0_{\mathrm{Y}\left[\left(\frac{1}{2}K(K+1)+1:K^2\right),\ p\ \right]} = \mathbb{E}\left[\mathrm{vecb}\left(D_p\left(\Gamma^0,\Omega_t^{zz}\right)f_t^0 f_t^{0\top} + f_t^0 f_t^{0\top} D_p^\top\left(\Gamma^0,\Omega_t^{zz}\right)\right)\right]$$

By Assumption F, $J^0_{\mathrm{Y}\left[\left(\frac{1}{2}K(K+1)+1:K^2\right),\ p\ \right]} = \mathrm{vecb}\left(D_p\left(\Gamma^0,\Omega^{zz}\right)V^{\mathrm{ff}} + V^{\mathrm{ff}}D_p^\top\left(\Gamma^0,\Omega^{zz}\right)\right)$. □

B.11 Proof of Theorem 4

Proof. (1) Given $\widehat{\Gamma}$ consistency to the two targets (Theorem 1), and that $\widetilde{f}_{t(\Gamma)}$ has bounded derivative around Γ^0, we have $\widetilde{f}_t(\widehat{\Gamma}) - \widetilde{f}_t\left(\Gamma^0(\Theta)\right) \xrightarrow{p} \mathbf{0}$. It remains to show the second term is $o_p(1)$. We have shown that $\frac{1}{N}\Gamma^\top Z_t^\top \widetilde{e}_{t(\Gamma)}$ is U-mean square converging and $N\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}$ is U-a.s. bounded (see intermediate steps in the proof of Lemma 6). Combined, $\left(\Gamma^\top Z_t^\top Z_t\Gamma\right)^{-1}\Gamma^\top Z_t^\top\widetilde{e}_{t(\Gamma)}$ is U-mean square converging. Because that is a uniform result, when evaluated at $\widehat{\Gamma}$ for the second term, it is converging (in m.s. which implies in probability).

(2) Given Theorem 2 and Proposition 3, $\widehat{\Gamma} - \Gamma^0(\Theta) = \mathcal{O}_p\left(1/\sqrt{NT}\right)$. So, the first term is $o_p\left(1/\sqrt{NT}\right)$. For the second term:

$$\sqrt{N}\left(\widehat{\Gamma}^\top Z_t^\top Z_t\widehat{\Gamma}\right)^{-1}\widehat{\Gamma}^\top Z_t^\top\widetilde{e}_t(\widehat{\Gamma}) = \sqrt{N}\left(\Gamma^{0\top}Z_t^\top Z_t\Gamma^0\right)^{-1}\Gamma^{0\top}Z_t^\top e_t + o_p(1) \tag{108}$$

$$\xrightarrow{d} \mathrm{Normal}\left(\mathbf{0},\mathbb{V}_t^{[2]}\right), \tag{109}$$

where $\mathbb{V}_t^{[2]} = \left(\Gamma^{0\top}\Omega_t^{zz}\Gamma^0\right)^{-1}\Gamma^{0\top}\Omega_t^{ze}\Gamma^0\left(\Gamma^{0\top}\Omega_t^{zz}\Gamma^0\right)^{-1}$, with Ω_t^{ze} from Assumption E(2). Notice neither Ω_t^{zz} or Ω_t^{ze} are affected by the rotation. So one only needs to plug in the value of Γ^0 as either $\Gamma^0(\Theta_{\mathrm{X}})$ or $\Gamma^0(\Theta_{\mathrm{Y}}^0)$ to evaluate $\mathbb{V}_t^{[2]}$ for the specific cases. □

C Results About Arrow C

C.1 Asymptotic Distribution of $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$

This subsection works out the asymptotic distribution of $\widehat{\Gamma}(\Theta_{\mathrm{Y}}) - \Gamma^0(\Theta_{\mathrm{Y}}^0)$, or Arrow C in Figure 2. Since the reference point $\Gamma^0(\Theta_{\mathrm{Y}}^0)$ is deterministic, this analysis provides the complete asymptotic

distribution of the estimator $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$. In contrast to Arrow B (studied in Section 5.2), Arrow C also depends on the randomness of the identification Θ_{Y}. We rely on the general result 2.a for the asymptotic distribution. Subsection 5.2 has already calculated three of the four right-hand side inputs: $H_{\mathrm{Y}}^0, J_{\mathrm{Y}}^0$ and the asymptotic distribution of $S\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right)$. The fourth piece that needs to be analyzed is the asymptotic distribution $I_{\mathrm{Y}}\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right)$.

Identification function $I_{\mathrm{Y}}(\Gamma)$, as defined in 4.2 is stacked up by two parts. The top $\frac{1}{2}K(K+1)$ entries are from ortho-normalizing Γ, which is irrelevant of sample information. Therefore, similar to the Θ_{X} case, the top part of $I_{\mathrm{Y}}\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right)$ is still deterministic. The bottom $\frac{1}{2}K(K-1)$ rows of $I_{\mathrm{Y}}(\Gamma)$ are from diagonalizing the sample second moment matrix of $\widehat{f}_t$, a process that introduces a rotational contamination with convergence rate $\sqrt{T}$. To see why it is this rate, suppose the true f_t^0 is observed, which follows a diagonal second moment matrix *in population*. However, its sample second moment would not be exactly diagonal, off by a sample error that is in the order of $1/\sqrt{T}$. Imposing the sample normalization requires rotating f_t^0 slightly just to offset that error. We make the following assumption to be specific about how much the sample second moment of f_t^0 is off.

Assumption G (Central Limit Theorems for Factor Second Moment)**.**
As $T \to \infty, \frac{1}{\sqrt{T}} \sum_t \mathrm{vecb}\left(f_t^0 f_t^{0\top} - V^{\mathrm{ff}}\right) \xrightarrow{d} \mathrm{Normal}\left(\mathbf{0}, \mathbb{V}^{[3]}\right)$*, where* $V^{\mathrm{ff}} := \mathbb{E}\left[f_t^0 f_t^{0\top}\right]$*, and* $\mathbb{V}^{[3]}$ *is a* $\frac{1}{2}K(K-1) \times \frac{1}{2}K(K-1)$ *matrix.*

With the assumption, we have the following result.

Lemma 12 (Asymptotic distribution of $I_{\mathrm{Y}}(\Gamma^0)$)**.** *Under Assumptions A–G,*

$$\sqrt{T} I_{\mathrm{Y}}\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{d} \mathrm{Normal}\left(\mathbf{0}_{K^2\times 1}, \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{V}^{[3]} \end{bmatrix}\right),$$

where $\mathbb{V}^{[3]}$ *is specified in Assumption G.*[41]

[41] In the variance matrix, the $\mathbf{0}$ block in the top left is $\frac{1}{2}K(K+1)\times\frac{1}{2}K(K+1)$, top right is $\frac{1}{2}K(K+1)\times\frac{1}{2}K(K-1)$, lower left is $\frac{1}{2}K(K-1)\times\frac{1}{2}K(K+1)$. Together with $\mathbb{V}^{[3]}$ of $\frac{1}{2}K(K-1)\times\frac{1}{2}K(K-1)$, they make up the variance matrix of $K^2 \times K^2$.

Lemma 12 implies the "contamination effect" that the normalization step introduces to $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ converges at the rate of $\sqrt{T}$. Surprisingly, this is even slower than that of the core optimization step, which converges at $\sqrt{NT}$ according to Proposition 3. Hence it solely shows up in the asymptotic distribution of $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ as the dominant term. In summary, the convergence rate of Arrow C is $\sqrt{T}$, while those of Arrows A and B are both $\sqrt{NT}$.

Theorem 5 (Asymptotic Distribution of Arrow C)**.** *Under Assumptions A–F and G, as* $N, T \to \infty$ *such that* $T/N \to 0$*,*

$$\sqrt{T}\left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{d} -\left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} J_{\mathrm{Y}}^{0\top} \mathrm{Normal}\left(\mathbf{0}, \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{V}^{[3]} \end{bmatrix}\right).$$

C.1.1 Proof of Lemma 12

Proof. The top $\frac{1}{2}K(K+1)$ rows of $I_{\mathrm{Y}}(\Gamma^0)$ are deterministic at $\mathbf{0}$: $I_{\mathrm{Y}_{\left[1:\frac{1}{2}K(K+1)\right]}}(\Gamma^0) = \mathbf{0}_{\frac{1}{2}K(K+1)\times 1}$. Next we analyze the four terms in Eq. 70 in the proof Lemma 8 First term has a plim, which in general is non-zero. When evaluated at Γ^0, since $\widetilde{f}_t(\Gamma^0) = f_t^0$, it is $\mathcal{O}_p\left(1/\sqrt{T}\right)$. According to the analysis in Lemma 8, second and third are $\mathcal{O}_p\left(1/\sqrt{NT}\right)$. Fourth is $\mathcal{O}_p\left(1/N\right)$. That is to say

$$\operatorname*{plim}_{N,T\to\infty} \sqrt{T} I_{[2]}(\Gamma^0) - \frac{1}{\sqrt{T}} \sum_t \left(f_t^0 f_t^{0\top} - V^{\mathrm{ff}}\right) = \mathbf{0}_{K\times K} \tag{110}$$

By the times-series CLT in Assumption G: $\sqrt{T}\mathrm{vecb}\left(I_{[2]}(\Gamma^0)\right) \xrightarrow{d} \mathrm{Normal}\left(\mathbf{0}_{\frac{1}{2}K(K-1)\times 1}, \mathbb{V}^{[3]}\right)$. In addition, the cross-covariances between the top and the bottom part are $\mathbf{0}$. □

C.1.2 Proof of Theorem 5

Proof. Apply Theorem 2, specifically (2.a), to the deterministic target $\gamma^0 = \gamma^0(\Theta_{\mathrm{Y}}^0)$ with the identification function I_{Y}. Specializing the generic Jacobians to the Θ_{Y} case gives

$$\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}}^0) = -\left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} \left(H_{\mathrm{Y}}^{0\top} \widehat{S}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) + J_{\mathrm{Y}}^{0\top} I_{\mathrm{Y}}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right)\right)$$

$$+ o_p\left(\widehat{S}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) + I_{\mathrm{Y}}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right)\right).$$

Multiply both sides by $\sqrt{T}$. By Proposition 3,

$$\sqrt{T}\,\widehat{S}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) = \frac{1}{\sqrt{N}}\sqrt{NT}\,\widehat{S}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{p} \mathbf{0},$$

because $T/N \to 0$. By Lemma 12,

$$\sqrt{T}\,I_{\mathrm{Y}}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{d} \text{Normal}\left(\mathbf{0}, \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{V}^{[3]} \end{bmatrix}\right).$$

The remainder term is also $o_p(1)$ after multiplying by $\sqrt{T}$, because $\widehat{S}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) = \mathcal{O}_p\left(1/\sqrt{NT}\right)$ by Proposition 3 and $I_{\mathrm{Y}}\left(\gamma^0(\Theta_{\mathrm{Y}}^0)\right) = \mathcal{O}_p\left(1/\sqrt{T}\right)$ by Lemma 12. Therefore Slutsky's theorem yields

$$\sqrt{T}\left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}}^0)\right) \xrightarrow{d} -\left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} J_{\mathrm{Y}}^{0\top} \text{Normal}\left(\mathbf{0}, \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{V}^{[3]} \end{bmatrix}\right).$$

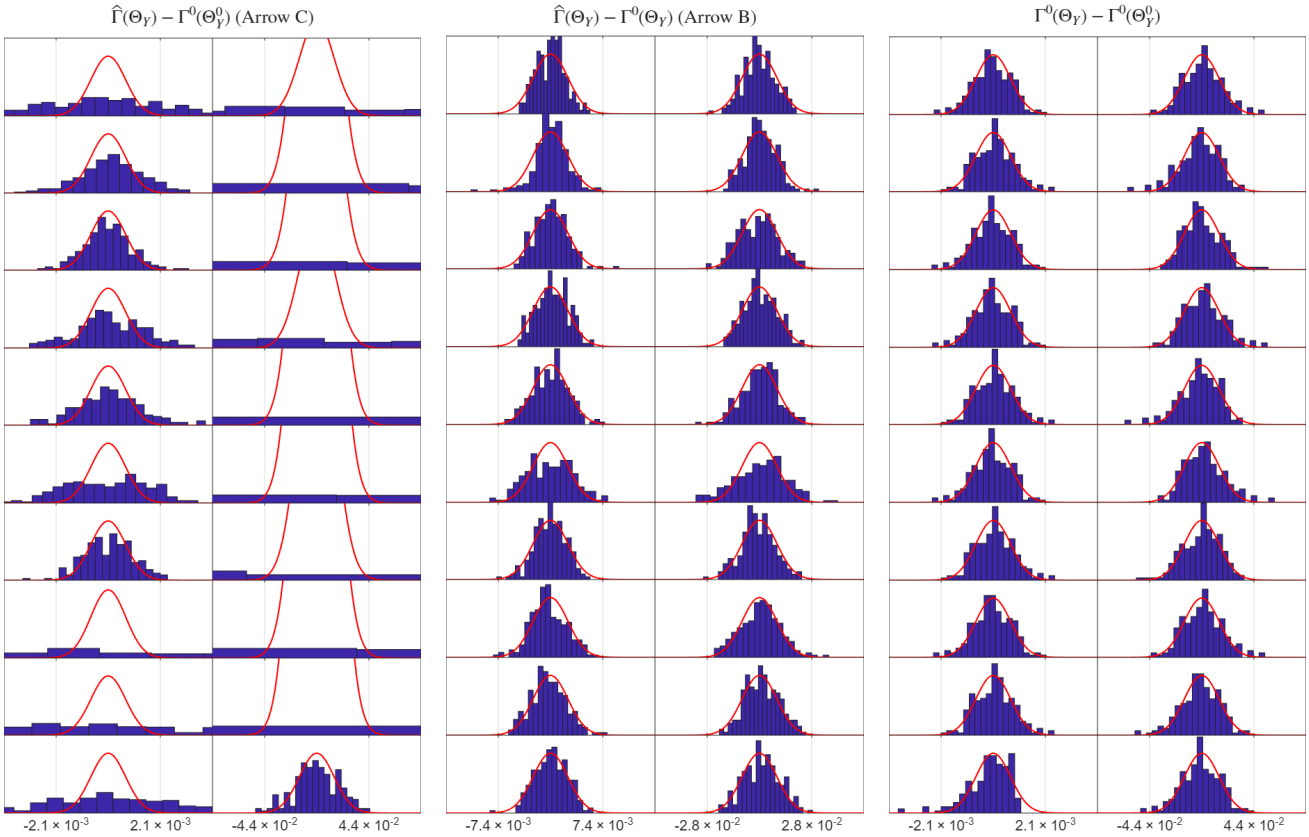
□

C.2 Simulation Results about Arrow C

Figure 7 presents the simulation results about Arrow C. The results are from the same simulations from Section 7. The first panel reports the simulated error of Arrow C and compare that with the theoretical ones from Theorem 5. Except for some entries that roughly match, the simulated small-sample distributions are wider than the theoretical distributions by orders of magnitude. This is purely a finite sample phenomenon; the asymptotics have no problem.

To diagnose the issue, we decompose Arrow C as $\widehat{\Gamma}(\Theta_{\mathrm{Y}}) - \Gamma^0(\Theta_{\mathrm{Y}}^0) = (\widehat{\Gamma}(\Theta_{\mathrm{Y}}) - \Gamma^0(\Theta_{\mathrm{Y}})) + (\Gamma^0(\Theta_{\mathrm{Y}}) - \Gamma^0(\Theta_{\mathrm{Y}}^0)) = \mathcal{O}_p\left(1/\sqrt{NT}\right) + \mathcal{O}_p\left(1/\sqrt{T}\right)$, and respectively plot the two terms in the second and third panels. Although the first term (arrow B) is asymptotically small, its level, as

Figure 7: Γ Estimation Errors — Simulated v.s. Asymptotic Approximation



Note: This figure is constructed with the same simulation exercises in Figure 3, and also follows the same format. The simulated distribution of a new set of random variables (written at the top of each panel) are reported. The corresponding theoretical distributions are from Theorems 5, 3.b, and 5 respectively. Additionally, the x-axes ticks are marked with numbers for comparison. Again, the x-axes range is ± 6 theoretical standard deviations, and the tick numbers mark ± 3 theoretical standard deviations.

reported in the second panel, is still large at $N = 200, T = 200$.[42] As a result, the first term messes up the small sample distribution of Arrow C, which theoretically is only driven by the second term. In the third panel, we ignore the first term and only look at the dominant second term. It conforms to the asymptotic theory from Theorem 5 well. In this sense, the third panel verifies the derivation of large sample asymptotics of the theorem.

So N, T-orders aside, why is the asymptotic variance of the first term (Arrow B) much larger than the second? Retracing the analyses, Arrow B hinges on the accuracy of the first order condition, which eventually depends on the N, T-CLT of $z^\top e f^{0\top}$ (Assumption E(1)). However, the second depends on the accuracy of the identification condition, which is about the time-series

[42]The second panel of Figure 7 is a copy of the second panel of Figure 3, except we now include the x-axis labels for comparing theoretical variances.

CLT of $f^0 f^{0\top}$ only (Assumption G). So the relative levels of the variances of e to f^0 is critical here, which can be roughly measured and calibrated to empirical data by R^2. As a check, scaling down the variance of e to an unrealistic level makes the first panel also converge well (detailed plot omitted).

C.3 Factor Estimation Error Measured Against $\{f_t^0\}$

Section 6 analyzed the factor estimation error against the normalized true $\widetilde{f}_t\left(\Gamma^0(\Theta_{\mathrm{Y}})\right)$. Now we measured factor estimation against f_t^0. We know $f_t^0 = \widetilde{f}_t\left(\Gamma^0\right)$. The analogue of Eq. (11) is

$$\widehat{f}_t\left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})\right) - f_t^0 \tag{111}$$

$$= \left(\widetilde{f}_t\left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})\right) - \widetilde{f}_t\left(\Gamma^0(\Theta_{\mathrm{Y}}^0)\right)\right) + \left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})^\top Z_t^\top Z_t \widehat{\Gamma}(\Theta_{\mathrm{Y}})\right)^{-1} \widehat{\Gamma}^\top Z_t^\top \widetilde{e}_t\left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})\right). \tag{112}$$

The only difference happens at the first term, which is now driven by the estimation error between $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ and $\Gamma^0(\Theta_{\mathrm{Y}}^0)$ (Arrow C), which has a convergence rate of $\sqrt{T}$ according to Subsection C.1. This implies the additional stochastic wedge introduced by a sample-based normalization might no longer be dominated by the second term. The following Theorem says $\widehat{f}_t\left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})\right)$ and f_t^0 converge the slower of $\sqrt{N}$ or $\sqrt{T}$.

Theorem 6 (Factor Estimation Error Measured Against $\{f_t^0\}$)**.** *Under Assumptions A–F, as* $N, T \to \infty$,

$$\widehat{f}_t\left(\widehat{\Gamma}(\Theta_{\mathrm{Y}})\right) - f_t^0 = \mathcal{O}_p\left(\max\left\{1/\sqrt{N}, 1/\sqrt{T}\right\}\right).$$

C.3.1 Proof of Theorem 6

Proof. Straightforward based on Theorems 4 and 5. □

D Convergence of ALS to a Stationary Point

Let $\{(x_t, Z_t)\}_{t=1}^T$ with $x_t \in \mathbb{R}^N$ and $Z_t \in \mathbb{R}^{N\times L}$ be given. For a fixed target rank $K \geq 1$, consider the least-squares objective

$$\Phi(\Gamma, \{f_t\}) := \sum_{t=1}^{T} \|x_t - Z_t \Gamma f_t\|_2^2, \qquad \Gamma \in \mathbb{R}^{L\times K},\ f_t \in \mathbb{R}^K. \tag{113}$$

Φ is quadratic and convex in each block separately, but jointly nonconvex due to bilinearity.

Alternating Least Squares (ALS). Fix any identification condition $\Theta \subset \Psi$, and let $\mathcal{N}_\Theta(\Gamma)$ denote the associated normalization mapping that sends any full-rank Γ to its unique rotationally equivalent representative in Θ. Starting from $\Gamma^{(0)} \in \Theta$, for $r = 0, 1, 2, \ldots$ perform:

$$\text{(F-step)} \quad \tilde{f}_t^{(r+1)} \in \arg\min_{f\in\mathbb{R}^K} \|x_t - Z_t \Gamma^{(r)} f\|_2^2, \quad t = 1, \ldots, T, \tag{114}$$

$$\text{(}\Gamma\text{-step)} \quad \tilde{\Gamma}^{(r+1)} \in \arg\min_{\Gamma\in\mathbb{R}^{L\times K}} \sum_{t=1}^{T} \|x_t - Z_t \Gamma \tilde{f}_t^{(r+1)}\|_2^2, \tag{115}$$

$$\begin{aligned} \text{(Normalization)} \quad \Gamma^{(r+1)} &:= \mathcal{N}_\Theta\big(\tilde{\Gamma}^{(r+1)}\big) = \tilde{\Gamma}^{(r+1)} R^{(r+1)}, \\ f_t^{(r+1)} &:= \big(R^{(r+1)}\big)^{-1} \tilde{f}_t^{(r+1)}, \quad t = 1, \ldots, T, \end{aligned}$$

where $R^{(r+1)}$ is the normalization matrix. This step is only a reparameterization, so

$$Z_t \Gamma^{(r+1)} f_t^{(r+1)} = Z_t \tilde{\Gamma}^{(r+1)} \tilde{f}_t^{(r+1)}, \qquad t = 1, \ldots, T.$$

Proposition 4 (ALS convergence and stationarity)**.** *Under Assumptions C–D, let* $\{(\Gamma^{(r)}, \{f_t^{(r)}\})\}_{r\geq 0}$ *be the normalized ALS sequence generated by* (114)–(115) *together with the normalization step above. Then:*

1. (Convergence of the ALS objective and existence of limits) *The sequence of objective values* $\{\Phi(\Gamma^{(r)}, \{f_t^{(r)}\})\}_{r\geq 0}$ *is nonincreasing and converges to a finite limit. Moreover, the sequence* $\{(\Gamma^{(r)}, \{f_t^{(r)}\})\}_{r\geq 0}$ *is bounded and thus admits cluster points.*

2. (Stationarity of limit points) *Every cluster point* $(\bar{\Gamma}, \{\bar{f}_t\})$ *of the ALS sequence satisfies*

$$\nabla_{f_t}\Phi(\bar{\Gamma}, \{\bar{f}_t\}) = 0, \quad t = 1, \ldots, T, \qquad \nabla_{\Gamma}\Phi(\bar{\Gamma}, \{\bar{f}_t\}) = 0,$$

i.e., every limit point is a stationary point of Φ.

Proof. (1) Convergence of the ALS objective and existence of limits: By Assumption D(1)–(3), for each r and t we have $G_t^{(r)} := (\Gamma^{(r)})^\top Z_t^\top Z_t \Gamma^{(r)} \succ 0$, so the F-step is a strictly convex quadratic with unique minimizer $\tilde{f}_t^{(r+1)} = \big(G_t^{(r)}\big)^{-1}(\Gamma^{(r)})^\top Z_t^\top x_t$, which satisfies the normal equations $(\Gamma^{(r)})^\top Z_t^\top (x_t - Z_t\Gamma^{(r)}\tilde{f}_t^{(r+1)}) = 0$. Solving the F-step exactly yields

$$\Phi(\Gamma^{(r)}, \{\tilde{f}_t^{(r+1)}\}) \;\le\; \Phi(\Gamma^{(r)}, \{f_t^{(r)}\}).$$

With $\{\tilde{f}_t^{(r+1)}\}$ fixed, the raw Γ-step is pooled OLS with positive-definite normal matrix $H^{(r+1)} = \sum_{t=1}^T (Z_t^\top Z_t) \otimes \big(\tilde{f}_t^{(r+1)}(\tilde{f}_t^{(r+1)})^\top\big)$, hence has a unique minimizer $\tilde{\Gamma}^{(r+1)}$ satisfying $\sum_{t=1}^T Z_t^\top (x_t - Z_t\tilde{\Gamma}^{(r+1)}\tilde{f}_t^{(r+1)})(\tilde{f}_t^{(r+1)})^\top = 0$, and

$$\Phi(\tilde{\Gamma}^{(r+1)}, \{\tilde{f}_t^{(r+1)}\}) \;\le\; \Phi(\Gamma^{(r)}, \{\tilde{f}_t^{(r+1)}\}).$$

Because the normalization step leaves fitted values unchanged,

$$\Phi(\Gamma^{(r+1)}, \{f_t^{(r+1)}\}) = \Phi(\tilde{\Gamma}^{(r+1)}, \{\tilde{f}_t^{(r+1)}\}).$$

Therefore $\{\Phi(\Gamma^{(r)}, \{f_t^{(r)}\})\}_{r\ge 0}$ is nonincreasing and bounded below by 0, hence convergent; this is the standard exact block-coordinate descent argument (see Tseng, 2001, Thm. 4.1; Grippo and Sciandrone, 2000).

By construction after each normalization step, $\Gamma^{(r)} \in \Theta \subset \Psi$. Since Ψ is compact by Assumption D(1), the sequence $\{\Gamma^{(r)}\}$ is bounded. Moreover, the convergent objective sequence is uniformly bounded above. Because $\Gamma^{(r)}$ stays in the compact set Θ and Assumptions D(1)–D(3) keep $\Gamma^\top Z_t^\top Z_t \Gamma$ uniformly positive definite on that set, the sublevel sets of the quadratic

map $f \mapsto \|x_t - Z_t \Gamma f\|_2^2$ are uniformly bounded over $\Gamma \in \Theta$. Hence $\{f_t^{(r)}\}_{r\geq 0}$ is bounded for each t, and thus $\{(\Gamma^{(r)}, \{f_t^{(r)}\})\}_{r\geq 0}$ is bounded and admits cluster points.

(2) Stationarity of limit points: Let $(\Gamma^{(r_j)}, \{f_t^{(r_j)}\}) \to (\bar{\Gamma}, \{\bar{f}_t\})$ be a convergent subsequence. By continuity of the F-step solution map on the positive-definite domain by Assumption D, $\tilde{f}_t^{(r_j+1)} = q_t(\Gamma^{(r_j)}) \to q_t(\bar{\Gamma}) =: \bar{f}_t$. The normalization step is an invertible reparameterization and preserves both fitted values and the block first-order conditions. Passing to the limit in the normal equations from the F-step and the raw Γ-step along the subsequence therefore yields

$$\bar{\Gamma}^\top Z_t^\top (x_t - Z_t \bar{\Gamma} \bar{f}_t) = 0, \quad t = 1, \ldots, T, \qquad \sum_{t=1}^{T} Z_t^\top (x_t - Z_t \bar{\Gamma} \bar{f}_t)\, \bar{f}_t^\top = 0,$$

which are precisely $\nabla_{f_t} \Phi(\bar{\Gamma}, \{\bar{f}_t\}) = 0$ and $\nabla_\Gamma \Phi(\bar{\Gamma}, \{\bar{f}_t\}) = 0$. Hence every cluster point is stationary. □

E Selecting the Number of Factors

The latent dimension K is a substantive modeling choice in IPCA, but a full model-selection theory for K is beyond the present paper's main aim. Our objective is to develop the estimation, identification, and inference theory of IPCA conditional on a given, finite K. Still, it is useful to indicate what a penalty-based analysis would look like and why, for IPCA, cross-validation is a particularly natural practical guide.

Under the asymptotic framework of this paper, L is fixed while $N, T \to \infty$. In that setting, adding one more factor increases model flexibility through one additional column of Γ and one additional factor coordinate in each period. Thus, relative to a K-factor specification, a $(K+1)$-factor specification adds one more L-dimensional loading direction and one more time series of factor realizations. A penalty-based criterion in the spirit of Bai and Ng (2002) would therefore balance the reduction in residual variance against a penalty of order $(L+T)\log(NT)/(NT)$, which, with L fixed, is effectively of order $\log(NT)/N$.

The difficulty is that, unlike PCA, IPCA is not an eigenvalue truncation problem. In PCA the effect of adding one more factor can be analyzed using the next eigenvalue of a residual covariance

matrix, and the spectral structure delivers the overfitting bound. In IPCA, after concentrating out the factors, the criterion involves time-varying matrices of the form $(\Gamma' Z_t' Z_t \Gamma)^{-1}$, so the $K \mapsto K+1$ comparison becomes a nonconvex re-optimization problem rather than a simple eigenvalue comparison. For that reason, the eigenvalue-based penalty theory used in PCA does not transfer cleanly here. We therefore use a more practical and, given the structure of IPCA, very reasonable approach based on cross-validation.

E.1 Cross-Validation

For practical work, cross-sectional J-fold cross-validation is especially well suited to IPCA. The key point is that IPCA estimates a common map Γ from observed characteristics to factor loadings. Once $(\widehat{\Gamma}, \{\widehat{f}_t\})$ are estimated on a training set of units, one can form predictions for a held-out unit i using only its observed characteristics:

$$\widehat{x}_{it}^{(-j,K)} \;=\; z_{i,t}^{\top} \widehat{\Gamma}_K^{(-j)} \widehat{f}_{t,K}^{(-j)}, \qquad i \in \mathcal{I}_j.$$

No unit-specific parameter needs to be re-estimated on the validation units. This is a major distinction from PCA. In a static PCA model, held-out units require their own loadings, so prediction for a held-out unit generally requires estimating that unit's loading from its own time series, blurring the separation between estimation and validation. For that reason, cross-sectional validation is especially natural for IPCA.

The procedure partitions the units into J folds, estimates the IPCA model on the training units for each candidate K, and evaluates the resulting predictions on the held-out units. The validation criterion is a pooled out-of-sample R^2 constructed from the held-out squared errors across folds. Figure 8 illustrates the design for five folds. .

We first validate this rule in a Monte Carlo environment. The data-generating process is calibrated to the empirically estimated IPCA model from the stock-return application, but the true number of factors is fixed at $K_0 = 4$. For each Monte Carlo draw we simulate a balanced panel with $N = 500$ assets and $T = 200$ periods, run 5-fold cross-validation over candidate values $K \in \{1, \ldots, 10\}$, and compute the pooled out-of-sample R^2. Figure 9 reports the mean pooled

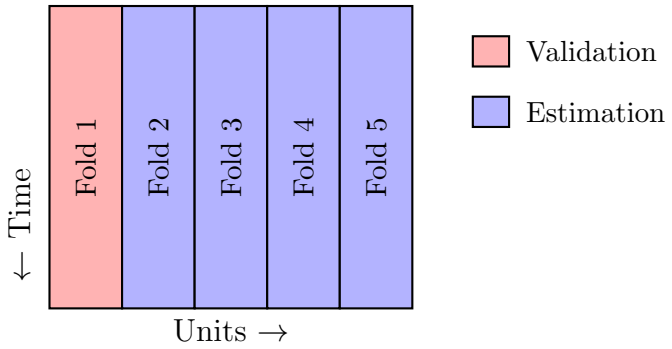


Figure 8: Cross-sectional J-fold cross-validation for IPCA. In each iteration one fold of units is held out for validation and the remaining folds form the estimation sample. All time periods are retained in both stages; only the cross section is partitioned.

out-of-sample R^2 across the Monte Carlo draws. The peak occurs at the true factor dimension $K_0 = 4$, and the validation fit declines once additional spurious factors are introduced. This is the finite-sample pattern one wants from a model-selection device: the criterion rewards genuine common structure but penalizes overfitting through deterioration of held-out performance.

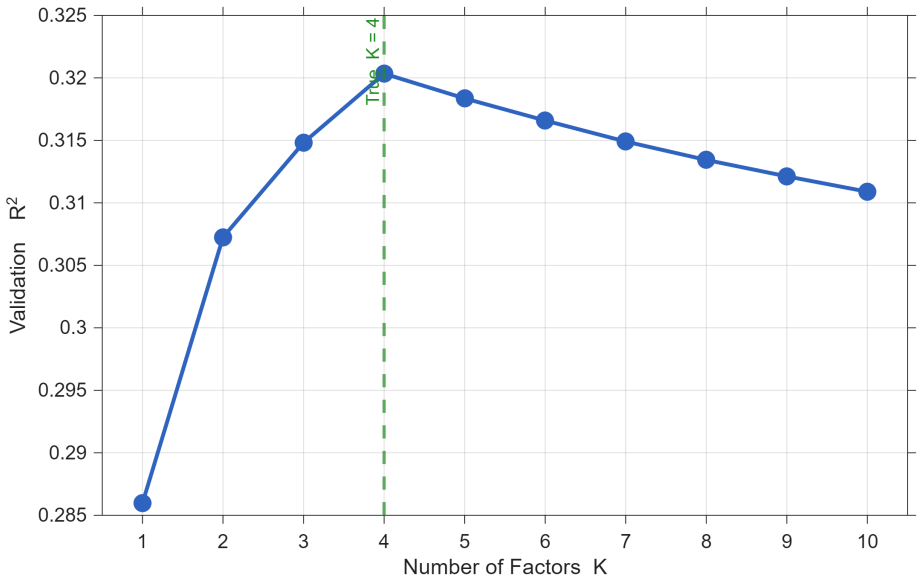


Figure 9: Monte Carlo validation of the IPCA cross-validation rule. Each simulated dataset has $N = 500$ units, $T = 200$ periods, and true factor dimension $K_0 = 4$. For each draw we run 5-fold cross-validation and compute the pooled out-of-sample R^2 for candidate values $K \in \{1, \ldots, 10\}$. The figure reports the mean pooled out-of-sample R^2 across 100 Monte Carlo draws. The dashed green line marks the true factor dimension.

We next apply the same logic to the international macroeconomic panel studied in Section 8.1. The sample contains 45 countries observed annually from 1961 to 2015, and the validation split

is performed across countries rather than across time. We use 10 folds and consider candidate dimensions $K \in \{1, \ldots, 5\}$. For each candidate K, the model is estimated on nine folds of countries and evaluated on the remaining fold using the held-out countries' observed characteristics and the factor path estimated from the training sample. Because GDP growth is demeaned at the country level before estimation, the pooled validation R^2 uses zero as the benchmark. Figure 10 reports the resulting validation curve. The purpose of the exercise is not to turn factor selection into a mechanical rule, but to provide a transparent out-of-sample assessment of whether the added factor dimension improves fit in a way that survives validation on unseen countries.

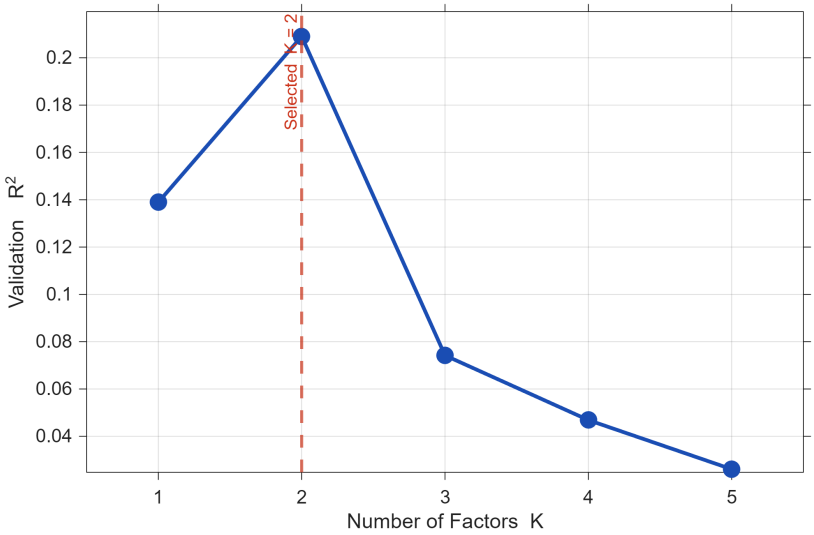


Figure 10: Cross-validation for the international macroeconomic application in Section 8.1. Countries are split into 10 folds. For each candidate $K \in \{1, \ldots, 5\}$, the IPCA model is estimated on the training folds and evaluated on the held-out countries. The plotted statistic is the pooled out-of-sample R^2, using zero as the benchmark because GDP growth is demeaned at the country level before estimation.

Taken together, these exercises provide practical guidance for selecting K in IPCA. A formal model-selection theory is left for future work, but cross-sectional cross-validation already exploits a central structural feature of IPCA , the common characteristic-loading map Γ, and therefore gives an operational and genuinely out-of-sample way to choose the number of factors.

F Inference on Γ under Θ_{X} and Θ_{Y}

In this section, we show that plugin variance estimation can be used to conduct asymptotically valid inference on Γ under both Θ_{X} and Θ_{Y}.

Theorem 3 establishes

$$\sqrt{NT}\,\left(\widehat{\gamma}(\Theta_{\mathrm{X}}) - \gamma^0(\Theta_{\mathrm{X}})\right) \xrightarrow{d} -m_{\mathrm{X}}\,\mathrm{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{X}}^{[1]}\right), \tag{116}$$

where $m_{\mathrm{X}} := \left(H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1} H_{\mathrm{X}}^{0\top}$ and $\mathbb{V}_{\mathrm{X}}^{[1]}$ is the score-limit covariance from Proposition 3. The asymptotic covariance of $\widehat{\gamma}(\Theta_{\mathrm{X}})$ is therefore

$$\Sigma_\gamma^{\mathrm{X}} \;=\; m_{\mathrm{X}}\,\mathbb{V}_{\mathrm{X}}^{[1]}\,m_{\mathrm{X}}^\top.$$

Under Θ_{X}, the identification function $I_{\mathrm{X}}(\Gamma)$ is deterministic, so the normalized true parameter $\gamma^0(\Theta_{\mathrm{X}})$ is not sample-dependent and the Jacobian $J_{\mathrm{X}}^0 = \left.\frac{\partial I_{\mathrm{X}}(\gamma)}{\partial \gamma^\top}\right|_{\gamma=\gamma^0(\Theta_{\mathrm{X}})}$ is a known constant selection matrix computable analytically without estimation. This makes Θ_{X} particularly convenient for feasible inference: unlike Θ_{Y}, where I_{Y} depends on $\widehat{f}_t(\Gamma)$ and introduces an additional random channel into the sandwich map, the Θ_{X} identification function introduces no estimation error into $\widehat{J}_{\mathrm{X}}$.

We first construct a plug-in variance estimator $\widehat{\Sigma}_\gamma^{\mathrm{X}}$ and prove its consistency for $\Sigma_\gamma^{\mathrm{X}}$, enabling asymptotically valid confidence intervals and tests for entries of Γ.

The following three assumptions supplement Assumptions A–F for the purposes of this section only.

Assumption H (Score Differentiability and Jacobian Uniform Convergence)**.** *There exists $\delta > 0$ such that $\widehat{S}(\Gamma)$ is twice continuously differentiable on the ball $\mathcal{B}(\Gamma^0(\Theta_{\mathrm{X}}), \delta)$. Moreover, the sample Jacobian converges uniformly to its limit:*

$$\sup_{\Gamma \in \mathcal{B}(\Gamma^0(\Theta_{\mathrm{X}}),\delta)} \left\| \frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top} - \frac{\partial S^0(\Gamma)}{\partial \gamma^\top} \right\| \xrightarrow{p} 0.$$

Assumption I (Bounded Fourth Moments)**.** *There exists $M < \infty$ such that*

$$\sup_{i,t} \mathbb{E}\left[\|z_{i,t}\|^4\right] \le M, \qquad \sup_{i,t} \mathbb{E}\left[\|e_{i,t}\|^4\right] \le M, \qquad \sup_{t} \mathbb{E}\left[\left\|f_t^0\right\|^4\right] \le M.$$

Assumption J (Invertibility of the Sandwich Denominator)**.** *The matrix $H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0$ is positive definite with smallest eigenvalue bounded away from zero:*

$$\lambda_{\min}\left(H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right) > c > 0.$$

Assumption H requires that the sample score Jacobian converges uniformly near the true parameter; this is analogous to the high-level differentiability condition already implicit in Theorem 2 and follows from primitive conditions on $z_{i,t}$ and $e_{i,t}$ analogous to those underlying Lemma 10. Assumption I imposes 4th-moment bounds, which are standard in panel LLN arguments. Assumption J is the standard identification (rank) condition that the composite linearization matrix is invertible; it is implicit in Theorem 2 and follows from Assumption D(1) together with the normalization structure of Θ_{X}.

Define the estimated score-influence contribution for observation (i,t):

$$\widehat{\psi}_{i,t} := \operatorname{vect}\left(\widehat{Q}\, z_{i,t}\, \widehat{e}_{i,t}\, \widehat{f}_t^{\top}\right),$$

where $\widehat{Q} = \mathbb{I}_L - \widehat{\Omega}_{zz}\widehat{\Gamma}\left(\widehat{\Gamma}^{\top}\widehat{\Omega}_{zz}\widehat{\Gamma}\right)^{-1}\widehat{\Gamma}^{\top}$, $\widehat{\Omega}_{zz} = \frac{1}{NT}\sum_{i,t} z_{i,t}^{\top} z_{i,t}$, and $\widehat{e}_{i,t} = x_{i,t} - z_{i,t}\widehat{\Gamma}\widehat{f}_t$. Define the plug-in outer-product covariance estimator:

$$\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} := \frac{1}{NT}\sum_{i,t} \widehat{\psi}_{i,t}\, \widehat{\psi}_{i,t}^{\top},$$

and let $\widehat{H}_{\mathrm{X}} = \left.\frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^{\top}}\right|_{\Gamma=\widehat{\Gamma}}$ be the sample Jacobian evaluated at $\widehat{\Gamma}$. The plug-in sandwich estimator of $\Sigma_{\gamma}^{\mathrm{X}}$ is then

$$\widehat{m}_{\mathrm{X}} := \left(\widehat{H}_{\mathrm{X}}^{\top}\widehat{H}_{\mathrm{X}} + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1}\widehat{H}_{\mathrm{X}}^{\top}, \qquad \widehat{\Sigma}_{\gamma}^{\mathrm{X}} := \widehat{m}_{\mathrm{X}}\, \widehat{\mathbb{V}_{\mathrm{X}}^{[1]}}\, \widehat{m}_{\mathrm{X}}^{\top}.$$

Note that J_{X}^0 enters exactly, not estimated, because $I_{\mathrm{X}}(\Gamma)$ is deterministic under Θ_{X}.

Proposition 5 (Consistency of the Plug-in Variance Estimator under Θ_{X})**.** *Under Assumptions A–F and H–J, as $N, T \to \infty$ with $T/N \to 0$,*

$$\widehat{\Sigma}_{\gamma}^{\mathrm{X}} \xrightarrow{p} \Sigma_{\gamma}^{\mathrm{X}}.$$

Consequently, for any $a \in \mathbb{R}^{LK}$ with $a^\top \Sigma_{\gamma}^{\mathrm{X}} a > 0$,

$$\frac{\sqrt{NT}\, a^\top \left(\widehat{\gamma}(\Theta_{\mathrm{X}}) - \gamma^0(\Theta_{\mathrm{X}})\right)}{\sqrt{a^\top \widehat{\Sigma}_{\gamma}^{\mathrm{X}} a}} \xrightarrow{d} \mathrm{Normal}\,(0,1)\,.$$

Proof. The claim $\widehat{\Sigma}_{\gamma}^{\mathrm{X}} \xrightarrow{p} \Sigma_{\gamma}^{\mathrm{X}}$ follows from $\widehat{m}_{\mathrm{X}} \xrightarrow{p} m_{\mathrm{X}}$ and $\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \xrightarrow{p} \mathbb{V}_{\mathrm{X}}^{[1]}$ via the continuous mapping theorem applied to the sandwich formula $\widehat{\Sigma}_{\gamma}^{\mathrm{X}} = \widehat{m}_{\mathrm{X}} \widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \widehat{m}_{\mathrm{X}}^\top$. We establish these two convergences in Parts 1 and 2 respectively, and conclude the last claim in Part 3.

Part 1: $\widehat{m}_{\mathrm{X}} \xrightarrow{p} m_{\mathrm{X}}$.

By Theorem 1, $\widehat{\Gamma} \xrightarrow{p} \Gamma^0(\Theta_{\mathrm{X}})$. By Assumption H, the sample Jacobian $\frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top}$ converges uniformly to $\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}$ over $\mathcal{B}(\Gamma^0(\Theta_{\mathrm{X}}), \delta)$. Since $\widehat{\Gamma} \in \mathcal{B}(\Gamma^0(\Theta_{\mathrm{X}}), \delta)$ eventually with probability approaching 1, and since $\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}$ is continuous by Assumption H, we have

$$\widehat{H}_{\mathrm{X}} = \left.\frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top}\right|_{\Gamma=\widehat{\Gamma}} \xrightarrow{p} \left.\frac{\partial S^0(\Gamma)}{\partial \gamma^\top}\right|_{\Gamma=\Gamma^0(\Theta_{\mathrm{X}})} = H_{\mathrm{X}}^0.$$

Hence $\widehat{H}_{\mathrm{X}}^\top \widehat{H}_{\mathrm{X}} + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0 \xrightarrow{p} H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0$. Under Assumption J the limit matrix is positive definite, so by continuity of matrix inversion on the set of positive-definite matrices:

$$\left(\widehat{H}_{\mathrm{X}}^\top \widehat{H}_{\mathrm{X}} + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1} \xrightarrow{p} \left(H_{\mathrm{X}}^{0\top} H_{\mathrm{X}}^0 + J_{\mathrm{X}}^{0\top} J_{\mathrm{X}}^0\right)^{-1}.$$

Multiplying by $\widehat{H}_{\mathrm{X}}^\top \xrightarrow{p} H_{\mathrm{X}}^{0\top}$ gives $\widehat{m}_{\mathrm{X}} \xrightarrow{p} m_{\mathrm{X}}$.

Part 2: $\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \xrightarrow{p} \mathbb{V}_{\mathrm{X}}^{[1]}$.

Let $\psi_{i,t} := \mathrm{vect}\left(Q^0 z_{i,t}^\top e_{i,t} f_t^{0\top}\right)$ denote the oracle influence contribution, where $Q^0 = \mathbb{I}_L -$

$\Omega^{zz}\Gamma^0(\Theta_{\mathrm{X}})\left(\Gamma^0(\Theta_{\mathrm{X}})^\top\Omega^{zz}\Gamma^0(\Theta_{\mathrm{X}})\right)^{-1}\Gamma^0(\Theta_{\mathrm{X}})^\top$. By Lemma 9, $\mathbb{V}_{\mathrm{X}}^{[1]} = (Q^0 \otimes \mathbb{I}_K)\,\Omega^{zef}\left(Q^{0\top} \otimes \mathbb{I}_K\right)$. We now show $\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \xrightarrow{p} \mathbb{V}_{\mathrm{X}}^{[1]}$ in two steps.

Step 1. We show

$$\frac{1}{NT}\sum_{i,t}\psi_{i,t}\psi_{i,t}^\top \xrightarrow{p} \mathbb{V}_{\mathrm{X}}^{[1]}.$$

Each entry of $\psi_{i,t}\psi_{i,t}^\top$ is a product of terms involving Q^0, $z_{i,t}$, $e_{i,t}$, and f_t^0. Since Q^0 is a fixed matrix of population parameters, $\|Q^0\| \le C_Q < \infty$ for some constant C_Q, and the instrument vector is bounded almost surely by Assumption D(2), there exists a constant $C < \infty$ such that

$$\mathbb{E}\left[\|\psi_{i,t}\|^2\right] \le C\,\mathbb{E}\left[\|z_{i,t}\|^2\|e_{i,t}\|^2\left\|f_t^0\right\|^2\right] \le C\,\mathbb{E}\left[\|e_{i,t}\|^2\left\|f_t^0\right\|^2\right] \le C\left(\mathbb{E}\left[\|e_{i,t}\|^4\right]\right)^{1/2}\left(\mathbb{E}\left[\left\|f_t^0\right\|^4\right]\right)^{1/2} < \infty,$$

where the last inequality uses Cauchy–Schwarz and Assumption I. Applying the panel LLN in Lemma 16 to the oracle outer-product array and using the stationarity and weak-dependence conditions of Assumptions A–F:

$$\frac{1}{NT}\sum_{i,t}\psi_{i,t}\psi_{i,t}^\top \xrightarrow{p} \left(Q^0 \otimes \mathbb{I}_K\right)\Omega^{zef}\left(Q^{0\top} \otimes \mathbb{I}_K\right) = \mathbb{V}_{\mathrm{X}}^{[1]}.$$

Step 2: Generated-regressor decomposition. We bound $\frac{1}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|^2$. Write the estimation errors in each component:

$$\begin{aligned}
\Delta Q &:= \widehat{Q} - Q^0, \\
\Delta e_{i,t} &:= \widehat{e}_{i,t} - e_{i,t} = z_{i,t}(\Gamma^0(\Theta_{\mathrm{X}}) - \widehat{\Gamma})f_t^0 + z_{i,t}\widehat{\Gamma}(f_t^0 - \widehat{f}_t), \\
\Delta f_t &:= \widehat{f}_t - f_t^0.
\end{aligned}$$

Since $\widehat{\Omega}_{zz} \xrightarrow{p} \Omega^{zz}$ (LLN for second moments of $z_{i,t}$) and $\widehat{\Gamma} \xrightarrow{p} \Gamma^0(\Theta_{\mathrm{X}})$ (Theorem 1), continuity of

the projection formula gives $\Delta Q = \mathcal{O}_p\left(1/\sqrt{NT}\right)$. Decompose:

$$\widehat{\psi}_{i,t} - \psi_{i,t} = \underbrace{\operatorname{vect}\left((\Delta Q)\, z_{i,t}^{\top}\, e_{i,t}\, f_t^{0\top}\right)}_{A_{i,t}} + \underbrace{\operatorname{vect}\left(Q^0\, z_{i,t}^{\top}\, (\Delta e_{i,t})\, f_t^{0\top}\right)}_{B_{i,t}} + \underbrace{\operatorname{vect}\left(Q^0\, z_{i,t}^{\top}\, e_{i,t}\, (\Delta f_t)^{\top}\right)}_{C_{i,t}} + R_{i,t}, \tag{117}$$

where $R_{i,t}$ collects all cross-product terms, each involving at least two error factors. We bound $\frac{1}{NT}\sum_{i,t}\|A_{i,t}\|^2$, $\frac{1}{NT}\sum_{i,t}\|B_{i,t}\|^2$, $\frac{1}{NT}\sum_{i,t}\|C_{i,t}\|^2$, and $\frac{1}{NT}\sum_{i,t}\|R_{i,t}\|^2$ in turn.

Bounding Term A.

$$\frac{1}{NT}\sum_{i,t}\|A_{i,t}\|^2 \;\le\; \|\Delta Q\|^2 \cdot \frac{1}{NT}\sum_{i,t}\|z_{i,t}\|^2\,\|e_{i,t}\|^2\,\left\|f_t^0\right\|^2.$$

The second factor is $\mathcal{O}_p(1)$ by Lemma 16 and Assumption I. Since $\|\Delta Q\|^2 = \mathcal{O}_p(1/(NT))$, the whole term is $\mathcal{O}_p(1/(NT)) = o_p(1)$.

Bounding Term B. Write $\Delta e_{i,t} = B_{1,it} + B_{2,it}$ with $B_{1,it} := z_{i,t}^{\top}(\Gamma^0(\Theta_{\mathrm{X}}) - \widehat{\Gamma})f_t^0$ and $B_{2,it} := z_{i,t}^{\top}\widehat{\Gamma}(f_t^0 - \widehat{f}_t)$. For B_1:

$$\frac{1}{NT}\sum_{i,t}\|z_{i,t}\|^2\,\|B_{1,it}\|^2 \;\le\; \left\|\widehat{\Gamma} - \Gamma^0(\Theta_{\mathrm{X}})\right\|^2 \cdot \frac{1}{NT}\sum_{i,t}\|z_{i,t}\|^4\,\left\|f_t^0\right\|^2 = \mathcal{O}_p\left(\frac{1}{NT}\right)\cdot\mathcal{O}_p(1) = o_p(1),$$

where we used $\left\|\widehat{\Gamma} - \Gamma^0(\Theta_{\mathrm{X}})\right\|^2 = \mathcal{O}_p(1/(NT))$ (Theorem 3) and the moment bound in Assumption I. For B_2, since $\left\|\widehat{\Gamma}\right\| = \mathcal{O}_p(1)$:

$$\frac{1}{NT}\sum_{i,t}\|z_{i,t}\|^2\,\|B_{2,it}\|^2 \;\le\; \left\|\widehat{\Gamma}\right\|^2 \cdot \frac{1}{T}\sum_{t}\left\|\widehat{f}_t - f_t^0\right\|^2 \cdot \frac{1}{N}\sum_{i}\|z_{i,t}\|^4.$$

By Theorem 4, $\frac{1}{T}\sum_t\left\|\widehat{f}_t - f_t^0\right\|^2 = \mathcal{O}_p(1/N)$. By Assumption I and Lemma 16, $\frac{1}{N}\sum_i\|z_{i,t}\|^4 = \mathcal{O}_p(1)$. Hence $\frac{1}{NT}\sum_{i,t}\|B_{2,it}\|^2 = \mathcal{O}_p(1/N) \to_p 0$.

Bounding Term C.

$$\frac{1}{NT}\sum_{i,t}\|C_{i,t}\|^2 \;\le\; \left\|Q^0\right\|^2\cdot\frac{1}{T}\sum_t\|\Delta f_t\|^2\cdot\frac{1}{N}\sum_i\|z_{i,t}\|^2\,\|e_{i,t}\|^2 = O(1)\cdot\mathcal{O}_p\left(\frac{1}{N}\right)\cdot\mathcal{O}_p(1) = \mathcal{O}_p\left(\frac{1}{N}\right)\to_p 0,$$

using that Q^0 is a fixed matrix of population parameters and hence $\|Q^0\|^2 = O(1)$, together with Theorem 4.

Bounding the remainder R. Each component of $R_{i,t}$ involves products of at least two error terms from $\{\Delta Q, \Delta e_{i,t}, \Delta f_t\}$. For the leading cross-product $(\Delta Q)\, z_{i,t}\, (\Delta e_{i,t})\, f_t^{0\top}$:

$$\frac{1}{NT}\sum_{i,t}\left\|(\Delta Q) z_{i,t} \Delta e_{i,t} f_t^{0\top}\right\|^2 \leq \|\Delta Q\|^2 \cdot \frac{1}{NT}\sum_{i,t}\|z_{i,t}\|^2 \|\Delta e_{i,t}\|^2 \left\|f_t^0\right\|^2 = \mathcal{O}_p\left(\frac{1}{NT}\right)\cdot\mathcal{O}_p\left(\frac{1}{N}\right) = o_p\left(\frac{1}{N}\right).$$

All other cross-products are bounded by at least one extra factor of $\mathcal{O}_p\left(1/\sqrt{N}\right)$ or $\mathcal{O}_p\left(1/\sqrt{NT}\right)$, giving $\frac{1}{NT}\sum_{i,t}\|R_{i,t}\|^2 = o_p(1/N) \to_p 0$.

Collecting the bounds. Applying $\|a+b+c+d\|^2 \leq 4(\|a\|^2 + \|b\|^2 + \|c\|^2 + \|d\|^2)$ to (117):

$$\frac{1}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|^2 \;\leq\; 4\left[\frac{1}{NT}\sum_{i,t}\|A_{i,t}\|^2 + \|B_{i,t}\|^2 + \|C_{i,t}\|^2 + \|R_{i,t}\|^2\right] = \mathcal{O}_p\left(\frac{1}{N}\right) \;\to_p\; 0. \tag{118}$$

Substep 2c: Cauchy–Schwarz conclusion. By the triangle inequality:

$$\begin{aligned}\left\|\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} - \frac{1}{NT}\sum_{i,t}\psi_{i,t}\psi_{i,t}^\top\right\| &= \left\|\frac{1}{NT}\sum_{i,t}\left(\widehat{\psi}_{i,t}\widehat{\psi}_{i,t}^\top - \psi_{i,t}\psi_{i,t}^\top\right)\right\| \\ &\leq \frac{2}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|\cdot\|\psi_{i,t}\| + \frac{1}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|^2. \end{aligned} \tag{119}$$

By the Cauchy–Schwarz inequality applied to the first term:

$$\frac{1}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|\cdot\|\psi_{i,t}\| \;\leq\; \sqrt{\frac{1}{NT}\sum_{i,t}\left\|\widehat{\psi}_{i,t} - \psi_{i,t}\right\|^2}\cdot\sqrt{\frac{1}{NT}\sum_{i,t}\|\psi_{i,t}\|^2} = \mathcal{O}_p\left(\frac{1}{\sqrt{N}}\right)\cdot\mathcal{O}_p(1) \to_p 0,$$

where we used (118) for the first factor and the oracle LLN in Substep 2a for the second. Together with (118) and (119), this gives:

$$\left\|\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} - \frac{1}{NT}\sum_{i,t}\psi_{i,t}\psi_{i,t}^\top\right\| \;\xrightarrow{p}\; 0.$$

Combining with Substep 2a: $\widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \xrightarrow{p} \mathbb{V}_{\mathrm{X}}^{[1]}$.

Part 3: Sandwich consistency and Wald pivot.

From Parts 1 and 2, $\widehat{\Sigma}_\gamma^{\mathrm{X}} = \widehat{m}_{\mathrm{X}} \widehat{\mathbb{V}_{\mathrm{X}}^{[1]}} \widehat{m}_{\mathrm{X}}^\top \xrightarrow{p} m_{\mathrm{X}} \mathbb{V}_{\mathrm{X}}^{[1]} m_{\mathrm{X}}^\top = \Sigma_\gamma^{\mathrm{X}}$ by the continuous mapping theorem.

For the Wald pivot, fix $a \in \mathbb{R}^{LK}$ with $a^\top \Sigma_\gamma^{\mathrm{X}} a > 0$. By Theorem 3:

$$\sqrt{NT}\, a^\top \left(\widehat{\gamma}(\Theta_{\mathrm{X}}) - \gamma^0(\Theta_{\mathrm{X}})\right) \xrightarrow{d} \text{Normal}\left(0,\ a^\top \Sigma_\gamma^{\mathrm{X}} a\right).$$

Since $a^\top \widehat{\Sigma}_\gamma^{\mathrm{X}} a \xrightarrow{p} a^\top \Sigma_\gamma^{\mathrm{X}} a > 0$, Slutsky's theorem gives:

$$\frac{\sqrt{NT}\, a^\top \left(\widehat{\gamma}(\Theta_{\mathrm{X}}) - \gamma^0(\Theta_{\mathrm{X}})\right)}{\sqrt{a^\top \widehat{\Sigma}_\gamma^{\mathrm{X}} a}} \xrightarrow{d} \text{Normal}\,(0,1)\,. \qquad \square$$

Theorem 3 also establishes

$$\sqrt{NT}\, \left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}})\right) \xrightarrow{d} -m_{\mathrm{Y}}\, \text{Normal}\left(\mathbf{0}, \mathbb{V}_{\mathrm{Y}}^{[1]}\right), \tag{120}$$

where $m_{\mathrm{Y}} := \left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} H_{\mathrm{Y}}^{0\top}$ and $\mathbb{V}_{\mathrm{Y}}^{[1]}$ is the score-limit covariance from Proposition 3 evaluated at $\Gamma^0(\Theta_{\mathrm{Y}}^0)$. The asymptotic covariance of $\widehat{\gamma}(\Theta_{\mathrm{Y}})$ around the normalized true parameter $\gamma^0(\Theta_{\mathrm{Y}})$ is therefore

$$\Sigma_\gamma^{\mathrm{Y}} \;=\; m_{\mathrm{Y}}\, \mathbb{V}_{\mathrm{Y}}^{[1]}\, m_{\mathrm{Y}}^\top.$$

Unlike the Θ_{X} case, the identification function $I_{\mathrm{Y}}(\Gamma)$ is sample-dependent because it depends on $\widehat{f}_t(\Gamma)$. Hence feasible inference under Θ_{Y} requires estimating not only the score Jacobian but also the identification Jacobian.

The next assumption is the Θ_{Y} counterpart of Assumption H, augmented to cover the identification Jacobian.

Assumption K (Score and Identification Jacobian Uniform Convergence under Θ_{Y})**.** *There exists $\delta > 0$ such that $\widehat{S}(\Gamma)$ and $I_{\mathrm{Y}}(\Gamma)$ are twice continuously differentiable on the ball $\mathcal{B}(\Gamma^0(\Theta_{\mathrm{Y}}^0), \delta)$*

with probability approaching one. Moreover,

$$\sup_{\Gamma \in \mathcal{B}(\Gamma^0(\Theta_{\mathrm{Y}}^0),\delta)} \left\| \frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top} - \frac{\partial S^0(\Gamma)}{\partial \gamma^\top} \right\| \xrightarrow{p} 0,$$

and

$$\sup_{\Gamma \in \mathcal{B}(\Gamma^0(\Theta_{\mathrm{Y}}^0),\delta)} \left\| \frac{\partial I_{\mathrm{Y}}(\Gamma)}{\partial \gamma^\top} - \frac{\partial I_{\mathrm{Y}}^0(\Gamma)}{\partial \gamma^\top} \right\| \xrightarrow{p} 0.$$

Assumption L (Invertibility of the Sandwich Denominator under Θ_{Y})**.** *The matrix* ${H_{\mathrm{Y}}^0}^\top H_{\mathrm{Y}}^0 + {J_{\mathrm{Y}}^0}^\top J_{\mathrm{Y}}^0$ *is positive definite with smallest eigenvalue bounded away from zero:*

$$\lambda_{\min}\left({H_{\mathrm{Y}}^0}^\top H_{\mathrm{Y}}^0 + {J_{\mathrm{Y}}^0}^\top J_{\mathrm{Y}}^0\right) > c > 0.$$

Reusing the estimated score-influence contribution $\widehat{\psi}_{i,t}$ from Section F, define

$$\widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}} := \frac{1}{NT} \sum_{i,t} \widehat{\psi}_{i,t} \widehat{\psi}_{i,t}^\top,$$

and let

$$\widehat{H}_{\mathrm{Y}} := \left. \frac{\partial \widehat{S}(\Gamma)}{\partial \gamma^\top} \right|_{\Gamma = \widehat{\Gamma}(\Theta_{\mathrm{Y}})}, \qquad \widehat{J}_{\mathrm{Y}} := \left. \frac{\partial I_{\mathrm{Y}}(\Gamma)}{\partial \gamma^\top} \right|_{\Gamma = \widehat{\Gamma}(\Theta_{\mathrm{Y}})}.$$

The plug-in sandwich estimator of $\Sigma_\gamma^{\mathrm{Y}}$ is then

$$\widehat{m}_{\mathrm{Y}} := \left(\widehat{H}_{\mathrm{Y}}^\top \widehat{H}_{\mathrm{Y}} + \widehat{J}_{\mathrm{Y}}^\top \widehat{J}_{\mathrm{Y}} \right)^{-1} \widehat{H}_{\mathrm{Y}}^\top, \qquad \widehat{\Sigma}_\gamma^{\mathrm{Y}} := \widehat{m}_{\mathrm{Y}} \widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}} \widehat{m}_{\mathrm{Y}}^\top.$$

Proposition 6 (Consistency of the Plug-in Variance Estimator under Θ_{Y})**.** *Under Assumptions A–F, I, K, and L, as* $N, T \to \infty$ *with* $T/N \to 0$*,*

$$\widehat{\Sigma}_\gamma^{\mathrm{Y}} \xrightarrow{p} \Sigma_\gamma^{\mathrm{Y}}.$$

Consequently, for any $a \in \mathbb{R}^{LK}$ *with* $a^\top \Sigma_\gamma^{\mathrm{Y}} a > 0$,

$$\frac{\sqrt{NT}\, a^\top \left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}})\right)}{\sqrt{a^\top \widehat{\Sigma}_\gamma^{\mathrm{Y}} a}} \xrightarrow{d} \mathrm{Normal}\,(0,1)\,,$$

so that Wald confidence intervals and tests for linear combinations of $\mathrm{vect}\,(\Gamma)$ *are asymptotically valid under* Θ_{Y} *as well.*

Proof. The proof parallels Proposition 5 but adds the estimation of the identification Jacobian. By the discussion in Section 4.3, $\Gamma^0(\Theta_{\mathrm{Y}}) \xrightarrow{p} \Gamma^0(\Theta_{\mathrm{Y}}^0)$, and Corollary 1 then implies $\widehat{\Gamma}(\Theta_{\mathrm{Y}}) \xrightarrow{p} \Gamma^0(\Theta_{\mathrm{Y}}^0)$. By Assumption K, both sample Jacobians converge uniformly to their deterministic limits over a neighborhood of $\Gamma^0(\Theta_{\mathrm{Y}}^0)$. Therefore,

$$\widehat{H}_{\mathrm{Y}} \xrightarrow{p} H_{\mathrm{Y}}^0, \qquad \widehat{J}_{\mathrm{Y}} \xrightarrow{p} J_{\mathrm{Y}}^0.$$

Under Assumption L, continuity of matrix inversion on the positive-definite cone gives

$$\widehat{m}_{\mathrm{Y}} = \left(\widehat{H}_{\mathrm{Y}}^\top \widehat{H}_{\mathrm{Y}} + \widehat{J}_{\mathrm{Y}}^\top \widehat{J}_{\mathrm{Y}}\right)^{-1} \widehat{H}_{\mathrm{Y}}^\top \xrightarrow{p} \left(H_{\mathrm{Y}}^{0\top} H_{\mathrm{Y}}^0 + J_{\mathrm{Y}}^{0\top} J_{\mathrm{Y}}^0\right)^{-1} H_{\mathrm{Y}}^{0\top} = m_{\mathrm{Y}}.$$

For $\widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}}$, the oracle LLN and generated-regressor decomposition used in the proof of Proposition 5 apply verbatim after replacing $\Gamma^0(\Theta_{\mathrm{X}})$, Q^0, and $\mathbb{V}_{\mathrm{X}}^{[1]}$ by their Θ_{Y} counterparts evaluated at $\Gamma^0(\Theta_{\mathrm{Y}}^0)$. Hence

$$\widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}} \xrightarrow{p} \mathbb{V}_{\mathrm{Y}}^{[1]}.$$

Combining the two convergences and applying the continuous mapping theorem to $\widehat{\Sigma}_\gamma^{\mathrm{Y}} = \widehat{m}_{\mathrm{Y}} \widehat{\mathbb{V}_{\mathrm{Y}}^{[1]}} \widehat{m}_{\mathrm{Y}}^\top$ gives

$$\widehat{\Sigma}_\gamma^{\mathrm{Y}} \xrightarrow{p} \Sigma_\gamma^{\mathrm{Y}}.$$

Finally, Theorem 3 gives

$$\sqrt{NT}\, a^\top \left(\widehat{\gamma}(\Theta_{\mathrm{Y}}) - \gamma^0(\Theta_{\mathrm{Y}})\right) \xrightarrow{d} \mathrm{Normal}\left(0,\ a^\top \Sigma_\gamma^{\mathrm{Y}} a\right).$$

Since $a^\top \widehat{\Sigma}^{\mathrm{Y}}_\gamma a \xrightarrow{p} a^\top \Sigma^{\mathrm{Y}}_\gamma a > 0$, Slutsky's theorem yields the stated Wald pivot. □

G Additional Details

G.1 Simulation Details

For given N, T, we generate a data panel consists of z, e, f^0 and use these to assemble the x panel. We calibrate the simulated data to the IPCA model (fixing $K = 2$ and $L = 10$) estimated from US monthly stock returns in KPS19. In particular, we use the ten most significant instruments from KPS19 as calibration targets. They are market capitalization, total assets, market beta, short-term reversal, momentum, turnover, price relative to its 52-week high, long-term reversal, unexplained volume, and idiosyncratic volatility with respect to the Fama-French three factor model. The simulation follows the following steps:

1. **Generate factors**. Fit a VAR(1) process to estimated IPCA factors from KPS19. Simulate f^0_t according to the estimated VAR employing normal innovations.
2. **Generate instruments**. For each stock, calculate the time-series averages of the instruments. Pool the demeaned characteristics into a panel and estimate a ten variable panel VAR(1). Next, for each i, generate the means of $z_{i,t}$ as an i.i.d. draw from the empirical distribution of the time series means. Then, simulate the dynamic component of $z_{i,t}$ from the estimated VAR with normal innovations.[43]
3. **Generate errors**. Elements of the error panel e are simulated from an i.i.d. normal distribution whose variance is calibrated so that the population R^2, defined as $1 - \mathbb{E}e^2/\mathbb{E}x^2$, equals 20%, matching the empirical R^2 in the estimated model in KPS19.
4. **Generate main panel**. We fix $\widehat{\Gamma}(\Theta_{\mathrm{Y}})$ at its empirically estimated value and calculate $x_{i,t}$ according to model equation (3).

[43] We generate $z_{i,t}$ with ex-ante i.i.d. means, so that each individual's time-series process is non-ergodic. This is deliberate so that $z_{i,t}$ admits the flexible property allowed by the assumptions and resembles real-world instrument data.

G.2 The Sign Issue in Normalization

As mentioned in footnote 14, identification conditions like Θ_{Y} with only [1] and [2] does not pin down the signs of each Γ column. To be precise, for any $\Gamma \in \Theta_{\text{Y}}$, let $\Gamma' = \Gamma \text{diag}\{s\}$, where $\text{diag}\{s\}$ is a $K \times K$ diagonal matrix with $+1$ or -1's on the diagonal. Then Γ and Γ' are obviously rotationally equivalent, because the corresponding factors can flip the signs accordingly. Meanwhile, both Γ, Γ' satisfy Θ_{Y}'s [1] and [2]. Without adding additional sign restrictions, this would violate the uniqueness property of an identification condition.

It is not hard to add restrictions for the signs to pin down this remaining bit of unidentification in theory. For example, one can restrict all the sample factors means $(\frac{1}{T}\sum_t \widehat{f}_{t(\Gamma)})$ to be positive (call this [3′]). Alternatively, restricting the first non-zero element in each column of Γ to be positive works as well (call this [3″]). Similar restrictions are seen in Stock and Watson (2002), Bai and Ng (2013).

However, we report that the sign issue is trickier in finite sample simulations. For example, if the true factors has zero (or close to zero) expectations ($\mathbb{E}[f_t^0] = \mathbf{0}$). Then, even if the true factors are observed, its finite sample averages are arbitrarily positive or negative, making [3′] unstable. Even if $\widehat{\Gamma}$ is estimated rather accurately, the small sign flipping of the factor mean makes a large difference in $\widehat{\Gamma} - \Gamma^0$, keeping it away from converging. Similarly, [3″] also runs into finite sample problems if Γ^0's first non-zero elements in some columns are close to zero.

So how to design a sign restriction such that the signal-to-noise ratio in picking the sign is always maximized, adapting to any potential peculiar model? One way is to make Γ's signs always align with those of Γ^0. So let [3‴] be Γ s.t. $\Gamma_k^\top \Gamma_k^0 > 0, \forall k$. Then, [3‴]'s obvious problem is it depends on population information (Γ^0), disqualifying it as an estimation's identification condition.

Finally, we give a sign restriction that is both theoretically sound and practically easy to use in simulation exercises. Let [3] be the set of Γ s.t. $\Gamma_k^\top \widehat{\Gamma}_k > 0, \forall k$, where $\widehat{\Gamma} = \arg\min_{\Gamma \in [1],[2],[3'']} \widehat{G}(\Gamma)$. The idea is for the Γ within the target minimizing set, we use [3″] (or [3′]) to pin down its signs. That gives the unique estimate. And for Γ outside of the set, we align its signs to the

estimate. This will normalize the true towards the direction of the estimate when constructing the reference point $\Gamma^0(\Theta)$.

This avoids the sign indeterminacy problem within a small neighborhood around $\widehat{\Gamma}$. The benefit is when calculating estimation error, $\left\|\widehat{\Gamma} - \Gamma^0(\Theta)\right\|$ always take the smallest possible value among all possible sign combinations. To calculate that quantity in simulation exercises, one can first ignore the sign issue and calculate $\widehat{\Gamma}$ and $\Gamma^0(\Theta)$ both up to sign unidentification. Then pick the signs to minimize that norm between the two:

$$\min_{s_1 \ldots s_K = \pm 1} \left\|\widehat{\Gamma} - \Gamma^0(\Theta)\text{diag}\left\{s\right\}\right\|. \tag{121}$$

G.3 International Macroeconomic Instruments Data

There are fifty-nine indicators (besides GDP growth itself) that we could use as instruments. Because we require a country-year observation to have GDP growth and all its instruments non-missing, we first restrict attention to only indicators that are 80% nonmissing in the panel; we choose 80% as a cut-off so as to be able to include import and export percentage, which are very natural instruments to include. We then drop indicators that *a priori* appear unconnected to global growth: adolescent fertility rates, fertility rates, life expectancy, mortality rates, total population, and surface area. We drop gross national income and gross national income per capita because of their high correlation with GDP and GDP growth, respectively. We drop Guinea, Haiti, and Tanzania because they don't have non-missing GDP growth before 1985. Because we want to include exports/imports as instruments, we drop Congo, Ethiopia, Paraguay, and Zambia because they don't have non-missing import/export data before 1985. We want to include Germany in our analysis and it has no CO2 emissions data before 1985, so we drop that indicator. We want to include Hong Kong in our analysis and it has no domestic credit data before 1985, so we drop that indicator. We want to include Belgium in our analysis and it has no population density data before 1985, so we drop that indicator. We want to include capital formation but Luxumbourg and Cabo Verde have no data before 1985, so we drop those countries. We drop the percentage of merchandise trade given its high correlation with imports

and exports which we do include. Urban population growth is quite correlated with population growth, so we exclude the former. This leaves us with the indicators discussed in the text, combined with an Industrial dummy. We then restrict attention to the available countries and Kose, Otrok, and Prasad (2012) labeled as Industrial and Emerging, leaving the countries in Table 2.

Table 2: List of Countries

Argentina	Hong Kong SAR, China	Norway
Australia	Iceland	Pakistan
Austria	India	Peru
Belgium	Indonesia	Philippines
Brazil	Ireland	Portugal
Canada	Israel	Singapore
Chile	Italy	South Africa
China	Japan	Spain
Colombia	Jordan	Sweden
Denmark	Korea, Rep.	Switzerland
Egypt, Arab Rep.	Malaysia	Thailand
Finland	Mexico	Turkey
France	Morocco	United Kingdom
Germany	Netherlands	United States
Greece	New Zealand	Venezuela, RB

H Persistence of Factor Loadings and Convergence Rates

The $\sqrt{NT}$ asymptotic normality result for $\hat{\Gamma}$ relies on a panel central limit theorem applied to the score

$$\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \sum_{t=1}^{T} \mathrm{vec}\big(z_{it}^{\top}\, e_{it}\, f_t^{\top}\big)\,,$$

where $e_{it} = \eta_{it} f_t + \mu_{it}$. A key component of this expression is

$$\frac{1}{\sqrt{NT}} \sum_{i,t} z_{it}^{\top}\, \eta_{it}\, f_t f_t^{\top}.$$

When η_{it} refreshes quickly across both i and t, the sum averages over NT effectively distinct contributions, which delivers the benchmark $\sqrt{NT}$ rate. When the loading noise is highly per-

sistent over time, however, the time dimension contributes less independent information, so the effective sample size moves toward the cross-sectional dimension.

A natural first robustness exercise is to model η_{it} as a highly persistent autoregressive process. This is useful empirically because it captures slowly moving loadings. At the same time, if the persistence parameter is fixed as T grows, then the process remains short-memory and the baseline $\sqrt{NT}$ asymptotics continue to apply. To obtain a genuine departure from that benchmark, persistence itself must strengthen with the sample size, so that the effective refresh rate slows as the time dimension expands.

To study this mechanism directly, we consider a blockwise-constant specification for η_{it}. Time is partitioned into non-overlapping blocks of length

$$b_T = T^{\delta},$$

and within each block η_{it} is held constant at a value drawn i.i.d. across units and blocks. The parameter $\delta \in [0, 1]$ governs the refresh rate. At $\delta = 0$, one has $b_T = 1$, so the loading noise refreshes every period and the benchmark $\sqrt{NT}$ rate applies. At $\delta = 1$, one has $b_T = T$, so η_{it} is time-invariant and the rate falls to $\sqrt{N}$. Intermediate values of δ generate intermediate degrees of persistence.

For square panels with $N = T$, the number of effective time blocks per unit is of order $T/b_T = T^{1-\delta}$. This yields the heuristic effective sample size $NT^{1-\delta}$ and therefore the convergence rate

$$\hat{\Gamma} - \Gamma_0 = O_p\big((NT^{1-\delta})^{-1/2}\big).$$

Equivalently, the root mean squared error should scale as $(NT)^{-(2-\delta)/4}$, so the slope of a $\log \mathrm{RMSE}$ versus $\log(NT)$ plot should equal $-(2-\delta)/4$. Figure 11 reports the simulation evidence.

The figure shows that the empirical slopes move smoothly between the benchmark $\sqrt{NT}$ and $\sqrt{N}$ cases as persistence increases. Across all values of δ, the RMSE declines with sample size, indicating that the estimator remains consistent even when Assumption E(1) fails. What

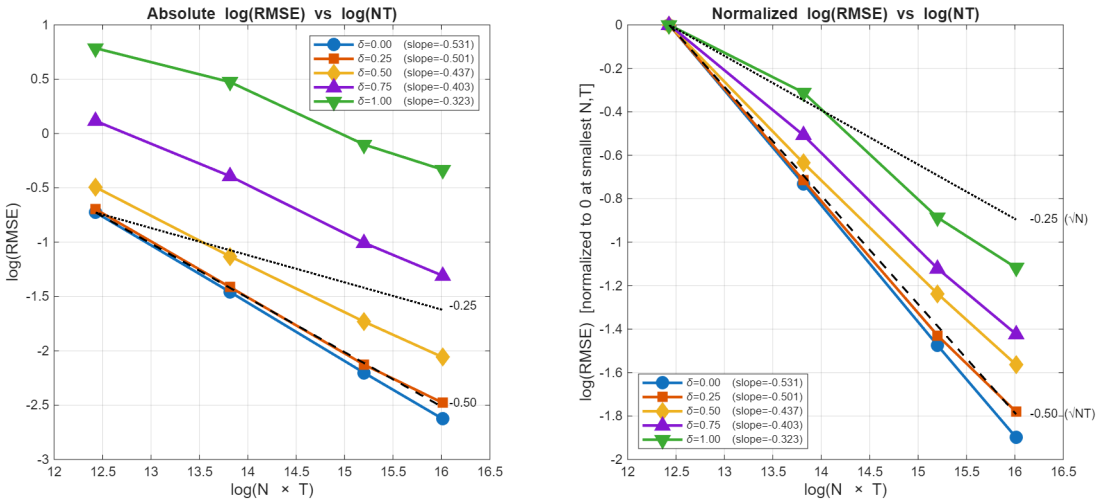


Figure 11: Convergence rates of $\hat{\Gamma}$ under blockwise-constant η_{it} with block length $b_T = T^{\delta}$. The left panel plots $\log \mathrm{RMSE}$ against $\log(NT)$ for each value of δ, together with reference lines for the benchmark $\sqrt{NT}$ and $\sqrt{N}$ rates. The right panel normalizes each curve to start at zero, highlighting how the slope changes as persistence increases. The results show a smooth transition from the fast-refreshing case to the time-invariant case.

changes is the convergence rate and, with it, the validity of $\sqrt{NT}$-based inference. In persistent environments, standardizing as if the benchmark rate continued to hold can understate uncertainty and overstate the precision of pricing-restriction tests.

I Low-level Conditions for Assumptions A.1–A.3

This last appendix section provides a construction that yields a two-dimensional panel structure that satisfies Assumptions A.1–A.3. It shows the lower-level underpinnings of the assumptions and illustrates the assumptions allows for a wide range of dynamics in the data generating process. This construction is otherwise tangential to the rest of the paper.

We construct what we call "stochastic panels" to model the generating process of a two-dimensional data panel, which has a time-series direction (denoted as [TS]) and a cross-sectional direction (denoted as [CS]). The construction involves *two* transformations that trace out a rectangular lattice of sample points. It imitates the underlying definition of the stochastic processes, which has *one* transformation that traces out a sequence of sample points.

Formally, a *stochastic panel* is defined as a random variable (vector) $X\colon \Omega \to \mathbb{R}^p$ on the probability space $\{\Omega, \mathfrak{F}, Pr\}$ that satisfies the following conditions with respect to two transformations $\mathbb{S}_{[\mathrm{d}]}\colon \Omega \to \Omega$ on the two different directions $[\mathrm{d}] \in \{[\mathrm{CS}], [\mathrm{TS}]\}$. By "on the probability space", we mean that X is $\mathfrak{F}$-measurable.

SP.1 **Measurable:** Both transformations are $\mathfrak{F}$-measurable, i.e., $\forall \Lambda \in \mathfrak{F}$, $\mathbb{S}_{[\mathrm{d}]}^{-1}(\Lambda) \in \mathfrak{F}$.[44]

SP.2 **Recombining:** $\mathbb{S}_{[\mathrm{d}]} \circ \mathbb{S}_{[-\mathrm{d}]} = \mathbb{S}_{[-\mathrm{d}]} \circ \mathbb{S}_{[\mathrm{d}]}$.[45]

SP.3 **Measure-preserving:** $\forall \Lambda \in \mathfrak{F}$, $Pr\left(\mathbb{S}_{[\mathrm{d}]}^{-1}(\Lambda)\right) = Pr(\Lambda)$, for either d.

SP.4 **Jointly ergodic:** Any event invariant to both transformations has a probability of either zero or one, i.e., $\forall \Lambda$ s.t. $\Lambda = \mathbb{S}_{[\mathrm{d}]}^{-1}(\Lambda), \forall d, Pr(\Lambda) = 0$ or 1.

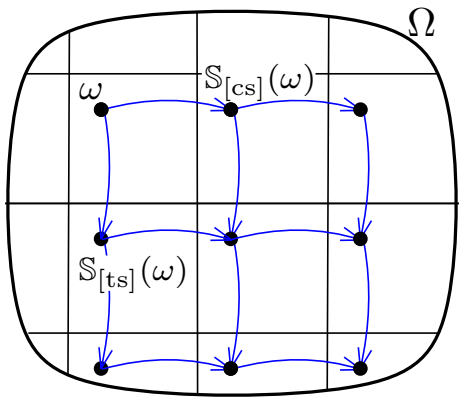
SP.5 **Cross-sectional exchangeable:** The sequence of random variables $\{X \circ \mathbb{S}_{[\mathrm{CS}]}^{i-1}, i = 1, 2, \dots\}$ is exchangeable.

Next, we discuss the conditions one by one and deduce the implied properties. Condition SP.1 is the familiar measurability condition from the definition of stochastic processes except applied

[44] $\mathbb{S}_{[\mathrm{d}]}^{-1}(\Lambda) := \{\omega \in \Omega : \mathbb{S}_{[\mathrm{d}]}(\omega) \in \Lambda\}$.

[45] $[-\mathrm{d}]$ means the other direction with respect to $[\mathrm{d}]$.

Figure 12: Sample Stochastic Panel Generated by Lattice Points



Notes: The horizontal blue arrows are $\mathbb{S}_{[CS]}$, vertical blue arrows are $\mathbb{S}_{[TS]}$. The random variables evaluated at the nine lattice points starting from ω constitute a sample stochastic panel with $N = 3, T = 3$. The square blocks represent partitions in $\mathfrak{F}$, the columns of blocks are $\mathcal{C}$'s partitions, and the rows blocks are $\mathcal{C}^{[CS]}$'s partitions.

in two directions. The interpretation of $\mathbb{S}_{[TS]}(\omega)$ is "the next period", and the interpretation of $\mathbb{S}_{[CS]}(\omega)$ is "the next individual" to be sampled, as illustrated in Figure 12. The condition guarantees that the probabilities of transformed sample points, e.g. $Pr\left\{\mathbb{S}_{[d]}(\omega) \in \Lambda\right\}$, are well-defined.

Condition SP.2 guarantees the commutative property that the next period's second individual is the same as the second individual in the next period. Therefore, starting from an $\omega \in \Omega$, following the two transformations N and T times, respectively, one can trace out an $N \times T$ rectangular lattice of sample points (again see Figure 12). The set of random variables evaluated at the lattice of sample points defines the sample stochastic panel conventionally represented with double subscripts:

$$X_{i,t} := X \circ \mathbb{S}_{[CS]}^{i-1} \circ \mathbb{S}_{[TS]}^{t-1}, \qquad \forall i = 1, \cdots, N,\ t = 1, \cdots, T.$$

Furthermore, define sub-σ algebra $\mathcal{C} \subseteq \mathfrak{F}$ as the collection of events invariant under $\mathbb{S}_{[CS]}$, the cross-sectional transformation. Symmetrically, sub-σ algebra $\mathcal{C}^{[CS]} \subseteq \mathfrak{F}$ as the collection of events invariant under $\mathbb{S}_{[TS]}$. For an intuitive illustration in Figure 12, imagine $\mathfrak{F}$ is generated with the square partitions, then $\mathcal{C}$ consists of the blocks that are in the same columns, and $\mathcal{C}^{[CS]}$ consists of the blocks that are in the same rows.

Let $X^{[\mathrm{TS}]}$ be a $\mathcal{C}$-measurable random variable, then by the definition of $\mathcal{C}$ above, $X^{[\mathrm{TS}]} = X^{[\mathrm{TS}]} \circ \mathbb{S}_{[\mathrm{CS}]}, \forall \omega \in \Omega$. That is to say, $X_{i,t}^{[\mathrm{TS}]}$ is constant for $i = 1, 2, \cdots$. Hence, the i subscript is redundant and can be dropped. The symmetric direction is also true for a $\mathcal{C}^{[\mathrm{CS}]}$-measurable random variable $X^{[\mathrm{CS}]}$. This provides a definition of random variables that are conventionally represented using only a single subscript.

$$
\begin{aligned}
X_t^{[\mathrm{TS}]} &:= X^{[\mathrm{TS}]} \mathbb{S}_{[\mathrm{TS}]}^{t-1}, && t = 1, \cdots, T, \\
X_i^{[\mathrm{CS}]} &:= X^{[\mathrm{CS}]} \mathbb{S}_{[\mathrm{CS}]}^{i-1}, && i = 1, \cdots, N.
\end{aligned}
$$

The t-subscript random variables are $\mathcal{C}$-measurable. They represent common or aggregate realization and contain distributional information about the "current" cross-sectional population. The factor process f_t is the main example. Symmetrically, $\mathcal{C}^{[\mathrm{CS}]}$-measurable i-subscript variables are about individual i's static characteristics. A static factor loading β_i would be of this sort.

Condition SP.3 implies that each direction itself defines a stationary stochastic process in the traditional sense. Stationary stochastic processes admit the one-directional law of large numbers (LLN).[46] It leads to a LLN result as in Assumption A.1.

Lemma 13 (One directional LLN)**.** *Under Conditions SP.1–3 and if* $\mathbb{E}\|X\|^2 < \infty$*, then*

$$
\frac{1}{N}\sum_{i=1}^{N} X_{i,t} \xrightarrow{L^2} \mathbb{E}\left[X_{\cdot,t} | \mathcal{C}\right], \forall t, \qquad \textit{and} \qquad \frac{1}{T}\sum_{t=1}^{T} X_{i,t} \xrightarrow{L^2} \mathbb{E}\left[X_{i,\cdot} \middle| \mathcal{C}^{[\mathrm{CS}]}\right], \forall i.
$$

Notice the right-hand sides are $\mathcal{C}$ or $\mathcal{C}^{[\mathrm{CS}]}$-measurable. That means, for example, the cross-sectional average converges to a time-specific aggregate, which can be written with a single-t subscript. This allows $X_{i,t}$ to be non-ergodic in either direction. Take the sizes of two firms for example, though $size_{i,t}$ and $size_{j,t}$ are both varying over t, one firm could be inherently larger than the others, such that their time-series averages can converge to different limits. That is to say fixing i, the time-series process $z_{i,t+1}, z_{i,t+2}, z_{i,t+3}, \ldots$ are not ergodic. Their time-series averages converge to different limits given by the expectations conditional on firm identity ($\mathcal{C}^{[\mathrm{CS}]}$).

[46] We write a mean-square convergence result in order to connect to Assumption A.1. An almost-sure convergence result is also possible.

Condition SP.4 introduces the "panel-wise" ergodicity when jointly considering both directions. First, it implies the single-subscript random variables are ergodic, and hence their averages converge to the unconditional expectations. This leads to a result that justifies Assumption A.2.

Lemma 14 (Single-subscript LLN)**.** *Under Conditions SP.1–4, let* $X^{[\mathrm{TS}]}$ *be* $\mathcal{C}$*-measurable and* $X^{[\mathrm{CS}]}$ *be* $\mathcal{C}^{[\mathrm{CS}]}$*-measurable (single-subscript) random vectors, and if* $\mathbb{E}\left\|X^{[\mathrm{TS}]}\right\|^2 < \infty$, $\mathbb{E}\left\|X^{[\mathrm{CS}]}\right\|^2 < \infty$*, then*

$$\frac{1}{N}\sum_{i=1}^{N} X_i^{[\mathrm{CS}]} \xrightarrow{L^2} \mathbb{E}X^{[\mathrm{CS}]}, \text{ as } N \to \infty \qquad \text{and} \qquad \frac{1}{T}\sum_{t=1}^{T} X_t^{[\mathrm{TS}]} \xrightarrow{L^2} \mathbb{E}X^{[\mathrm{TS}]}, \text{ as } T \to \infty.$$

The measure-preserving structure (SP.3) also yields an time series stationarity result that matches Assumption A.3

Lemma 15 (Time-series Stationarity)**.** *Under Conditions SP.1–3, let* X *be an* $\mathfrak{F}$*-measurable random variable (or vector), fixing a finite* N*, the distribution of* $X_{i,t}, X_{2,t}, \ldots, X_{N,t}$ *is stationary over* t*.*

The two results above imply panel-wise averages to converge to deterministic limits, even as we intentionally allow non-deterministic one-directional limits. The convergence result to deterministic limits paves the foundation for frequentist inference on stochastic panel-based models.

Lemma 16 (Panel LLN)**.** *Under Conditions SP.1–4, and if* $\mathbb{E}\|X\|^2 < \infty$*, then as* $N, T \to \infty$*,*

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} X_{i,t} \xrightarrow{L^1} \mathbb{E}X.$$

The four conditions discussed so far are symmetric in the two directions. However, we expect the cross section to be independent in some sense, while the time series evolutions are serially dependent. SP.5 formalizes these properties by strengthening stationarity in SP.3 to exchangeability only for the [CS] direction.[47] Under this condition, $\mathcal{C}^{[\mathrm{CS}]}$-measurable single i-subscript variables (for instance the fixed loadings β_i in PCA) are cross-sectionally i.i.d. And,

[47] We note SP.5 is not required for consistency results, but only for asymptotic normality.

the double-subscript variables are i.i.d. *conditional on* $\mathfrak{F}^{[\text{TS}]}$.[48] For example, *book-to-market*$_{i,t}$ is i.i.d. across stocks *conditional* on the common time-specific information. Unconditionally, it is not independent (only exchangeable) due to aggregate fluctuations of the value ratio.

I.1 Proofs

Proof of Lemma 13:

Proof. This is a direct application of Birkhoff Law of Large Numbers (Hansen and Sargent, 2013, Theorem 2.5.1). Notice in the two right-hand sides, the dot "$\cdot$" can take any natural number without changing the value of the conditional expectation. □

Proof of Lemma 14

Proof. Apply Lemma 13, we have $\frac{1}{N}\sum_{i=1}^{N} X_i^{[\text{CS}]} \xrightarrow{L^2} \mathbb{E}\left[X^{[\text{CS}]}\middle|\mathcal{C}\right]$. It remains to show $X_i^{[\text{CS}]}$ is ergodic. We know $X^{[\text{CS}]}$ is measured by $\mathcal{C}^{[\text{CS}]}$, within which every event is $\mathbb{S}_{[\text{TS}]}$-invariant. So an $\mathbb{S}_{[\text{CS}]}$-invariant event within $\mathcal{C}^{[\text{CS}]}$ must be invariant to both transformations. By condition SP.4, it has probability either 0 or 1. That is to say, $\left\{X_i^{[\text{CS}]}\right\}$ is an ergodic stochastic process (in the traditional one-directional sense). That implies $\mathbb{E}\left[X^{[\text{CS}]}\middle|\mathcal{C}\right] = \mathbb{E}X^{[\text{CS}]}$ w.p. 1, which in turn gives the desired m.s. convergence.

The other direction is symmetric. □

Proof of Lemma 15

Proof. Since X is $\mathfrak{F}$-measurable, $\mathbb{S}_{[\text{CS}]}$ is $\mathfrak{F}$-measurable (SP.1), we know $X_{i,t}, X_{2,t}, \ldots, X_{N,t}$ as a long random vector is $\mathfrak{F}$-measurable too. Since $\mathbb{S}_{[\text{TS}]}$ is measure-preserving with respect to Pr for any $\Lambda \in \mathfrak{F}$, any $\mathfrak{F}$-measurable random variable is time-series stationary (i.e. after repeated $\mathbb{S}_{[\text{TS}]}$ transformations). That includes $X_{i,t}, X_{2,t}, \ldots, X_{N,t}$. □

Proof of Lemma 16:

[48] The first claim is because ergodic exchangeable processes are i.i.d. The second is de Finetti decomposition of exchangeable processes onto sub-σ-algebras of invariant events (see Hansen and Sargent, 2013).

Proof.

$$\frac{1}{NT}\sum_t\sum_i X_{i,t} = \frac{1}{T}\sum_t\left(\frac{1}{N}\sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]\right) + \frac{1}{T}\sum_t \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right] \tag{122}$$

In Lemma 17, we show the first term $\xrightarrow{L^1} \mathbf{0}$, as $N \to \infty$, uniformly across T. This uniform convergence result stated rigorously is: $\forall\epsilon$, $\exists N_\epsilon$ such that $\forall T$, the expected L^1 norm of the first term is less than ϵ.

For the second term, notice the summand $\mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]$ is $\mathcal{C}$-measurable. Therefore, Lemma 14 implies L^2 convergence towards the unconditional expectation, which in turn implies L^1 convergence. That is to say the second term $\xrightarrow{L^1} \mathbb{E}X$, as $T \to \infty$. Also, notice the second term is irrelevant of N. That means, $\forall\epsilon$, $\exists T_\epsilon$ such that $\forall N$, the expected L^1 norm of the second term is less than ϵ.

Combining the results of the two terms, we reach the conclusion of simultaneous L^1 convergence as both $N, T \to \infty$. The reason is for any increasing path of N, T pair, so long as it reaches the region of $N > N_\epsilon, T > T_\epsilon$, the L^1 norm is bounded by 2ϵ. □

I.1.1 Lemma 17 and its Proof

Lemma 17. *Under the conditions of Lemma 16,*

$$\frac{1}{T}\sum_t\left(\frac{1}{N}\sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]\right) \xrightarrow{L^1} 0$$

as $N \to \infty$ uniformly across T.

Proof. Apply $\mathbb{E}\left\|\cdot\right\|$ to the left-hand side, by the triangular inequality,

$$\mathbb{E}\left\|\frac{1}{T}\sum_t\left(\frac{1}{N}\sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]\right)\right\| \tag{123}$$

$$\leq \frac{1}{T}\sum_t \mathbb{E}\left\|\frac{1}{N}\sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]\right\|. \tag{124}$$

We argue given an N, the norm $\left\|\frac{1}{N}\sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t}|\mathcal{C}\right]\right\|$ on the right-hand side is also a

random variable that satisfies SP.1–5. Hence, its unconditional expectation is constant in t. The key is to notice the stochastic panel's conditions SP.1–5 are about $\mathbb{S}_{[\text{TS}]}, \mathbb{S}_{[\text{CS}]}$ with respect to the probability space $\{\Omega, \mathfrak{F}, Pr\}$. Any random variable X on the probability space carries over the same properties. All we need to do is verifying the random variable is indeed $\mathfrak{F}$-measurable. To do that, define random variable $X^{[n]} : \Omega \to \mathbb{R}$ as $X^{[n]} := \left\| \frac{1}{N} \sum_i X_{i,1} - \mathbb{E}\left[X_{\cdot,1} | \mathcal{C}\right] \right\|$. In this expression, $X_{i,1} = X \circ \mathbb{S}_{[\text{CS}]}^{i-1}$ and $\mathbb{S}_{[\text{TS}]}$ is a $\mathfrak{F}$-measurable transformation; $\mathbb{E}\left[X_{\cdot,1} | \mathcal{C}\right]$ is $\mathfrak{F}$-measurable; all other functions (average, norm etc.) are also $\mathfrak{F}$-measurable. Therefore, $X^{[n]}$ is $\mathfrak{F}$-measurable. Notice $X^{[n]} \circ \mathbb{S}_{[\text{TS}]}^{t-1} = \left\| \frac{1}{N} \sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t} | \mathcal{C}\right] \right\|$. It is a stationary stochastic process according to Lemma 15.

As a result, "$\frac{1}{T} \sum_t$" can be dropped from (124):

$$\mathbb{E}\left\| \frac{1}{T} \sum_t \left(\frac{1}{N} \sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t} | \mathcal{C}\right] \right) \right\| \leq \mathbb{E}\left\| \frac{1}{N} \sum_i X_{i,t} - \mathbb{E}\left[X_{\cdot,t} | \mathcal{C}\right] \right\|. \tag{125}$$

On the right-hand side, t can take any value as we have argued the expectation is constant in t. As $N \to \infty$, the right-hand side $\to 0$, by the LLN (or equivalently Lemma 13). That means $\forall T$, the left-hand side is bounded by an object converging to zero as $N \to \infty$. Importantly, the converging bound is irrelevant of T. Therefore, we have the desired uniform convergence result of the left-hand side. □

J Implementation of the ALS Algorithm

This section provides the pseudocode for the Alternating Least Squares (ALS) algorithm used to estimate the IPCA model. The data consist of a panel of dependent variables $x_t \in \mathbb{R}^N$ and instruments $Z_t \in \mathbb{R}^{N \times L}$ for $t = 1, \ldots, T$, together with a pre-specified number of factors K, a chosen identification condition Θ, and a convergence tolerance $\epsilon > 0$.

Algorithm 1 IPCA Alternating Least Squares (ALS) Estimation

% —— Initialization ——

for $t = 1, \ldots, T$ **do**

  $y_t \leftarrow \frac{1}{N} Z_t^\top x_t$

end for

Form $Y = [y_1, \ldots, y_T] \in \mathbb{R}^{L \times T}$

Compute the top-K eigenvectors of $YY^\top \;\rightarrow\; \Gamma^{(0)}$

% —— Main loop ——

for $m = 1, 2, \ldots$ **do**

  % F-step: update raw factors given $\Gamma^{(m-1)}$

  for $t = 1, \ldots, T$ **do**

    $\tilde{f}_t^{(m)} \leftarrow \big(\Gamma^{(m-1)\top} Z_t^\top Z_t \, \Gamma^{(m-1)}\big)^{-1} \Gamma^{(m-1)\top} Z_t^\top x_t$

  end for

  % Γ-step: update raw loadings given $\{\tilde{f}_t^{(m)}\}$

  $A \leftarrow \sum_{t=1}^{T} \big(Z_t^\top Z_t\big) \otimes \big(\tilde{f}_t^{(m)} \, \tilde{f}_t^{(m)\top}\big)$

  $b \leftarrow \sum_{t=1}^{T} \mathrm{vec}\big(Z_t^\top x_t \, \tilde{f}_t^{(m)\top}\big)$

  $\mathrm{vec}(\tilde{\Gamma}^{(m)}) \leftarrow A^{-1} \, b$

  % Normalization step: rotate to the chosen representative in Θ

  Normalize according to Θ

  $\Gamma^{(m)} \leftarrow \tilde{\Gamma}^{(m)}, \quad f_t^{(m)} \leftarrow \tilde{f}_t^{(m)}$ for all t

  % Convergence check

  $S^{(m)} \leftarrow \frac{1}{NT} \sum_{t=1}^{T} \mathrm{vec}\big(Z_t^\top (x_t - Z_t \, \Gamma^{(m)} f_t^{(m)}) \, f_t^{(m)\top}\big)$

  if $\|S^{(m)}\|_\infty < \epsilon$ **then**

    break

  end if

end for

Set $\bar{\Gamma} \leftarrow \Gamma^{(m)}, \quad \bar{f}_t \leftarrow f_t^{(m)}$ for all t

The initialization forms characteristic-managed portfolios $y_t = Z_t^\top x_t/N$ and uses their leading eigenvectors as a starting point; this is a good approximation when $Z_t^\top Z_t/N$ does not vary too much across t. Each iteration consists of an F-step, which is a cross-sectional OLS for every t, a raw Γ-step, which is a pooled panel OLS, and then a normalization step that rotates the raw pair $(\tilde{\Gamma}^{(m)}, \{\tilde{f}_t^{(m)}\})$ to the representative associated with the chosen identification condition Θ. Convergence is monitored via the score function after normalization; an equivalent alternative is to stop when the change in the objective $|\widehat{G}(\Gamma^{(m)}) - \widehat{G}(\Gamma^{(m-1)})| < \epsilon$.